

Tournament score sequences and weighted component asymptotics

Explicit corrections and uniform inverse models

Research report 2 October 2026

Part II: conditional phase coexistence 3 October 2026

Part III: beyond criticality 4 October 2026

Abstract

The exact generating function for tournament score sequences permits a clean separation between an explicit square-root singularity and an analytic divisor remainder. We use that separation to derive computable asymptotic corrections for all and strong score sequences, and a uniform expansion for the generating function weighted by the number of irreducible components. The latter is valid to every fixed half-power order on a specified logarithmic subcritical window, with an absolute remainder. Its first two terms yield an explicit coexistence profile with a $\log \log n$ centering correction. We also give inverse-location results for specified one-variable models and, when n is known, for component-weight calibration. Exact small-size enumeration, symbolic coefficient identities, numerical comparisons, and portable replay code accompany the proof.

The leading enumeration theorem, exact generating function, and general renewal and heavy-traffic mechanisms are established prior work. The emphasis here is the explicit local lattice specialization, correction generators, uniform error bounds, and carefully delimited inverses. No exhaustive novelty claim is made.

Part II (added 3 October 2026) answers the first further question of Part I. Under a component weight approaching criticality at the coexistence rate, the score sequence is asymptotically a logistic mixture of a giant-block regime and a regime with a linear number of blocks. On the giant event the remaining blocks are, up to total variation $O(1/\log n)$, two independent geometric renewal strings, which gives $\text{Gamma}(2, 1)$ laws for the block count and the giant's deficit, stable residual fluctuations, a mixed Fréchet law for the second-largest block and every fixed inverse-logarithmic correction; on the other event the centered block count has a $3/2$ -stable limit jointly with a Fréchet largest block. Its coefficient input re-proves Part I's uniform two-term theorem; the probabilistic results are new to the repository.

Part III (added 5 October 2026) treats the supercritical side that Parts I and II exclude, on a fixed small interval of weights above the critical one. The weighted count is an exact moving-pole residue plus a branch-cut expansion to every fixed order, uniformly through criticality, with computable kernels resolving the coalescence at $nt = O(1)$, where t is the excess growth rate. The block statistics change on the different scale $tn^{2/3} = O(1)$: the centred block count has a tempered $3/2$ -stable limit, Gaussian beyond that scale, and the large blocks form a tempered Poisson process whose maximum passes from the critical Fréchet-type law to a Gumbel law. Convergent Puiseux inversion of the free energy and Lambert- W centrings complete the inverse direction. Its critical-endpoint terms re-prove Part I's critical expansion, adding its $n^{-3/2}$ coefficient; the rest is new to the repository.

Results and scope at a glance

- All scores are OEIS A000571; strong scores are A351822. Strongly connected *labeled tournaments*, A054946, are a different counting problem.

- At each fixed order K , all and strong counts have explicit relative corrections through n^{-K} with remainder $O(n^{-K-1})$.
- For $0 \leq \tau \leq C \log n$ with fixed C , the normalized weighted count has computable terms $n^{-m/2} \widehat{C}_m(\tau)$ and an absolute $O_{M,C}(n^{-(M+1)/2})$ remainder at every fixed M .
- Two coefficient contributions coexist at $\tau = \frac{1}{2} \log n + 2 \log \log n + \log(\mu\sqrt{\pi}/2) + s$. This is a coefficient asymptotic, not a proved conditional distributional mixture.
- The inverses specify the model, normalization, and known quantities. They do not certify finite- n integer rounding or finite- n confidence intervals.

The results do not cover $\tau < 0$, a supercritical moving pole, truncation order growing with n , or convergence of the full asymptotic series. All numerical decimal evaluations are non-interval computations.

[write] **Added 2 October 2026, batch 77.** This report is a single manuscript, printed in full: manuscript 68 of batch 77 of ProveIt’s incoming-reports intake, delivered as the archive `tournament-score-sequences-source.zip` (wrapper directory `tournament-score-sequences/`) in commit `096ee7b87`, and placed in commit `34f1acd4b` in the research-report collection, among the OEIS-sequence asymptotics reports (main file `report.tex`, now `article.tex`). The manuscript names no ProveIt commit and continues no repository path, so it has no pin; it names neither an author nor a tool, and its PDF metadata leave the author empty. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. No other report of the repository treats tournament score sequences or A000571, A351822, A145855, A054946. The writing step prefixed every label with `tss:`, typeset the six binomial coefficients with `\binom` instead of `\choose` (same output, no amsmath warning), set the bibliography ragged-right, and added these [write] notes; no statement, proof or number of the manuscript was altered.

Credit, kept. The exact generating function (3) and the block decomposition are Claesson–Dukes–Franklín–Stefánsson’s [1] (with the divisor evaluation attributed there to Alekseyev), the leading asymptotics (8) are Kolesnik’s [2] (shorter proof [3]), and the renewal and logarithmic heavy-traffic/heavy-tail mechanisms are Blanchet–Glynn’s and Olvera-Cravioto–Blanchet–Glynn’s [4, 5]. The report’s contribution is the explicit local refinement; no exhaustive novelty search is claimed.

Notation. The manuscript reuses several letters. Each symbol is printed as delivered; the table fixes its meaning by section, with the tempting false readings.

symbol	meaning here (and what it is <i>not</i>)
W, W_{-1}	$W(z, u)$ is the component-weighted generating function and $W_n(u)$ its coefficients; W_{-1} in (51) is the lower branch of Lambert’s function.
s	$s = nw$ is the Hankel variable (Sections 4–5); s is also the shift of the coexistence centre, $\tau = b_n + s$ (Sections 6–7).
b_j, b_n, B	b_j are the odd Puiseux coefficients of $B(t)$ (15); b_n is the coexistence centre (47); $B_j(x)$ are Bernoulli polynomials.
c, c_m^σ, C	$c = e^{-\lambda}\mu$; c_m^σ the fixed-order corrections; c also a contour constant in $\exp(-cn w)$. C is the fixed window constant ($0 \leq \tau \leq C \log n$); $C_m(\tau)$ the uniform terms; $C_\sigma = e^{\sigma\lambda}/(2\sqrt{\pi})$.
D_j^σ, D, d_j, d	D_j^σ the odd local coefficients (21); $D(w)$ the denominator series (40) with coefficients d_j ; d a divisor in (2). $d_4 = \ell$.
E, e_j, E_n	$E(v) = \sum e_j v^j$ is the even part (14); $E_n(w)$ the resolvent error (35). e_2 is a constant, not e^2 .
$H, H_j, \mathcal{H}, h$	$H(\tau)$ the entire kernel (32); H_j harmonic numbers in (16); $\mathcal{H}$ the Hankel contour; $h = \log 4$ (Section 7).
k, K	$k = 2/(3\mu)$; k is also an index. K is a fixed truncation order (Section 3) and a generic constant; K_s the path constant (54).
p, q, q_j	$p = 1 - e^{-\lambda}$; $q = \tau/n$; q_j the renewal mass function.
t	$t = \sqrt{1 - 4z}$ (Sections 2–3); t is the free weight parameter of $\mathcal{M}_{n,M}(t)$ in Section 7.
F, R, r_n	$F(z)$ the central-binomial part and $R(z)$ the divisor remainder (10), $r_n = [z^n]R$; $F_{K,\sigma}(x)$ the fixed-order inverse model.
a, a_c, a_n	$a = a_n(\tau) = u_n(\tau)c$; $a_c = c/p$ its critical value; $a_x(\tau_x)$ its real-variable extension in (52).
m, M	m counts strong blocks in $S_{n,m}$ and indexes c_m^σ, C_m ; M is a truncation order and also a summation cutoff in (20).
I, S, N_n	$I(z), I_n$ strong sequences; $S(z), S_n$ all sequences, $S_{n,m}$ by number of blocks; N_n the divisor sum (A145855). I_n omits A351822’s empty object.

[write] **Added 3 October 2026, batch 85.** Since 3 October 2026 this report has two Parts. Part I is the batch-77 manuscript described in the preceding note, printed as before: its sections, equations, theorems and labels keep their numbers, so “Section 6” and “Section 9” still mean what the manuscript of Part II cites. The preceding note’s “single manuscript, printed in full” now describes Part I; its statement that no other report treats tournament score sequences remains true. Part II (Sections 10–20 and two appendices) is manuscript 04 of batch 85, an AI-assisted, unrefereed and unformalized manuscript pinned at commit `d68b65ea0`. It answers Part I’s first further question: at the weight $u_n(b_n + s)$ the structure is a logistic mixture, with probability tending to $(1 + e^{-s})^{-1}$ of one block of size $n - O_{\mathbb{P}}(n/\log n)$ and otherwise $n/a_c + O_{\mathbb{P}}(n^{2/3})$ blocks and no macroscopic block, and it derives conditional Gamma, stable, Fréchet and point-process laws, residual approximations to every fixed inverse-logarithmic order, and an $O(1/\log n)$ rate (Theorems 12.1–12.4, Proposition 18.1). The scope bullet above that calls the coexistence profile “a coefficient asymptotic, not a proved conditional distributional mixture” remains true of Corollary 6.1 by itself; the mixture is Part II’s theorem. Part II’s symbols m, a_n, r_n, k, K, d and B do not mean what they mean here; a separate table at the start of Part II fixes them.

[write] **Added 5 October 2026, batch 98.** Since 5 October 2026 this report has three Parts. Part III (Sections 21–32 and Appendices C–E, printed after Part II’s two appendices) is manuscript 03 of batch 98, an AI-assisted, unrefereed and unformalized manuscript pinned at commit `106eb191a`. It answers Part I’s second further question, the supercritical crossover, on a fixed small interval of weights $u \geq u_c$, and Part II’s RQ6 in part: an exact pole residue plus a branch-cut expansion to every fixed order (Theorem 24.1), tempered-stable and Gaussian block counts (Theorems 27.2, 27.3), a tempered Poisson process of large blocks and a Gumbel maximum (Theorems 28.2, 29.1). The sentence before the first note above, that the results do not cover $\tau < 0$ or a supercritical moving pole, remains true of Part I and of Part II; Part III’s argument $a_n(t) \geq 0$ is $-\tau$ (Section 24). The joint theory across $\tau = 0$ is still open (Part III’s RQ4). Part III’s symbols $b_n, a_n(t), t, \tau, H, B, \mathcal{K}_j, \ell, L, M_n$ and s do not mean what they mean in Parts I or II; a table at the start of Part III fixes them.

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Part I

Explicit corrections and uniform inverse models

1 The counting problem and established identities

A tournament is an orientation of a complete graph. Its score sequence is the nondecreasing list of vertex outdegrees. Let S_n count distinct score sequences of length n , and set $S_0 = 1$. A sequence $(s_1, \dots, s_n)$ is a score sequence precisely when

$$0 \leq s_1 \leq \dots \leq s_n \leq n-1, \quad \sum_{i=1}^j s_i \geq \binom{j}{2} \ (j < n), \quad \sum_{i=1}^n s_i = \binom{n}{2}. \quad (1)$$

The sequence is strong or irreducible when each proper-prefix inequality is strict. This includes the singleton sequence. We use I_n for the number of nonempty strong sequences, with $I_0 = 0$. The OEIS entry A351822 includes an empty-object value 1 at index zero; its full listed generating function is consequently $1 + I(z)$ under our notation.

Equality indices in (1) give the unique ordered decomposition into strong blocks. When a prefix of length j is removed, subtract j from each subsequent score. Conversely, concatenating two blocks after adding the first length to the second block gives a score sequence. Thus a block is an object of the score-sequence class, not an underlying labeled tournament. The first values are

n	0	1	2	3	4	5	6	7	8	9	10
S_n	1	1	1	2	4	9	22	59	167	490	1486
I_n	0	1	0	1	1	3	7	21	61	184	573

Let φ be Euler's totient function and define

$$N_n = \frac{1}{2n} \sum_{d|n} (-1)^{n+d} \varphi(n/d) \binom{2d}{d}, \quad A(z) = \sum_{n \geq 1} \frac{N_n}{n} z^n. \quad (2)$$

The auxiliary N_n is OEIS A145855. Claesson, Dukes, Franklin, and Stefánsson (CDFS) prove the score-sequence identity [1, Corollaries 12–13]

$$S(z) = \sum_{n \geq 0} S_n z^n = \exp A(z). \quad (3)$$

Their paper attributes the divisor evaluation in (2) to Alekseyev's 2008 proof of a conjecture of Jovović. The signs $(-1)^{n-d}$ and $(-1)^{n+d}$ are equal. These established exact identities are inputs to the analytical results below; we do not reproduce the full combinatorial proof of (3).

The ordered block decomposition now gives

$$I(z) = 1 - \exp(-A(z)), \quad W(z, u) = \frac{1}{1 - uI(z)} = \sum_{n, m \geq 0} S_{n,m} z^n u^m. \quad (4)$$

Here $S_{n,m}$ counts sequences with m strong blocks, $S_{0,0} = 1$, and $W_n(u) = [z^n]W(z, u)$. For integer $u > 0$, the factor u^m colors the blocks; for real $u > 0$ it is a positive component weight. In particular $W_n(1) = S_n$ and $S_{n,1} = I_n$.

Exact computation uses

$$nS_n = \sum_{k=1}^n N_k S_{n-k}, \quad (5)$$

$$I_n = S_n - \sum_{k=1}^{n-1} I_k S_{n-k}, \quad (6)$$

$$W_0(u) = 1, \quad W_n(u) = u \sum_{k=1}^n I_k W_{n-k}(u). \quad (7)$$

The first two recurrences can be evaluated with integers. Independently enumerating the solutions of (1) checks both the totals and the number of equality indices, hence the component statistic itself.

1.1 Prior asymptotics and the intended refinement

Kolesnik [2] proved the leading asymptotics for S_n , I_n , and each fixed component count. In the notation introduced below, these include

$$S_n \sim \frac{e^\lambda}{2\sqrt{\pi}} \frac{4^n}{n^{5/2}}, \quad I_n \sim \frac{e^{-\lambda}}{2\sqrt{\pi}} \frac{4^n}{n^{5/2}}, \quad \frac{S_{n,m}}{S_n} \rightarrow mp^{m-1}e^{-2\lambda}. \quad (8)$$

Bassan, Donderwinkel, and Kolesnik give a shorter proof of the leading result [3]. We do not present the leading enumeration problem as an open conjecture solved here.

Uniform renewal expansions and geometric-sum approximations have an established literature, including Blanchet and Glynn [4]. Olvera-Cravioto, Blanchet, and Glynn [5] explicitly study a logarithmic heavy-traffic/heavy-tail transition. Their queueing and survival-probability setting differs from the local lattice coefficient problem here. We credit the transition mechanism to that literature and prove the needed coefficient estimates directly, rather than importing an unchecked theorem. The contribution of this report is an explicit refinement for the score-sequence model, with constants, fixed-order generators, and inverse scopes. A bounded primary-source review supports this focus; it cannot certify that no equivalent refinement exists elsewhere.

2 An exact analytic and singular splitting

Throughout, let

$$\rho = \frac{1}{4}, \quad \lambda = A(\rho), \quad \mu = \rho A'(\rho), \quad p = 1 - e^{-\lambda}, \quad c = e^{-\lambda}\mu, \quad u_c = \frac{1}{p}, \quad a_c = \frac{c}{p}. \quad (9)$$

All these constants except ρ are derived from the exact series, and λ, μ, p, c, a_c are positive.

Separate the divisor $d = n$ in (2):

$$A(z) = F(z) + R(z), \quad F(z) = \frac{1}{2} \sum_{n \geq 1} \binom{2n}{n} \frac{z^n}{n^2}. \quad (10)$$

Writing $r_n = [z^n]R(z)$ gives the exact rational formula

$$r_n = \frac{2nN_n - \binom{2n}{n}}{2n^2}. \quad (11)$$

For a proper divisor d of n , $d \leq n/2$. Since $\varphi(n/d) \leq n$ and $\binom{2d}{d} \leq 4^d \leq 2^n$, the crude bound using at most n divisors gives $|r_n| \leq 2^{n-1}$. Therefore R is analytic in $|z| < 1/2$. Analyticity on the boundary $|z| = 1/2$ is not needed or asserted.

Put $t = \sqrt{1-4z}$, choosing the principal branch near the real interval $z < \rho$, and set $v = t^2$. Differentiation of the central-binomial series yields

$$\frac{dF}{dt} = -\frac{2t}{1-t^2} \log \frac{2}{1+t}. \quad (12)$$

Integrating the even and odd parts, with $F(\rho) = \pi^2/12 - \log^2 2$, gives the convergent local decomposition

$$A(z) = E(t^2) + B(t), \quad (13)$$

$$E(v) = \frac{\pi^2}{12} - \log^2 2 + \log 2 \log(1-v) - \frac{1}{4} \log^2(1-v) + R\left(\frac{1-v}{4}\right), \quad (14)$$

$$B(t) = \sum_{j \geq 1} b_j t^{2j+1}, \quad b_j = \frac{2}{2j+1} \sum_{k=0}^{j-1} \frac{1}{2k+1}. \quad (15)$$

For example, $b_1 = 2/3$, $b_2 = 8/15$, and $b_3 = 46/105$. Write $E(v) = \sum e_j v^j$. Then

$$e_0 = \lambda, \quad e_j = -\frac{\log 2}{j} - \frac{H_{j-1}}{2j} + \frac{(-1)^j R^{(j)}(\rho)}{4^j j!} \quad (j \geq 1), \quad (16)$$

where $H_0 = 0$. In particular $e_1 = -\mu$ and

$$e_2 = -\frac{\log 2}{2} - \frac{1}{4} + \frac{R''(\rho)}{32}. \quad (17)$$

Thus, with $w = 1 - 4z$,

$$A(z) = \lambda - \mu w + \frac{2}{3} w^{3/2} + e_2 w^2 + O(w^{5/2}), \quad (18)$$

$$I(z) = p - cw + \frac{2}{3} e^{-\lambda} w^{3/2} + O(w^2). \quad (19)$$

These are analytic Puiseux expansions on a fixed slit neighborhood, not merely real-axis estimates.

The split also accelerates computation:

$$\lambda = \frac{\pi^2}{12} - \log^2 2 + \sum_{n \geq 1} \frac{r_n}{4^n}, \quad \mu = \log 2 + \sum_{n \geq 1} \frac{nr_n}{4^n}. \quad (20)$$

After terms through M , the absolute tail bounds are respectively 2^{-M-1} and $(M+2)2^{-M-1}$. Derivative tails are similarly bounded by a polynomial times 2^{-M} . Exact-integer subtraction in (11) avoids cancellation before the conversion to floating point.

Constant	Approximate value	Constant	Approximate value
λ	0.3302375439859293	μ	0.6739526018515949
e_2	-0.5989545674814332	u_c	3.5555932000209863
a_c	1.7223486864283872	$k = 2/(3\mu)$	0.9891892469

The displayed decimals are numerical evaluations, not certified enclosures.

2.1 The renewal interpretation

Define $q_j = I_j 4^{-j}/p$ for $j \geq 1$. This is a probability mass function, and its generating function is $I(y/4)/p$. Its mean is $\sum j q_j = \rho I'(\rho)/p = c/p = a_c$. Equation (8) gives a mass tail proportional to $j^{-5/2}$, hence finite mean and infinite variance. At $u = u_c$, $W(y/4, u_c)$ is the renewal generating function. The limiting constant a_c^{-1} is the classical finite-mean renewal behavior. This interpretation motivates the two mechanisms but is not used in place of the contour proof.

3 Fixed weights and explicit enumeration corrections

For $\sigma \in \{+1, -1\}$, define $T_n^+ = S_n$ and $T_n^- = I_n$ for $n \geq 1$. Introduce

$$D_j^\sigma = [t^{2j+3}] \exp(\sigma(E(t^2) - \lambda)) \sinh B(t), \quad j \geq 0. \quad (21)$$

Both signs use the same $\sinh B$: the odd part of $\exp(\sigma B)$ is $\sigma \sinh B$, and the minus sign in $-\exp(-A)$ cancels it for I_n .

Define the standard gamma-ratio coefficients by

$$\frac{\Gamma(n - \alpha)}{\Gamma(n + 1)} \sim n^{-\alpha-1} \sum_{\ell \geq 0} g_\ell(\alpha) n^{-\ell}. \quad (22)$$

They result by exponentiating the formal series

$$\sum_{j \geq 1} \frac{(-1)^{j+1} (B_{j+1}(-\alpha) - B_{j+1}(1))}{j(j+1)n^j}, \quad (23)$$

where $B_j(x)$ is the Bernoulli polynomial. In particular $g_0 = 1$ and $g_1(\alpha) = \alpha(\alpha + 1)/2$.

Theorem 3.1 (All fixed orders for all and strong scores). *For every fixed integer $K \geq 0$,*

$$T_n^\sigma = \frac{e^{\sigma\lambda}}{2\sqrt{\pi}} 4^n n^{-5/2} \left(\sum_{m=0}^K \frac{c_m^\sigma}{n^m} + O_K(n^{-K-1}) \right), \quad (24)$$

where

$$c_m^\sigma = 2\sqrt{\pi} \sum_{j=0}^m \frac{D_j^\sigma g_{m-j}(j + 3/2)}{\Gamma(-j - 3/2)}. \quad (25)$$

The first three coefficients are

$$\begin{aligned} c_0^\sigma &= 1, \\ c_1^\sigma &= -\frac{1}{8} + \sigma \frac{5}{2} \mu, \\ c_2^\sigma &= \frac{1}{128} + \frac{35}{8} \mu^2 + \sigma \left(\frac{63}{16} \mu + \frac{35}{4} e_2 \right). \end{aligned} \quad (26)$$

Every finite collection is computable from finitely many derivatives of R at ρ .

Proof. The exact split continues to a dented neighborhood of $z = \rho$, while F is analytic in the plane slit along $[\rho, \infty)$ and R is analytic for $|z| < 1/2$. Thus ρ is the only singularity on the convergence

circle. The odd local part of either generating function is $e^{\sigma\lambda} \sum_{j \geq 0} D_j^\sigma t^{2j+3}$. For noninteger α the exact coefficient identity is

$$[z^n](1-4z)^\alpha = 4^n \frac{\Gamma(n-\alpha)}{\Gamma(-\alpha)\Gamma(n+1)}. \quad (27)$$

Apply (22) to each retained odd power and collect equal relative powers of n^{-1} ; this gives (25).

For the error, retain additional terms of the convergent Puiseux expansion beyond the requested order and use the same dented coefficient contour. The next nonanalytic term and its gamma-ratio expansion give $O(4^n n^{-K-7/2})$ after terms through relative order K . The locally analytic even part is handled by analytic continuation of the full expansion and contour deformation. One must not infer an exponentially small error from a merely local Taylor polynomial, or apply an unqualified transfer rule to an integer-power remainder. Keeping enough further Puiseux terms and the common continuation domain avoids both errors.

Finally $D_0 = 2/3$ and $D_1^\sigma = 8/15 - (2/3)\sigma\mu$. Expanding one term further and using (23) yields (26). $\square$

The second corrections are approximately $c_2^+ = -0.5921736164232970$ and $c_2^- = 4.5821545749204735$. This is a fixed-order asymptotic expansion; it makes no assertion that the infinite series in $1/n$ converges.

Proposition 3.2 (Fixed subcritical and critical weights). *For fixed $0 < u < u_c$, uniformly on compact subsets of that interval,*

$$W_n(u) \sim \frac{ue^{-\lambda}}{(1-up)^2} \frac{4^n}{2\sqrt{\pi}n^{5/2}}. \quad (28)$$

There is a full expansion in relative integer powers of $1/n$ at every fixed order, generated by the odd t coefficients of

$$\frac{1}{1-u+u\exp\{-E(t^2)-B(t)\}}$$

and (27). At the critical weight,

$$4^{-n}W_n(u_c) = \frac{1}{a_c} + \frac{2}{3\mu a_c \sqrt{\pi}} n^{-1/2} + O(n^{-3/2}). \quad (29)$$

There is no n^{-1} term.

Proof. For subcritical u , $|uI(z)| \leq up < 1$ on $|z| \leq \rho$, so the analytic composition has no zero denominator there. Compactness and the continuation in Section 2 provide a common dented neighborhood on compact weight sets. The first odd term is $\frac{2}{3}ue^{-\lambda}(1-up)^{-2}t^3$; transfer gives (28). At u_c , Laurent expansion gives

$$W(z, u_c) = a_c^{-1}t^{-2} + \frac{2}{3\mu a_c}t^{-1} + O(1).$$

The exact t^{-2} coefficient is 4^n . The next nonanalytic term after t^{-1} is t , of order $4^n n^{-3/2}$. Nonnegative even powers are locally analytic and produce no algebraic n^{-1} term under the full continuation argument. $\square$

[write] **Added 5 October 2026, batch 98.** Part III re-derives (29) as the endpoint $t = 0$ of its uniform expansion and supplies the next coefficient: (164) reads $4^{-n}W_n(u_c) = 1/a_c + \frac{2}{3\mu a_c \sqrt{\pi}} n^{-1/2} + \frac{\kappa \mathcal{K}_1(0)}{\pi a_c^2} n^{-3/2} + O(n^{-5/2})$, with $\kappa = 2e^{-\lambda}/(3p)$ and $\mathcal{K}_1(0)$ explicit in (165), and only the powers $n^{-j-1/2}$ occur after the constant. Intake computations from exact coefficients (at placement and at the write; not shipped) found $n^{5/2}$ times the remainder after these three terms to be $-0.0109, -0.0103, -0.0101, -0.0100$ at $n = 10, 20, 40, 80$, consistent with the stated $O(n^{-5/2})$. Part III is unrefereed, prepared with AI assistance and not formalized.

4 A uniform logarithmic near critical expansion

Let

$$u_n(\tau) = \frac{n}{np + \tau c}, \quad a_n(\tau) = u_n(\tau)c, \quad k = \frac{2}{3\mu}. \quad (30)$$

Thus $0 \leq \tau$ corresponds to $u_n(\tau) \leq u_c$. We sometimes write u, a for $u_n(\tau), a_n(\tau)$.

Theorem 4.1 (First two uniform terms). *Fix $C > 0$. Uniformly for $0 \leq \tau \leq C \log n$,*

$$4^{-n}W_n(u_n(\tau)) = \frac{1}{a_n(\tau)} \left\{ e^{-\tau} + kn^{-1/2}H(\tau) \right\} + O_C(n^{-1}), \quad (31)$$

where the entire function

$$H(\tau) = \frac{1}{\sqrt{\pi}} {}_1F_1(2; 1/2; -\tau) = \sum_{j \geq 0} \frac{(-1)^j (j+1) \tau^j}{\Gamma(j+1/2)}. \quad (32)$$

The remainder is absolute on the 4^{-n} normalization. The displayed model is eventually positive throughout this window, and its relative error is $O_C((\log n)^2 n^{-1/2})$. Replacing $e^{-\tau}$ by the exact carrier $(1 + \tau/n)^{-n-1}$ preserves the absolute bound.

4.1 One common coefficient contour

Use $y = 4z$, $w = 1 - y$, and $q = \tau/n$. By (19) and (30),

$$1 - uI(y/4) = a\{w + q - kw^{3/2} + O(w^2)\}. \quad (33)$$

The remainder constant is uniform for u in a compact positive neighborhood of u_c , which contains the whole parameter window for large n .

Fix $\pi/2 < \theta < \pi$. On the rays $w = re^{\pm i\theta}$ and throughout the joining arc $|\arg w| \leq \theta$,

$$|w + q| \geq \cos(\theta/2)(|w| + q), \quad q \geq 0. \quad (34)$$

The perturbation in (33) is therefore less than half the leading denominator in a single sufficiently small neighborhood, independent of q . This excludes local zeros on the indented sector. At $q = 0$, the critical endpoint singularity is encircled by the indentation, not omitted.

Away from $y = 1$ on $|y| = 1$, positivity and aperiodicity give $|I(y/4)| < p$. Indeed, equality would require the positive terms at degrees 1, 3, 4 to have the same phase; $y = y^3 = y^4$ forces $y = 1$. A compact arc away from 1 has a strict nonvanishing margin. Analytic continuation from $F + R$ thickens this arc slightly while retaining that margin uniformly as $u \rightarrow u_c$.

Combining this compact continuation with the local sector yields a common dented contour: two local rays joined to an indentation $|w| = 1/n$, and outer parts with $|y| \geq 1 + \delta$ for some fixed $\delta > 0$ away from the local pieces. Outer contributions are exponentially small. This argument asserts a suitable zero-free contour neighborhood, not an unjustified zero-free disk beyond the original circle.

4.2 Resolvent expansion and its error

On the local contour, a geometric-resolvent expansion gives

$$W(y/4, u) = \frac{1}{a(w+q)} + \frac{k}{a} \frac{w^{3/2}}{(w+q)^2} + E_n(w), \quad (35)$$

where

$$|E_n(w)| \leq K \left\{ \frac{|w|^2}{(|w|+q)^2} + \frac{|w|^3}{(|w|+q)^3} \right\} \leq 2K. \quad (36)$$

For sufficiently small ray radius, $|(1-w)^{-n-1}| \leq K \exp(-cn|w|)$. Thus the ray integral of (36) is $O(n^{-1})$; the indentation has length $O(n^{-1})$ and bounded coefficient kernel. The common outer contour adds only exponentially small terms.

The first model coefficient is exact:

$$[y^n](w+q)^{-1} = (1+q)^{-n-1}. \quad (37)$$

Its difference from $e^{-\tau}$ is $O(n^{-1})$ uniformly on the stated logarithmic window: the exponential factor absorbs the polynomial in τ from Taylor expansion.

For the fractional model set $s = nw$. Let $\mathcal{H}$ be the fixed-angle Hankel contour starting at infinity on the lower ray, following the unit right-hand arc from angle $-\theta$ to θ , and returning to infinity on the upper ray. The principal power of s is used. On this contour, $|s+\tau| \geq c(|s|+\tau)$ and $|s^{3/2}/(s+\tau)^2| \leq K|s|^{-1/2}$. Hence

$$\begin{aligned} [y^n] \frac{w^{3/2}}{(w+q)^2} &= n^{-1/2} \frac{1}{2\pi i} \int_{\mathcal{H}} e^s \frac{s^{3/2}}{(s+\tau)^2} ds + O(n^{-3/2}) \\ &= n^{-1/2} H(\tau) + O(n^{-3/2}). \end{aligned} \quad (38)$$

To justify replacing $(1-s/n)^{-n-1}$ by e^s , first restrict to $|s| \leq n^\eta$ with fixed $\eta < 1/2$. The difference is bounded by $O((|s|+|s|^2)/n)$ times a decaying exponential. This is uniformly integrable against the preceding bound. The remaining rays are exponentially suppressed. For $|\tau| < 1$, expanding $(s+\tau)^{-2}$ on the unit indentation and using the reciprocal-gamma Hankel formula gives (32); analytic continuation extends the identity along $\tau \geq 0$. The model pole at $s = -\tau$ is on the excluded cut, so no residue has been silently removed. Equations (35)–(38) prove the absolute part of Theorem 4.1.

4.3 Positivity and relative error

Repeated expansion of the inverse-Laplace integral gives

$$H(\tau) = \frac{3}{4\sqrt{\pi}\tau^2} \left(1 + \frac{5}{\tau} + \frac{105}{4\tau^2} + O(\tau^{-3}) \right) \quad (\tau \rightarrow +\infty). \quad (39)$$

For instance expand $(s+\tau)^{-2}$ on $|s| \leq \tau/2$ and bound the rest of the fixed rays exponentially. Reciprocal gamma values give the displayed coefficients. In particular $H(\tau) > 0$ eventually and $H(\tau) \asymp \tau^{-2}$ there. On a fixed initial interval, $e^{-\tau}$ dominates the bounded $n^{-1/2}H(\tau)$ term; no global positivity of H alone is required. The full model is consequently bounded below by a positive constant times $n^{-1/2}/(\log n)^2$ on the window. Dividing the absolute error by this bound proves the relative statement.

5 The half power generator to every fixed order

The exact local denominator series is

$$D(w) = \frac{p - I((1-w)/4)}{c} = \sum_{j \geq 2} d_j w^{j/2}, \quad d_2 = 1, \quad d_3 = -k, \quad d_4 = \ell = \frac{\mu^2/2 - e_2}{\mu}. \quad (40)$$

In particular $\ell \simeq 1.225695427220266$. The sign before e_2 is minus, because $p - I = \exp(-A) - \exp(-\lambda)$.

5.1 Entire kernel functions

For the finite monomials arising below, define

$$\mathcal{K}_{\alpha,b}(\tau) = \frac{1}{2\pi i} \int_{\mathcal{H}} e^s \frac{s^\alpha}{(s+\tau)^b} ds = \sum_{j \geq 0} \frac{(-1)^j (b)_j \tau^j}{j! \Gamma(b+j-\alpha)}. \quad (41)$$

The contour identity holds for $\tau \geq 0$, and the convergent series supplies the entire continuation. Here $(b)_j$ denotes the rising factorial and $1/\Gamma$ is its entire reciprocal, including its zeros. Equivalently this is ${}_1\tilde{F}_1(b; b-\alpha; -\tau)$, the *regularized* confluent hypergeometric function. If $b-\alpha$ is a nonpositive integer, the unregularized product ${}_1F_1(b; b-\alpha; -\tau)/\Gamma(b-\alpha)$ is not a literal defined expression. Formula (41) avoids this degeneracy.

Theorem 5.1 (Uniform expansion at every fixed order). *Put $\varepsilon = n^{-1/2}$ and define $C_m(\tau)$ by the finite formal coefficient extraction*

$$C_m(\tau) = \frac{1}{2\pi i} \int_{\mathcal{H}} e^s [\varepsilon^m] \frac{\exp \left\{ \sum_{j \geq 1} \varepsilon^{2j} \left(\frac{s^j}{j} + \frac{s^{j+1}}{j+1} \right) \right\}}{s + \tau + \sum_{j \geq 1} d_{j+2} \varepsilon^j s^{1+j/2}} ds. \quad (42)$$

For each fixed m , this is a finite linear combination of the entire kernels in (41). For every fixed integer $M \geq 0$ and fixed $C > 0$,

$$4^{-n} W_n(u_n(\tau)) = \frac{1}{a_n(\tau)} \sum_{m=0}^M n^{-m/2} C_m(\tau) + O_{M,C}(n^{-(M+1)/2}) \quad (43)$$

uniformly on $0 \leq \tau \leq C \log n$. In particular $C_0 = e^{-\tau}$, $C_1 = kH(\tau)$, and

$$C_2(\tau) = e^{-\tau} \left\{ \frac{\tau^2}{2} - \tau - \ell(\tau^2 - 2\tau) + k^2 \left(-3\tau + 3\tau^2 - \frac{\tau^3}{2} \right) \right\}. \quad (44)$$

For $M \geq 1$ the model is eventually positive, and its relative error is $O_{M,C}((\log n)^2 n^{-M/2})$.

Proof. On the contour of Section 4.1, write $D(w) = w + V(w)$ with $V(w) = O(w^{3/2})$. Inequality (34) gives $V(w)/(w+q) = O(|w|^{1/2})$ uniformly in $q \geq 0$. Expanding the resolvent through M such factors leaves a remainder bounded by

$$K \frac{|w|^{(M+1)/2}}{|w|+q} \leq K |w|^{(M-1)/2}. \quad (45)$$

Since the local Puiseux series is genuinely convergent, truncating the numerators at the corresponding total half-power order has the same or a smaller error. Integration along the rays and the radius-1/ n indentation makes (45) an $O(n^{-(M+1)/2})$ coefficient error, including $M = 0$.

After $s = nw$, the exact coefficient kernel has the identity

$$(1 - \varepsilon^2 s)^{-\varepsilon^{-2}-1} = e^s \exp \left\{ \sum_{j \geq 1} \varepsilon^{2j} \left(\frac{s^j}{j} + \frac{s^{j+1}}{j+1} \right) \right\}. \quad (46)$$

For the prescribed fixed M , choose a sufficiently small fixed $\eta > 0$ and expand on $|s| \leq n^\eta$. Taylor remainders are bounded by the required power of $n^{-1/2}$ times a fixed polynomial in $|s|$ and a decaying exponential. The denominator bound gives uniform integrable domination for each finite coefficient and remainder. Outside this range both the exact kernel and the finitely many model terms are exponentially small. The common outer contour also contributes exponentially small terms. This proves (42) and (43), with constants depending on M, C but not on τ, n .

At order ε^2 , the three kernels are

$$\frac{s + s^2/2}{s + \tau}, \quad -\ell \frac{s^2}{(s + \tau)^2}, \quad k^2 \frac{s^3}{(s + \tau)^3}.$$

Their time-one inverse Laplace values yield (44). Polynomial parts in s have zero Hankel integral (or are distributions at time zero), so they do not contribute at time one. In particular $C_2(0) = 0$, in agreement with (29).

Each fixed C_m is bounded on $\tau \geq 0$ by its fixed-contour integral. The extra terms for $m \geq 2$ therefore total $O(n^{-1})$, which is negligible compared with the positive lower bound from Theorem 4.1. Positivity and the relative error follow. $\square$

The quantifiers are essential: choose the truncation order and logarithmic window constant *before* taking $n \rightarrow \infty$. The theorem does not claim uniformity in growing M , convergence of its infinite formal expansion, or validity for a supercritical weight.

6 The explicit two contribution coexistence profile

Set

$$b_n = \frac{1}{2} \log n + 2 \log \log n + \log \frac{\mu \sqrt{\pi}}{2}, \quad \tau_n(s) = b_n + s. \quad (47)$$

For s in a fixed compact real interval, this parameter is nonnegative and lies in a fixed logarithmic window for sufficiently large n .

Corollary 6.1 (Coexistence). *Uniformly for s in compact real intervals,*

$$a_c \sqrt{n} (\log n)^2 4^{-n} W_n(u_n(\tau_n(s))) \longrightarrow \frac{2}{\mu \sqrt{\pi}} (1 + e^{-s}). \quad (48)$$

Proof. The exponential term satisfies $\sqrt{n} (\log n)^2 e^{-b_n-s} = 2e^{-s}/(\mu \sqrt{\pi})$. For the second term, (39) and $\tau_n(s)/\log n \rightarrow 1/2$ give $k(\log n)^2 H(\tau_n(s)) \rightarrow 2/(\mu \sqrt{\pi})$. Also $a_n(\tau_n(s)) \rightarrow a_c$. The normalized absolute error in Theorem 4.1, multiplied by $\sqrt{n} (\log n)^2$, tends to zero. $\square$

The balance is between the near-critical exponential coefficient contribution and the heavy-tail correction. The $2 \log \log n$ term is necessary because the heavy-tail contribution contains τ^{-2} . A compact τ window cannot see this coexistence balance. Logarithmic heavy-traffic/heavy-tail transitions are established phenomena [5]; (48) supplies the model-specific centering and local-coefficient profile.

Nothing in the coefficient decomposition alone identifies the conditional law of the number or sizes of blocks in either contribution. In particular, (48) is not a proved distributional mixture, and the factors 1 and e^{-s} are not asserted to be mixture probabilities.

[write] **Added 3 October 2026, batch 85.** Part II supplies the separate uniform estimates that this paragraph says the coefficient decomposition alone cannot give. It computes the weight of the event that one block exceeds $n/2$ exactly (101), shows that this weight is asymptotic to the heavy-tail term ((102), by the giant-deletion Theorem 14.2) and that the complementary event carries the exponential term ((112), exhausted by local-limit windows). Hence, at $\tau = b_n + s$, the factor 1 in (48) is the limiting relative weight of the giant-block event and the factor e^{-s} that of the event with a linear number of blocks, and $1/(1 + e^{-s})$ and $e^{-s}/(1 + e^{-s})$ are mixture probabilities (Theorem 12.1). The two sentences above remain true of (48) on its own. Part II is unrefereed, prepared with AI assistance and not formalized.

7 What can be inverted and with what error

An inverse requires a specified one-variable map. We give three such choices. In every case, a real inverse-location error is distinct from a guarantee about rounding to an integer threshold.

7.1 Fixed family size from an all or strong count

Let $h = \log 4$ and $C_\sigma = e^{\sigma\lambda}/(2\sqrt{\pi})$. For fixed K , define the eventually positive model

$$F_{K,\sigma}(x) = C_\sigma 4^x x^{-5/2} \left(1 + \sum_{j=1}^K c_j^\sigma x^{-j} \right). \quad (49)$$

It is eventually strictly increasing, with logarithmic derivative $h + O(1/x)$. Denote its eventual real inverse by $x_{K,\sigma}(Y)$. Theorem 3.1 and the mean value theorem then imply, whenever $Y = T_n^\sigma$ and n is sufficiently large,

$$n = x_{K,\sigma}(Y) + O(x_{K,\sigma}(Y)^{-K-1}). \quad (50)$$

For $K = 0$ the model inverse is explicit:

$$x_{0,\sigma}(Y) = -\frac{5}{2h} W_{-1} \left(-\frac{2h}{5} \left(\frac{C_\sigma}{Y} \right)^{2/5} \right). \quad (51)$$

The branch W_{-1} selects the large positive solution. Higher-order inverses can be found by substitution or Newton iteration on $\log F_{K,\sigma}(x) - \log Y$.

7.2 Size along a specified coexistence path

Fix $s \in \mathbb{R}$. Extend (47) to large real x and define

$$G_s(x) = \frac{4^x}{a_x(\tau_x)} \left\{ e^{-\tau_x} + kx^{-1/2} H(\tau_x) \right\}, \quad \tau_x = \frac{1}{2} \log x + 2 \log \log x + \log \frac{\mu\sqrt{\pi}}{2} + s. \quad (52)$$

This is eventually positive with logarithmic derivative tending to h . If $x_s(Y)$ is its eventual inverse and $Y = W_n(u_n(\tau_n(s)))$, then

$$n = x_s(Y) + O\left(\frac{(\log x_s(Y))^2}{\sqrt{x_s(Y)}}\right). \quad (53)$$

This uses the explicit derivative of the model, not differentiation of an uncontrolled enumeration remainder. An elementary first inverse center, with $L = \log Y$, is

$$x = \frac{L + \frac{1}{2} \log L + 2 \log \log L - \frac{1}{2} \log h - \log K_s}{h} + o(1), \quad K_s = \frac{2(1 + e^{-s})}{a_c \mu \sqrt{\pi}}. \quad (54)$$

The discrete weighted sequence along this fixed path is eventually increasing because its consecutive ratio tends to 4. An exact threshold still requires bracketing: small inverse error does not resolve an integer boundary without a certified separation.

7.3 Known size and an unknown component weight

Fix a compact real interval J . Suppose n is known, $\tau = b_n + s$ with $s \in J$, and the target is specified as

$$Y = 4^{-n} W_n(u_n(\tau)). \quad (55)$$

For $n \geq 1$, $W_n(u)$ is a nonconstant polynomial with nonnegative coefficients and is strictly increasing on $u > 0$. Since $u_n(\tau)$ decreases strictly, the exact target determines τ uniquely on the specified window.

For fixed $M \geq 1$, define

$$\mathcal{M}_{n,M}(t) = \left(\frac{p}{c} + \frac{t}{n}\right) \sum_{m=0}^M n^{-m/2} C_m(t). \quad (56)$$

For all sufficiently large n , this model is strictly decreasing on the padded interval $b_n + (J + [-1, 1])$ and has derivative magnitude at least a positive constant times $n^{-1/2}/(\log n)^2$. Indeed, the C_0 derivative is negative of that size, the C_1 derivative is smaller by $O(1/\log n)$ using $H'(t) = O(t^{-3})$, and the higher derivatives are bounded by their contour integrals. The derivative $1/n$ of the prefactor is negligible. Only the explicit model is differentiated.

The padding ensures the model brackets the exact target even when s is at an endpoint of J : the model change over a unit of padding dominates the absolute error in (43). There is therefore one root $\hat{\tau}$ of $\mathcal{M}_{n,M}(\hat{\tau}) = Y$ on that padded interval, and the mean value theorem gives

$$|\hat{\tau} - \tau| = O_{M,J}((\log n)^2 n^{-M/2}). \quad (57)$$

Since $|u'_n(t)| = nc/(np + tc)^2 = O(n^{-1})$ there, the inferred weight $\hat{u} = n/(np + \hat{\tau}c)$ satisfies

$$|\hat{u} - u_n(\tau)| = O_{M,J}((\log n)^2 n^{-1-M/2}). \quad (58)$$

The displayed C_2 thus improves weight localization from $O(\log^2 n/n^{3/2})$ to $O(\log^2 n/n^2)$. These are asymptotic localization bounds for a known- n model, not finite- n certified error bars or an unspecified inverse of a bivariate sequence.

[write] **Added 2 October 2026, batch 77.** The general steps of this section are instances of results of the repository's transseries volume ([Analysis/Transseries/docs/series-and-transseries/](#)

`Transseries_And_Inversion/`), and no novelty is claimed for them. The explicit inverse (51) is `p0:thm:lambert-core` for $hx - \frac{5}{2} \log x = \log(Y/C_\sigma)$, that is $a = h = \log 4$, $b = -5/2$, with the branch W_{-1} that its branch rule selects for the large solution. The three localization statements (50), (53) and (57) divide a model error by a model slope, which is the residual-to-root conversion `p0:thm:backward-error`. The remark that an exact threshold still needs bracketing is the separation condition, part (2) of `p0:thm:staircase`; no exact-rounding rule (its part (1)) is used. The generic staircase arithmetic is formalized in Lean, in the file `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`, as `Fabius.staircase_ceil` and `Fabius.staircase_separation`; those lemmas concern an arbitrary monotone function, not S_n or W_n , and give this report no formal status.

8 Reproducible checks and numerical interpretation

The accompanying archive contains this editable T_EX source, the compiled PDF, a portable Python verification program, generated JSON data, a replay script, and a SHA-256 manifest. It does not require a connection to OEIS or another external data service to run the mathematical checks. The code independently generates the integer recurrences rather than loading an OEIS term list.

The check layers have different evidentiary roles:

1. **Exact enumeration.** Enumerate all nondecreasing candidates through a small size, apply Landau’s inequalities, and count equality indices. Compare the resulting $S_n, I_n, S_{n,m}$ with the divisor and renewal recurrences. Integer-weight evaluations test the component polynomial directly.
2. **Exact algebra.** Verify the Puiseux and gamma-ratio formulas for c_1^σ, c_2^σ , the denominator coefficient $d_4 = (\mu^2/2 - e_2)/\mu$, and the three-kernel formula for C_2 . Such assertions are algebraic identity checks.
3. **Finite generator.** Construct the formal half-power coefficients through $M = 4$, evaluating integer-exponent kernels by polynomial/rational decomposition and nondegenerate half-integer kernels by Kummer functions. This avoids undefined ordinary hypergeometric parameters.
4. **Numerical recurrence and inverse comparisons.** Compute normalized weighted coefficients, compare fixed-order models, and solve the explicit known- n inverse. These finite checks can detect implementation mistakes; they do not prove uniform asymptotics, certify a finite- n bound, or establish novelty.

For illustration, at the coexistence center $\tau = b_n$, let $Q = 4^{-n}W_n(u_n(b_n))$. The following values use the corrected C_2 and the relative-error convention $Q/\mathcal{M}_{n,M}(b_n) - 1$:

n	b_n	Q	$M = 1$ error	$M = 2$ error
100	4.841567	0.00618023445	−6.722%	1.585%
300	5.818773	0.00261743283	−5.724%	0.792%
1000	6.803789	0.00105627898	−3.461%	0.235%

These examples illustrate a finite-size benefit of the extra term, not a universal monotonic-error claim. At $\tau = 0$, $C_2(0) = 0$, so the $M = 1$ and $M = 2$ models coincide. Higher generator terms are needed to improve that special point.

For the fixed families, the residual $n^2(T_n^\sigma/(C_\sigma 4^n n^{-5/2}) - 1 - c_1^\sigma/n)$ approaches c_2^σ . Representative values are

n	All-score residual	Strong-score residual
100	−0.6704789062	4.4569099700
200	−0.6311256921	4.5186728913
1000	−0.5999299576	4.5693162876
Limit coefficient	−0.5921736164	4.5821545749

All decimal computations, including values computed at high working precision, remain non-interval checks. The proof of each remainder is analytical. A validated finite- n certificate would require interval evaluation and explicit constants throughout the contour estimates.

8.1 How to replay

From the extracted archive, run `bash replay.sh`. The script runs exact and symbolic assertions, regenerates numerical JSON data, and invokes `pdflatex` twice to rebuild the PDF. Python requirements are listed in `requirements.txt`; a working L^AT_EX installation is also needed. Use `python3 scripts/verify_manifest.py` before replay to check the packaged file hashes. The README specifies versions and commands. Generated results are written inside the extracted directory, and the source code has no absolute workspace paths.

[write] **Added 2 October 2026, batch 77.** This subsection describes the delivered archive. In the repository the files keep their bytes but not their places: `scripts/` is `code/`, `results/` is `data/`, `replay.sh` and `build_pdf.sh` are in `code/`, `requirements.txt` is in `data/`, and `report.tex` became `article.tex`. The delivered PDF and the manifest SHA256SUMS were verified at placement and not shipped (both survive in the arrival commit), so `verify_manifest.py` cannot run as shipped. `run_checks.py` writes into `results/` beside `scripts/`, overwriting the recorded outputs; the report’s README.md therefore runs it on a scratch copy with the delivered layout. At intake (2 October 2026) that run reported status PASS and all seven JSON files equal to the recorded ones up to line endings. An independent intake computation (exact S_n to $n = 1200$ from the totient formula) gave $\lambda = 0.33023754398592927\dots$ and $\mu = 0.67395260185159494\dots$, $n(S_n/(C_+ 4^n n^{-5/2}) - 1) = 1.55938$ at $n = 1200$ against $c_1^+ = 1.55988$, and second-order residuals -0.59993 ($n = 1000$) and -0.59864 ($n = 1200$) approaching $c_2^+ = -0.59217$ at rate $1/n$, consistent with Theorem 3.1.

9 Further questions

The results suggest several concrete extensions without settling them:

1. **Conditional component laws at coexistence.** Identify which component-count and component-size ranges contribute to the two coefficient terms. A genuine mixture theorem needs separate uniform estimates for those ranges, rather than only the sum of their weights.
2. **Supercritical crossover.** For $\tau < 0$, the denominator can develop a moving positive-real pole. Its residue and interaction with the branch point require a revised contour analysis; the present zero-exclusion proof does not apply unchanged.
3. **Wider parameter windows and explicit constants.** Determine the maximal useful growth of τ for a chosen truncation and turn the asymptotic O constants into practical, interval-certified bounds. This would support reliable finite-size thresholds and certified weight inversion.
4. **Growing component counts or truncation orders.** Fixed- m laws and fixed- M expansions

do not cover m or M growing with n . Uniformity in those directions may involve additional scales and optimal truncation questions.

5. **Comparison with general renewal theorems.** Locate the strongest existing local lattice theorem that specializes to the present half-power structure, and compare its hypotheses and remainder constants with the explicit generator. This is necessary before claiming priority for any general mechanism.

[write] **Added 3 October 2026, batch 85.** Question 1 is answered, for the weight $u_n(b_n + s)$ with s in a compact interval, by Part II (unrefereed, AI-assisted, not formalized): the heavy-tail contribution comes from configurations with a unique block of size $n - O_{\mathbb{P}}(n/\log n)$, around which the remaining blocks form two nearly independent geometric renewal strings, and the exponential contribution from configurations with $n/a_c + O_{\mathbb{P}}(n^{2/3})$ blocks and largest block of order $n^{2/3}$ (Theorems 12.1–12.4). Growing shifts $s_n \rightarrow \pm\infty$ are not covered (Part II’s question RQ5). Question 2 is restated, still open, as Part II’s RQ6. For question 5, Part II compares its mechanism with the Gibbs-partition, balls-in-boxes and big-jump literature (Section 10); that comparison is a bounded source check, so the question stays open. Questions 3 and 4 are unchanged: Part II’s block counts grow with n , but under the moving weight $u_n(b_n + s)$, not in the fixed- m law of (8) or the fixed- M expansions.

[write] **Added 5 October 2026, batch 98.** Question 2 is answered by Part III (unrefereed, AI-assisted, not formalized) on a fixed small supercritical interval: for $r = up = 1/Q(e^{-t})$ with $0 \leq t \leq t_0$ ($t_0 > 0$ not made effective) the moving pole is the simple real pole $z = e^{-t}$, separated from the branch point on a common slit disk (Lemma 24.2), its residue is e^{nt}/m_t , and the branch integral has an expansion to every fixed order uniformly through the coalescence $nt = O(1)$ (Theorem 24.1). Under the parametrization $u = u_n(\tau)$ of (30) that side is $\tau = -a_n(t) \leq 0$. Part III’s RQ4 (one analytic family across $\tau = 0$, uniformly) remains open; a dated note in Section 31 shows that Part III’s first two terms, re-expanded at bounded $a = -\tau$, reproduce (31) formally, through an exact identity for $H(-a)$, and records the continuation of Theorem 5.1 to $\tau \in [-C, 0]$ as a further question. The sentence after Theorem 5.1 that it claims no “validity for a supercritical weight” remains true of that theorem. Questions 3–5 are unchanged by Part III; its RQ7 restates the effective-constants part of question 3 for the supercritical side.

Part II

Conditional phase coexistence: giant blocks, Gamma laws, and stable fluctuations

Summary. We resolve the conditional component-law question left open in Part I, the ProveIt report on tournament score sequences, OEIS A000571 and A351822. When each irreducible block receives a weight approaching its critical value at a specified logarithmic rate, we prove a logistic mixture of two structurally different regimes. One has a unique block of size $n - O_{\mathbb{P}}(n/\log n)$; the other has $n/m + O_{\mathbb{P}}(n^{2/3})$ blocks and no block of macroscopic size. In the first regime, removing the giant block leaves two independent geometric renewal strings up to a total variation error $O(1/\log n)$. This gives a Gamma(2, 1) law for both the rescaled block count and the giant’s deficit, a stable law at the next scale, and an explicit mixed Fréchet law for the second-largest block. A positive polynomial tilt of the residual block count improves the approximation to every fixed inverse-logarithmic order. In the second regime, the centered number of blocks converges to the negative of a spectrally positive 3/2-stable variable, jointly with an explicit Fréchet limit for the largest block.

The exact enumeration identity, the leading asymptotic counts, and the general phenomenon of condensation have established precedents. The contribution here is a proved conditional description in the size-dependent tournament model, together with an elementary total variation argument and all normalization constants. The analytic input from the repository is reproved, and the stable local limit theorem is proved directly by Fourier inversion. Exact enumeration checks and floating-point diagnostics accompany the article. No priority claim is made for general Gibbs-partition or heavy-tail mechanisms.

Main result in one display. Let K_n be the number of irreducible blocks and $M_{1,n}$ the largest block size. At the weight defined precisely in Section 12,

$$\left(\frac{mK_n}{n}, \frac{M_{1,n}}{n}\right) \xrightarrow{d} \frac{1}{1+e^{-s}} \delta_{(0,1)} + \frac{e^{-s}}{1+e^{-s}} \delta_{(1,0)}.$$

Here $m = 1.7223486864\dots$ is an explicitly defined mean block size. The first atom describes one giant block and a random sublinear number of remaining blocks; the second describes a linear number of blocks.

[write] **Added 3 October 2026, batch 85: provenance of Part II.** Part II is a single manuscript, printed in full: manuscript 04 of batch 85 of ProveIt’s incoming-reports intake, *Conditional phase coexistence for tournament score sequences: Giant blocks, Gamma laws, and stable fluctuations* (“Research manuscript: OEIS A000571 and A351822”, dated 3 October 2026; a 1,687-line source and a 25-page PDF; its PDF metadata name the author as “Research manuscript prepared with AI assistance”). It arrived as the archive `tournament_coexistence_research.zip` (wrapper directory `tournament_coexistence/`, main file `tournament_coexistence.tex`) in commit 317c1ce2e and was placed in commit 713149ded as an addition to this report. It pins the repository at commit d68b65ea064084fb121df9bb431b21d086f7be81 and continues this report’s `article.tex`; the report was not changed between the pin and the placement, so the manuscript read exactly the text printed as Part I. Its code, data and figures are shipped with the prefix `02-coex-`; its own L^AT_EX source, README, PDF and checksum list are not shipped and survive in the archive of the arrival commit. The manuscript is unrefereed and was prepared with AI assistance; its proofs are ordinary mathematical proofs, nothing in it is formalized in Lean or Rocq, and its place in this report confers no formal status.

What it answers. Part II answers the first further question of Part I (“Conditional component laws at coexistence”, Section 9) and supplies the separate estimates that the caveat at the end of Section 6 says the coefficient decomposition alone does not give: at $\tau = b_n + s$ the heavy-tail term (the factor 1 in (48)) is the weight of the event $\mathcal{C}_n$ that one block exceeds $n/2$, and the exponential term (the factor e^{-s}) is the weight of its complement $\mathcal{F}_n$ (Theorem 12.1). Dated notes at those two places of Part I say so.

Overlap with Part I, printed and credited. Proposition 13.2 and Corollary 13.3 re-prove Theorem 4.1 and Corollary 6.1 in the same normalization (a note in Section 13 gives the dictionary). The tail $q_j \sim dj^{-5/2}$ of Lemma 13.1 re-derives the mass tail of Part I’s renewal interpretation (Section 2, from (8)) with the constant d made explicit, and its aperiodicity argument is the one of Section 4.1. The constants (63)–(64) extend (9) ($m = a_c$), the centre (67) is (47), the split (80) is (10), the recurrences (124) are (5)–(7), and (126) is (20). The coefficients 1, 5, 105/4 that open Proposition 17.1 are those of (39): both expand the same heavy-tail kernel, a consistency check rather than a duplicate. Part II’s exact enumeration program is an independent reimplement of the Landau check of Part I’s `code/check_exact_counting.py`, extended to the giant-event polynomial. Everything else in Part II is new to the repository: Theorems 12.1–12.4, 14.2, Corollaries 14.3, 15.1, 17.2, the direct local limit Lemma 16.1, Propositions 17.1 and 18.1, and the point-process statements. The inversion (121) is an instance of a theorem of the repository’s transseries volume (note after it).

Editorial changes. Section k of the manuscript is Section $k + 9$ here, and its Appendices A and B are Appendices A and B; every label carries the prefix `tss:cx:`. The manuscript’s references to “the pinned ProveIt report” (its citation `Repo`, whose pin and file path this note records) now point to Part I, in its abstract (the summary above), in Section 10 and in its status table; the bibliographies are merged, so the shared entries are Part I’s and the four new ones (Białas–Bogacz–Burda–Johnston, Stuffer, Denisov–Dieker–Shneer, Armendáriz–Loulakis) are added. The two displayed figures are the shipped `02-coex-` files, and the shipped phase-profile figure, which the manuscript does not display, is added as Figure 3 with a caption taken from the delivered code README. The two theorem titles of Section 13 name the Part I results they restate, and the `[write]` notes are new. No statement, proof or number of the manuscript was altered.

[write] Added 3 October 2026, batch 85: notation of Part II. Part II keeps the manuscript’s symbols, and several of them clash with Part I’s. Within Part II every symbol has the meaning below; the right column gives Part I’s meaning, the tempting false reading. Shared notions keep one symbol: $\lambda, \mu, \rho, p, u_c, A, N_j, F(z), R(z), H(\tau), \mathcal{H}, W, W_n(u), W_{-1}, b_n, \tau, s$ (the coexistence shift), $q_j, S_n, I_n, E_n(w)$ (the resolvent error) and M (a fixed truncation order) mean the same in both Parts. The manuscript’s own symbol list is Appendix A.

symbol	meaning in Part II	not: meaning in Part I
m	mean critical block length $e^{-\lambda}\mu/p = 1.72234\dots$, which is Part I’s a_c	a number of strong blocks ($S_{n,m}$), or the index of c_m^σ, C_m
k	a block-count index ($S_{n,k} = [z^n]I^k$, sums over k)	$k = 2/(3\mu)$, which is Part II’s κ/m
κ	$2e^{-\lambda}/(3p) = d\Gamma(-3/2)$, the Puiseux and stable coefficient	(not used)
$K_n, K_{L,n}, K_{R,n}, K(x)$	numbers of blocks: in all, before and after the giant, of a sequence x	K a fixed truncation order or a generic constant
r, r_n, R_j	the normalized weight $r = up$, $r_n = (1 + m\tau_n/n)^{-1}$, and $rm = a_n(\tau)$ of Part I; $R_j = [z^j]R$	$r_n = [z^n]R$ (Part II’s R_n)
u_n	r_n/p , which is Part I’s $u_n(\tau_n)$	—
$a_n, a(k)$	the stable scales $(\kappa n/m)^{2/3}$ and $(\kappa k)^{2/3}$	$a_n(\tau) = u_n(\tau)c$, which is Part II’s rm
d	tail amplitude, $q_j \sim dj^{-5/2}$, $d = e^{-\lambda}/(2\sqrt{\pi}p)$; $d \mid j$ in (61) is a divisor	a divisor, or d_j the coefficients of $D(w)$
D_n, D	the coexistence scale (91); D its implicit-centre analogue in the proof of Proposition 18.1	$D(w), D_j^\sigma$

symbol	meaning in Part II	not: meaning in Part I
$B_n(r)$, B_n , $B(\delta)$	$B_n(r)$ the unnormalized weight of $\mathcal{C}_n$; B_n <i>without argument</i> the second-maximum scale (77), $B(\delta)$ the same scale as a function of δ (two meanings inside the manuscript, which its appendix discloses)	$B(t)$ the odd series, b_j , Bernoulli B_j
$Z_n(r)$, Z	$Z_n(r) = 4^{-n} W_n(r/p)$; Z the standard spectrally positive 3/2-stable variable (73)	—
$h_n(T)$, h	the density ratio in (96); $h > 0$ a small Chernoff parameter in Section 15	$h = \log 4$
δ , δ_n	$1 - r$, $1 - r_n \sim m\tau_n/n$	$\delta > 0$ a fixed contour margin (Section 4.1)
E , E_L , E_R , $\mathbb{E}$	exponential variables of mean one; $\mathbb{E}$ expectation	$E(v)$ the even part, e_j
G	a Gamma(2, 1) variable	$G_s(x)$ the path model (52)
L	$\log n$ in Proposition 18.1; a cutoff in the strong-law argument of Section 14	$L = \log Y$ in (54)
c_0	$\log(m/(4d))$ in Proposition 18.1	$c_0^\sigma = 1$, the leading fixed-order coefficient
$\mathcal{C}_n$, $\mathcal{F}_n$; C , C_0 , $C_1, \dots$	the events $\{M_{1,n} > n/2\}$ and $\{M_{1,n} \leq n/2\}$; generic or window constants	$C_m(\tau)$ the uniform terms, C_σ
$F(x)$	$\mathbb{P}(X \leq x)$ in Section 17 (elsewhere $F(z)$ as in Part I)	—
t	$\sqrt{1 - 4z}$ in Section 13; also a Fourier or Laplace variable, a size, and a ray parameter	the free weight parameter of $\mathcal{M}_{n,M}(t)$
z	the generating-function variable; in the proof of Proposition 13.2 also the Hankel variable nw	Part I's Hankel variable is $s = nw$
q , Q , X , S_k^X	$q = \tau/n$ inside the proof of Proposition 13.2 (as in Part I); Q the probability generating function of q_j ; X , X_i block lengths and S_k^X their partial sums	—
N_δ , N , T , $T_{L,n}$	string counts and total lengths	T_n^σ (S_n or I_n)
J_s , J_f , J	a Bernoulli variable, a Laplace exponent, a block count	J a compact interval
g , g_u	the stable density and its defective version	$g_\ell(\alpha)$ the gamma-ratio coefficients
ϕ , φ	$\phi(t) = Q(e^{it})$; φ Euler's totient as in Part I	—
ν , ν_δ , π_s	the Lévy intensity, the geometric-string law, the logistic weight $(1 + e^{-s})^{-1}$	—
$\mathcal{P}$, P_M , R_r , $L_{n,r}$, U , V	Poisson process, tilt polynomial, reference and actual residual laws, a uniform variable, the largest Poisson point	$R(z)$ is not R_r
I	an interval of count coordinates in Section 16.3 (elsewhere $I(z)$)	—
$M_{j,n}$	the j th largest block size	M a truncation order

10 Problem, provenance, and contribution

10.1 The particular open question addressed

The pinned ProveIt report *Tournament score sequences and weighted component asymptotics*, now Part I of this report, proves a two-term coefficient expansion and a coexistence profile, then explicitly leaves the conditional laws of the component counts and sizes open. Its Section 9, first research question, asks which structural ranges produce the two terms. Its Section 6 emphasizes that the two terms alone were not yet identified as probabilities of two events.

This article supplies that missing identification. The events are

$$\mathcal{C}_n = \{M_{1,n} > n/2\}, \quad \mathcal{F}_n = \{M_{1,n} \leq n/2\}.$$

The threshold $n/2$ makes the large block unique and gives an exact factorization. It does not impose an artificial asymptotic boundary: we prove that the maximum divided by n tends to 1 on $\mathcal{C}_n$ and to 0 on $\mathcal{F}_n$. Any fixed threshold strictly between zero and one therefore yields asymptotically the same classification.

The repository was inspected at commit `d68b65ea064084fb121df9bb431b21d086f7be81`. The continued report is under `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a000571-tournament-score-sequences/`, the directory of this report; its text at that commit is the text printed as Part I. The research index also records extensive work on other OEIS sequences; the choice here advances an explicit remaining question instead of repeating the classical enumeration theorem.

10.2 What belongs to the literature

The score-sequence identity used below is due to Claesson, Dukes, Franklín, and Stefánsson [1]; a shorter explanation is given by Bassan, Donderwinkel, and Kolesnik [3]. Kolesnik proved the leading all-score and strong-score asymptotics and the limiting law of the number of blocks at the unweighted value $u = 1$ [2]. Those are inputs and precedents, not open problems solved here.

There is also substantial prior work on the probability mechanism. Variable-box-number balls-in-boxes models already exhibit a competition between a condensed state and an extensive number of boxes [7]. Stuffer’s phase diagram for Gibbs partitions contains mixture regimes, stable count fluctuations, and descriptions of the remaining components after deletion of a maximum [8]. The large-deviation and conditional independence theory of Denisov–Dieker–Shneer and Armendáriz–Loulakis gives a general framework for a single large summand [9, 10]. Logarithmic transitions between geometric-sum and heavy-tail contributions occur in the queueing literature [5].

Stuffer’s mixture theorem uses fixed inner and outer coefficient sequences with matched polynomial tails. Its condensed component count has a finite limit. The balls-in-boxes studies establish the ensemble and its phase transition. Neither comparison is replaced by a claim that mixture limits or stable fluctuations were previously unknown.

The distinction relevant here is the changing outer weight $r_n^{K_n}$: $r_n \uparrow 1$ at distance of order $\log n/n$. The condensed number of blocks grows on the scale $n/\log n$, rather than staying bounded. We prove the explicit conditional laws in this window for tournament weights. The elementary renewal-string argument is presented in full, so none of the general large-deviation theorems is invoked with unchecked hypotheses. The literature comparison is a bounded source check, not a proof of publication priority.

Statement	Status in this article
Exact count identity and block decomposition	Established input, with the elementary decomposition recalled.
Leading $4^n n^{-5/2}$ asymptotics	Established result; the needed stronger local expansion is derived.
Two-term logarithmic coefficient law	Repository result (Theorem 4.1 and Corollary 6.1 of Part I), reproved in the manuscript; restated in Section 13.
Logistic probabilities of structural events	Proved here relative to the pinned report's open question.
Geometric-string total variation approximation	Proved directly, with an $O(1/\log n)$ error in this model.
Conditional Gamma, stable, and second-maximum laws	Proved here for the moving tournament weight.
General existence of condensation or stable limits	Credited to prior theory; no claim of a new general phenomenon.

This is an unrefereed research manuscript prepared with AI assistance. The proofs are ordinary mathematical proofs, not Lean or Rocq formalizations. The exact finite checks support the formulas but are not substitutes for the limiting arguments.

11 Score sequences and the exact renewal model

11.1 Irreducible blocks

A tournament is an orientation of a complete graph. A score sequence is its nondecreasing list of out-degrees. Landau's characterization, as stated in [1], says that a sequence of integers $(s_1, \dots, s_n)$ is a score sequence if and only if

$$0 \leq s_1 \leq \dots \leq s_n \leq n-1, \quad \sum_{i=1}^j s_i \geq \binom{j}{2} \quad (j < n), \quad \sum_{i=1}^n s_i = \binom{n}{2}. \quad (59)$$

It is *strong* or *irreducible* if every proper-prefix inequality is strict. A singleton is irreducible.

If equality holds at a proper prefix of length j , split the sequence there and subtract j from each entry of the second part. Both parts satisfy (59). For example, the prefix sum condition for the translated second part follows from

$$\binom{j+\ell}{2} - \binom{j}{2} - j\ell = \binom{\ell}{2}.$$

The first translated entry is nonnegative because the next prefix inequality implies $s_{j+1} \geq j$. Conversely, concatenating a score sequence of length j with one whose entries have all been increased by j preserves (59). Cutting at every equality therefore gives a unique ordered list of irreducible blocks.

Let S_n count all score sequences and I_n count nonempty strong ones, with $S_0 = 1$ and $I_0 = 0$. These are respectively A000571 and the positive-index part of A351822 [6]. The latter OEIS entry gives the empty object the value one; our generating function omits it. Write

$$S(z) = \sum_{n \geq 0} S_n z^n, \quad I(z) = \sum_{n \geq 1} I_n z^n, \quad S_{n,k} = [z^n] I(z)^k.$$

Unique decomposition gives

$$S(z) = \frac{1}{1 - I(z)}, \quad W(z, u) = \frac{1}{1 - uI(z)}, \quad W_n(u) = \sum_{k=1}^n S_{n,k} u^k \quad (n \geq 1). \quad (60)$$

The probability law studied here assigns to each score sequence x of length n the probability $u^{K(x)}/W_n(u)$. Thus the random objects are score sequences, not uniformly random labeled tournaments.

11.2 Exact enumerative input and constants

Define

$$N_j = \frac{1}{2j} \sum_{d|j} (-1)^{j+d} \varphi(j/d) \binom{2d}{d}, \quad A(z) = \sum_{j \geq 1} \frac{N_j}{j} z^j. \quad (61)$$

Here N_j is A145855. The established identity is

$$S(z) = \exp A(z), \quad I(z) = 1 - \exp(-A(z)). \quad (62)$$

We use (62) as a cited combinatorial theorem [1, 3]; we do not reprove its nontrivial bijection.

Set

$$\rho = \frac{1}{4}, \quad \lambda = A(\rho), \quad \mu = \rho A'(\rho), \quad p = 1 - e^{-\lambda}, \quad (63)$$

$$u_c = p^{-1}, \quad m = \frac{e^{-\lambda} \mu}{p}, \quad d = \frac{e^{-\lambda}}{2\sqrt{\pi} p}, \quad \kappa = \frac{2e^{-\lambda}}{3p} = d\Gamma(-3/2). \quad (64)$$

The probability generating function of a critical block length is

$$Q(y) = \frac{I(\rho y)}{p} = \sum_{j \geq 1} q_j y^j, \quad q_j = \frac{I_j 4^{-j}}{p}. \quad (65)$$

In particular, $Q(1) = 1$ and $\mathbb{E}X = Q'(1) = m$ for X with law q .

For $r = up$, the normalized weighted count is exactly

$$Z_n(r) = 4^{-n} W_n(u) = [y^n] \frac{1}{1 - rQ(y)} = \sum_{k=1}^n r^k \mathbb{P}(X_1 + \cdots + X_k = n). \quad (66)$$

For any ordered positive lengths $\ell_1 + \cdots + \ell_k = n$, their total unnormalized weight is $r^k \prod_i q_{\ell_i}$. Conditioning this measure on total length n is identical to the score-sequence law after the internal blocks have been summed out.

12 Precise statements of the conditional limits

Use the constants in (63)–(64), and put

$$b_n = \frac{1}{2} \log n + 2 \log \log n + \log \frac{m}{4d} = \frac{1}{2} \log n + 2 \log \log n + \log \frac{\mu \sqrt{\pi}}{2}. \quad (67)$$

Fix $s \in \mathbb{R}$ and, for all sufficiently large n , let

$$\tau_n = b_n + s, \quad r_n = \frac{1}{1 + m\tau_n/n}, \quad u_n = \frac{r_n}{p}, \quad \delta_n = 1 - r_n = \frac{m\tau_n}{n + m\tau_n}. \quad (68)$$

The probability law below always uses the weight u_n . Let $M_{1,n} \geq M_{2,n} \geq \dots$ be the ranked block sizes; missing sizes are zero. On $\mathcal{C}_n$, let $K_{L,n}, K_{R,n}$ be the numbers of blocks before and after the giant, and let $T_{L,n}, T_{R,n}$ be their total lengths. Then

$$K_n = K_{L,n} + K_{R,n} + 1, \quad n - M_{1,n} = T_{L,n} + T_{R,n}.$$

Theorem 12.1 (Structural coexistence). *With $\pi_s = (1 + e^{-s})^{-1}$,*

$$\mathbb{P}(\mathcal{C}_n) \longrightarrow \pi_s, \quad \mathbb{P}(\mathcal{F}_n) \longrightarrow 1 - \pi_s. \quad (69)$$

Moreover,

$$\left(\frac{mK_n}{n}, \frac{M_{1,n}}{n} \right) \xrightarrow{d} \pi_s \delta_{(0,1)} + (1 - \pi_s) \delta_{(1,0)}. \quad (70)$$

These conclusions hold uniformly for s in a fixed compact interval in the sense of bounded continuous test functions.

Theorem 12.2 (Complete residual law on the giant event). *For each side independently, draw a count G_δ with*

$$\mathbb{P}(G_\delta = k) = \delta(1 - \delta)^k, \quad k \geq 0,$$

and then draw that many independent block lengths with law q . The joint law of the actual left and right length strings, conditional on $\mathcal{C}_n$, is within $O(1/\tau_n)$ in total variation of the product of these two string laws with $\delta = \delta_n$.

Consequently, if E_L, E_R are independent exponential variables of mean one and $G = E_L + E_R$, then, conditionally on $\mathcal{C}_n$,

$$(\delta_n K_{L,n}, \delta_n K_{R,n}, \delta_n T_{L,n}/m, \delta_n T_{R,n}/m) \xrightarrow{d} (E_L, E_R, E_L, E_R), \quad (71)$$

and

$$\left(\delta_n K_n, \frac{\delta_n}{m}(n - M_{1,n}) \right) \xrightarrow{d} (G, G), \quad G \sim \text{Gamma}(2, 1). \quad (72)$$

The density of G is $xe^{-x}\mathbf{1}_{\{x>0\}}$.

Theorem 12.3 (Stable block-count fluctuations on the other event). *Let Z be the real random variable with characteristic function*

$$\mathbb{E}e^{itZ} = \exp\{(-it)^{3/2}\}, \quad t \in \mathbb{R}, \quad (73)$$

using the principal power. Put $a_n = (\kappa n/m)^{2/3}$. Conditionally on $\mathcal{F}_n$,

$$\frac{m(K_n - n/m)}{a_n} \xrightarrow{d} -Z, \quad \frac{M_{1,n}}{n} \xrightarrow{\mathbb{P}} 0. \quad (74)$$

In fact, the convergence is joint with

$$\left(\frac{m(K_n - n/m)}{a_n}, \frac{M_{1,n}}{a_n} \right) \xrightarrow{d} (-Z, V), \quad \mathbb{P}(V \leq y) = \exp\left(-\frac{1}{2\sqrt{\pi}y^{3/2}}\right) \quad (y > 0). \quad (75)$$

The pair (Z, V) is constructed from the same Poisson point process in Section 16.3; its coordinates are dependent. In particular $K_n/n \xrightarrow{\mathbb{P}} 1/m$ on $\mathcal{F}_n$. The stable variable is centered and spectrally positive; its negative appears because a positive excess in a sum of block lengths decreases the number of blocks needed to reach n .

Two further fluctuation theorems sharpen the description of $\mathcal{C}_n$. They are proved in Section 17.

Theorem 12.4 (Residual fluctuation and second maximum). *Conditionally on $\mathcal{C}_n$, jointly,*

$$\left(\delta_n K_n, \frac{(n - M_{1,n}) - m(K_n - 1)}{(\kappa/\delta_n)^{2/3}} \right) \xrightarrow{d} (G, G^{2/3}Z), \quad (76)$$

where $G \sim \text{Gamma}(2, 1)$ and Z from (73) are independent on the right. If

$$B_n = \left(\frac{2d}{3\delta_n} \right)^{2/3}, \quad (77)$$

then, for $x > 0$,

$$\mathbb{P}(M_{2,n}/B_n \leq x \mid \mathcal{C}_n) \longrightarrow (1 + x^{-3/2})^{-2}. \quad (78)$$

More precisely, for $t \geq 0$ and $x > 0$,

$$\mathbb{E} \left[e^{-t\delta_n K_n} \mathbf{1}_{\{M_{2,n} \leq B_n x\}} \mid \mathcal{C}_n \right] \longrightarrow (1 + t + x^{-3/2})^{-2}. \quad (79)$$

These statements distinguish three scales on the giant event: n for the giant, $n/\log n$ for its deficit and the total number of remaining blocks, and $(n/\log n)^{2/3}$ for the largest remaining blocks and the centered fluctuation of their total length. On the other event the count and the largest block fluctuate on the larger scale $n^{2/3}$.

13 Analytic foundations and the uniform coefficient estimate

13.1 The singular part comes from one divisor

Separating $d = j$ in (61) gives

$$A(z) = F(z) + R(z), \quad F(z) = \frac{1}{2} \sum_{j \geq 1} \binom{2j}{j} \frac{z^j}{j^2}. \quad (80)$$

Every proper divisor of j is at most $j/2$. Bounding each binomial coefficient by $4^{j/2}$ and using at most j divisors shows that $|[z^j]R(z)| \leq 2^{j-1}$. Hence R is analytic for $|z| < 1/2$. The function F continues across the circle $|z| = 1/4$ except at $z = 1/4$, with a slit along the positive real direction.

For $t = \sqrt{1 - 4z}$, differentiation of the binomial series gives

$$\frac{dF}{dt} = -\frac{2t}{1 - t^2} \log \frac{2}{1 + t}.$$

The derivative has a convergent power series at $t = 0$. Its first odd contribution after integration is $2t^3/3$. With $w = 1 - y$ this gives

$$A(y/4) = \lambda - \mu w + \frac{2}{3}w^{3/2} + O(w^2), \quad (81)$$

$$Q(y) = 1 - mw + \kappa w^{3/2} + O(w^2). \quad (82)$$

The expansions are convergent local Puiseux expansions on a slit neighborhood, not just asymptotic statements along the real axis.

Lemma 13.1 (Tail and aperiodicity). *The critical law has finite mean m , satisfies*

$$q_j = dj^{-5/2}(1 + O(j^{-1})), \quad \mathbb{P}(X > x) \sim \frac{2d}{3}x^{-3/2}, \quad (83)$$

and has infinite variance. Also $q_1, q_3, q_4 > 0$, and $|Q(e^{it})| < 1$ for $0 < |t| \leq \pi$.

Proof. For noninteger β ,

$$[y^j](1-y)^\beta = \frac{\Gamma(j-\beta)}{\Gamma(-\beta)\Gamma(j+1)} = \frac{j^{-\beta-1}}{\Gamma(-\beta)}(1 + O(j^{-1})).$$

Apply this identity to the first two nonanalytic terms of the convergent Puiseux expansion of Q . A contour dented around $y = 1$, with its remaining part outside the unit circle, bounds the omitted terms at the corresponding higher powers. Analytic integer powers contribute no leading tail. Thus the leading coefficient is $\kappa/\Gamma(-3/2) = d$, and the next possible relative order is $1/j$. Summing the tail gives the second assertion. The moment claims follow by comparison with powers of j .

The counts $I_1 = I_3 = I_4 = 1$ give the three positive probabilities. Equality in the triangle inequality for $Q(e^{it})$ would force $e^{it} = e^{3it} = e^{4it}$. Their exponent differences have greatest common divisor one, so $t = 0$ modulo 2π . $\square$

13.2 Two terms with an absolute remainder

The next proposition is the part of the earlier report needed here. We give the proof to keep the dependency on an unrefereed analytic argument reviewable.

[write] **Added 3 October 2026, batch 85: a re-proof of Part I.** Proposition 13.2 and Corollary 13.3 are not new results. Proposition 13.2 is Theorem 4.1 of Part I, with (85) equal to (32) and (86) the first two terms of (39) (note $1/\Gamma(-3/2) = 3/(4\sqrt{\pi})$); the manuscript's "earlier report" is Part I. The normalizations agree exactly: with $r = (1 + m\tau/n)^{-1}$ and $u = r/p$ one has $u = u_n(\tau)$ of (30) (because $pm = c$), so $Z_n(r) = 4^{-n}W_n(u_n(\tau))$, $rm = a_n(\tau)$ and $\kappa/m = k = 2/(3\mu)$. Corollary 13.3 at $\tau = b_n + s$ is Corollary 6.1: $m = a_c$ and $4d/m^2 = 2/(a_c\mu\sqrt{\pi})$, so (92) is (48) divided by $a_c\sqrt{n}(\log n)^2$; its first display (90), for any $\tau \asymp \log n$ in the window, is the same two-term reading of (31). The manuscript's proof below follows Part I's Section 4 step for step ((87) is (34), (88) is (35), (89) is (38), and the zero exclusion on the circle is the same aperiodicity argument); it is printed, as delivered, so that Part II can be checked on its own, and it is not a second route.

Proposition 13.2 (Uniform logarithmic expansion; Theorem 4.1 restated). *For fixed $C > 0$, put $r = (1 + m\tau/n)^{-1}$. Uniformly for $0 \leq \tau \leq C \log n$,*

$$Z_n(r) = \frac{1}{rm} \left\{ e^{-\tau} + \frac{\kappa}{m} n^{-1/2} H(\tau) \right\} + O_C(n^{-1}), \quad (84)$$

where

$$H(\tau) = \sum_{j \geq 0} \frac{(-1)^j (j+1) \tau^j}{\Gamma(j+1/2)} = \frac{{}_1F_1(2; 1/2; -\tau)}{\sqrt{\pi}}. \quad (85)$$

As $\tau \rightarrow +\infty$,

$$H(\tau) = \frac{1}{\Gamma(-3/2)\tau^2} \left(1 + \frac{5}{\tau} + O(\tau^{-2}) \right). \quad (86)$$

Proof. Let $w = 1 - y$ and $q = \tau/n$. Equation (82) gives

$$1 - rQ(y) = rm \left(w + q - \frac{\kappa}{m} w^{3/2} + O(w^2) \right).$$

Fix θ with $\pi/2 < \theta < \pi$. On $|\arg w| \leq \theta$,

$$|w + q| \geq \cos(\theta/2)(|w| + q) \quad (q \geq 0). \quad (87)$$

Consequently the $w^{3/2}$ perturbation is uniformly smaller than half the leading term in a sufficiently small fixed sector. Lemma 13.1 excludes denominator zeros on the unit circle away from 1. Compactness and analytic continuation thicken that part of the contour a little outside the circle, uniformly for $r \rightarrow 1$. Together these observations supply a common coefficient contour: local rays $w = te^{\pm i\theta}$, a joining arc of radius $1/n$ on their right, and outer pieces on which $|y| > 1 + \epsilon$. No assertion about a whole zero-free disk beyond the circle is needed.

Expanding the resolvent on this contour gives

$$\frac{1}{1 - rQ(y)} = \frac{1}{rm(w + q)} + \frac{\kappa}{rm^2} \frac{w^{3/2}}{(w + q)^2} + E_n(w), \quad (88)$$

where

$$|E_n(w)| \leq C_1 \left(\frac{|w|^2}{(|w| + q)^2} + \frac{|w|^3}{(|w| + q)^3} \right) \leq 2C_1.$$

On each local ray the coefficient kernel is bounded by $C_2 \exp(-cn|w|)$. Its integral against E_n is $O(n^{-1})$; the joining arc has the same order of length. The outer contribution is exponentially small.

The first coefficient in (88) is exactly $(1 + \tau/n)^{-n-1}$. Its replacement by $e^{-\tau}$ costs $O(n^{-1})$ uniformly on the stated window. For the second coefficient, use $z = nw$. With a fixed-angle Hankel contour $\mathcal{H}$ around the negative real direction, indented by a unit arc to its right,

$$[y^n] \frac{w^{3/2}}{(w + q)^2} = n^{-1/2} \frac{1}{2\pi i} \int_{\mathcal{H}} e^z \frac{z^{3/2}}{(z + \tau)^2} dz + O(n^{-3/2}). \quad (89)$$

Indeed (87) bounds the integrand apart from its exponential by $C|z|^{-1/2}$. On $|z| \leq n^\eta$, $\eta < 1/2$, replacing $(1 - z/n)^{-n-1}$ by e^z has an integrable relative error $O((|z| + |z|^2)/n)$, and the rest of the rays decays exponentially. For $|\tau| < 1$, expand $(z + \tau)^{-2}$ and integrate term by term; the reciprocal-gamma Hankel identity gives (85). Continuation along $\tau \geq 0$ is valid by (87). The pole at $z = -\tau$ remains on the excluded negative direction throughout.

Finally expand $(z + \tau)^{-2}$ on the part $|z| \leq \tau/2$ of the same contour. The complementary rays have exponential decay. The first two integrals are $\tau^{-2}/\Gamma(-3/2)$ and $-2\tau^{-3}/\Gamma(-5/2)$; their ratio gives the coefficient 5. This proves (86) and the proposition. $\square$

Corollary 13.3 (The two positive contributions; Corollary 6.1 restated). *If $\tau \asymp \log n$ within a fixed logarithmic window, then*

$$Z_n(r) = \frac{e^{-\tau}}{m} (1 + o(1)) + \frac{d}{m^2} \frac{n^{-1/2}}{\tau^2} (1 + o(1)). \quad (90)$$

At $\tau = b_n + s$, set

$$D_n = \frac{4d}{m^2} \frac{n^{-1/2}}{(\log n)^2}. \quad (91)$$

Then, uniformly for bounded s ,

$$Z_n(r_n) = D_n(1 + e^{-s} + o(1)), \quad \frac{e^{-\tau_n}}{m} = D_n e^{-s}. \quad (92)$$

Proof. Use $r = 1 + O(\log n/n)$ and $d = \kappa/\Gamma(-3/2)$ in Proposition 13.2. The absolute error $O(n^{-1})$ is $o(n^{-1/2}/\log^2 n)$. Substitution of (67) proves the second statement. $\square$

14 An elementary total variation theorem for a giant block

This section is independent of the analytic form of Q . It isolates the reason that two exponential variables, and hence a Gamma law, appear.

Definition 14.1 (Geometric renewal string). For a probability mass function q on the positive integers and $0 < \delta < 1$, a geometric renewal string has law

$$\nu_\delta(x_1, \dots, x_k) = \delta(1 - \delta)^k \prod_{i=1}^k q_{x_i}, \quad k \geq 0. \quad (93)$$

The empty string is included. Formula (93) is normalized because its sum over strings of length k is $\delta(1 - \delta)^k$.

Theorem 14.2 (Giant deletion for a changing geometric weight). *Let $q_j = dj^{-1-\alpha}(1 + O(j^{-1}))$, where $d > 0$ and $\alpha > 1$, and let its mean m be finite. Give an ordered string of total size n weight $r^k \prod_i q_{x_i}$, where $r = 1 - \delta$ may depend on n . Suppose $\delta \rightarrow 0$ and $n\delta \rightarrow \infty$. Let C be the event that a block exceeds $n/2$, and let $B_n(r)$ be its unnormalized weight. Then*

$$B_n(r) = \frac{rq_n}{\delta^2} \left(1 + O\left(\frac{1}{n\delta} + \frac{1}{n}\right) \right). \quad (94)$$

The law of the two strings obtained by deleting this unique block, conditional on C , has total variation distance

$$O\left(\frac{1}{n\delta} + \frac{1}{n}\right) \quad (95)$$

from $\nu_\delta \otimes \nu_\delta$.

Proof. Choose independent left and right strings with this product law. Let T be their combined total size and J their combined number of blocks. Inserting a block of size $n - T$ between them gives an ordered string in C precisely when $T < n/2$. The location of the large block is then determined uniquely. Hence

$$B_n(r) = \frac{rq_n}{\delta^2} \mathbb{E}h_n(T), \quad h_n(T) = \frac{q_{n-T}}{q_n} \mathbf{1}_{\{T < n/2\}}. \quad (96)$$

Furthermore, the conditional residual law has density $h_n(T)/\mathbb{E}h_n(T)$ with respect to the product string law. These are exact identities, before any limiting argument.

For $0 \leq t < n/2$, the assumed tail gives, uniformly in t ,

$$\frac{q_{n-t}}{q_n} = (1 - t/n)^{-1-\alpha}(1 + O(n^{-1})), \quad \left| \frac{q_{n-t}}{q_n} - 1 \right| \leq C \left(\frac{t}{n} + \frac{1}{n} \right).$$

Each string has expected size mr/δ , so $\mathbb{E}T = 2mr/\delta$. On $T \geq n/2$, $|h_n - 1| = 1$ and Markov's inequality applies. Therefore

$$\mathbb{E}|h_n(T) - 1| \leq C' \left(\frac{\mathbb{E}T}{n} + \frac{1}{n} \right) = O\left(\frac{1}{n\delta} + \frac{1}{n}\right). \quad (97)$$

This proves (94). Finally,

$$d_{\text{TV}}\left(\frac{h_n}{\mathbb{E}h_n}\nu_\delta^{\otimes 2}, \nu_\delta^{\otimes 2}\right) = \frac{1}{2}\mathbb{E}\left|\frac{h_n}{\mathbb{E}h_n} - 1\right| \leq \frac{\mathbb{E}|h_n - 1|}{\mathbb{E}h_n},$$

which proves (95). $\square$

Only regular variation, rather than the displayed $O(j^{-1})$ error, is needed for an $o(1)$ total variation statement: on $t < n/2$ the ratio is bounded, and $T/n \xrightarrow{\mathbb{P}} 0$. The stronger local tail is used to give the explicit rate.

14.1 Gamma laws and the location of the giant

Let N_δ be the count in one geometric string, and let T_δ be its total size. Then

$$\delta N_\delta \xrightarrow{d} E, \quad E \sim \text{Exp}(1). \quad (98)$$

Also

$$\delta(T_\delta - mN_\delta) \xrightarrow{\mathbb{P}} 0. \quad (99)$$

For completeness, choose $A > 0$. The probability $N_\delta > A/\delta$ can be made uniformly small by increasing A . By the ordinary strong law, $\max_{k \leq L} |\sum_{i \leq k} (X_i - m)|/L \rightarrow 0$ almost surely as $L \rightarrow \infty$. Apply this with $L = \lceil A/\delta \rceil$ to obtain (99). The argument uses only a finite first moment.

Theorem 14.2, with $\alpha = 3/2$ and $n\delta_n \asymp \tau_n$, now proves Theorem 12.2. It proves convergence of the entire strings, so every bounded statistic of them can be transferred; the statistic may itself depend on n .

Corollary 14.3 (A uniform split around the giant). *Conditional on $\mathcal{C}_n$, jointly with $\delta_n K_n \xrightarrow{d} G$,*

$$\frac{K_{L,n}}{K_n - 1} \xrightarrow{d} U, \quad \frac{T_{L,n}}{n - M_{1,n}} \xrightarrow{d} U, \quad (100)$$

where U is uniform on $(0, 1)$ and independent of G . Ratios with zero denominator can be assigned any value.

Proof. The four-coordinate limit (71) reduces both ratios to $E_L/(E_L + E_R)$. The change of variables $(E_L, E_R) = (GU, G(1 - U))$ has Jacobian G and density Ge^{-G} on $G > 0$, $0 < U < 1$. Thus U is uniform and independent of G . The limiting denominator is positive almost surely. $\square$

15 Identification of the phase probabilities

For the tournament law, the exact marked-block identity in (96) is equivalently

$$B_n(r) = r \sum_{t < n/2} q_{n-t}[y^t] \frac{1}{(1 - rQ(y))^2}. \quad (101)$$

It sums over the unique block exceeding half the total size. It does not confuse a count of marked blocks with the probability of an event containing several possible marks.

Theorem 14.2 and (68) give

$$B_n(r_n) \sim \frac{r_n q_n}{\delta_n^2} \sim \frac{d}{m^2} \frac{n^{-1/2}}{\tau_n^2} \sim D_n. \quad (102)$$

Divide this by (92). The result is $\mathbb{P}(\mathcal{C}_n) \rightarrow \pi_s$, proving (69). On $\mathcal{C}_n$, Theorem 12.2 implies $K_n/n \xrightarrow{\mathbb{P}} 0$ and $M_{1,n}/n \xrightarrow{\mathbb{P}} 1$ because $n\delta_n \rightarrow \infty$.

15.1 A short independent check from a probability generating function

The limiting block-count mixture can also be read from a controlled perturbation of the parameter. For fixed $t \geq 0$,

$$\mathbb{E}e^{-tK_n/n} = \frac{Z_n(r_n e^{-t/n})}{Z_n(r_n)}. \quad (103)$$

Write $r_n e^{-t/n} = (1 + m\tau'_n/n)^{-1}$. Exactly,

$$\tau'_n = \frac{n}{m} \left[(1 + m\tau_n/n) e^{t/n} - 1 \right] = \tau_n + t/m + o(1).$$

Uniformity in Corollary 13.3 yields

$$\mathbb{E}e^{-tK_n/n} \longrightarrow \frac{1 + e^{-s-t/m}}{1 + e^{-s}}. \quad (104)$$

This is the Laplace transform of atoms at 0 and $1/m$. Since $0 \leq K_n/n \leq 1$, it identifies the weak limit and also all its moments. The argument needs the expansion for nearby parameters; an uncontrolled decomposition at one value would not suffice.

The giant event already accounts for the whole limiting mass at $K_n/n = 0$. Subtracting its bounded-test-function limits from (104) proves $K_n/n \xrightarrow{\mathbb{P}} 1/m$ on $\mathcal{F}_n$.

15.2 Exclusion of intermediate macroscopic blocks

Fix $\epsilon > 0$, and choose $\eta > 0$ so small that $m\eta < \epsilon/4$. For

$$k \geq n/m - \eta n,$$

a specified block of size at least ϵn leaves at most $(1 - \epsilon)n$ for the other $k - 1$ blocks. This is a fixed proportion below their mean for large n . Since $X \geq 1$, its negative exponential moment exists. The finite mean expansion

$$\log \mathbb{E}e^{-hX} = -mh + o(h) \quad (h \downarrow 0)$$

shows, by choosing one sufficiently small fixed $h > 0$, that

$$\mathbb{P}(X_1 + \cdots + X_{k-1} \leq (1 - \epsilon)n) \leq e^{-c_\epsilon n} \quad (105)$$

uniformly over these $k \leq n$. Marking the exceptional block, summing over its size and location, and over k , gives an unnormalized weight at most $n^2 e^{-c_\epsilon n}$, since $r_n^k \leq 1$ and $\sum_j q_j = 1$. Division by the polynomially small $Z_n(r_n)$ still gives a vanishing probability.

On $\mathcal{F}_n$, the complementary range of k has probability tending to zero by the preceding count limit. Thus $M_{1,n}/n \xrightarrow{\mathbb{P}} 0$ on $\mathcal{F}_n$. This completes the proof of Theorem 12.1. The same estimates, with parameters on a compact s -interval, give the asserted uniformity.

Corollary 15.1 (Macroscopic moments). *Let J_s be Bernoulli with $\mathbb{P}(J_s = 1) = 1 - \pi_s$. Then $mK_n/n \xrightarrow{d} J_s$ and $M_{1,n}/n \xrightarrow{d} 1 - J_s$, jointly. Consequently,*

$$\begin{aligned} \frac{\mathbb{E}K_n}{n} &\longrightarrow \frac{1 - \pi_s}{m}, & \frac{\text{Var } K_n}{n^2} &\longrightarrow \frac{\pi_s(1 - \pi_s)}{m^2}, \\ \frac{\text{Cov}(K_n, M_{1,n})}{n^2} &\longrightarrow -\frac{\pi_s(1 - \pi_s)}{m}. \end{aligned} \quad (106)$$

Proof. The normalized variables are bounded, so weak convergence implies convergence of the indicated moments. This does not assert moment convergence for the unbounded stable or Gamma rescalings. $\square$

16 A direct stable local limit theorem and the fluid law

16.1 Local limit by Fourier inversion

Lemma 16.1 (Lattice 3/2-stable local limit). *Let $S_k^X = X_1 + \dots + X_k$ and $a(k) = (\kappa k)^{2/3}$. The variable Z in (73) has a continuous density g , and*

$$\sup_{\ell \in \mathbb{Z}} \left| a(k) \mathbb{P}(S_k^X = \ell) - g\left(\frac{\ell - mk}{a(k)}\right) \right| \longrightarrow 0. \quad (107)$$

Proof. Put $\phi(t) = Q(e^{it})$. The expansion (82) implies

$$\log \phi(t) = imt + \kappa(-it)^{3/2} + O(t^2). \quad (108)$$

Indeed $1 - e^{it} = -it + O(t^2)$, and its principal 3/2 power has the indicated leading term. For real t ,

$$(-it)^{3/2} = -\frac{|t|^{3/2}}{\sqrt{2}}(1 + i \operatorname{sgn} t).$$

It follows that $|\phi(t)| \leq \exp(-c|t|^{3/2})$ near zero. Away from zero Lemma 13.1 gives a strict uniform bound below one.

The characteristic functions of $(S_k^X - mk)/a(k)$ converge pointwise to $\exp\{(-it)^{3/2}\}$, by (108); their limit is continuous at zero and hence is a characteristic function. It is integrable, so its inverse Fourier transform is a continuous density g of total integral one.

Lattice inversion, followed by $v = a(k)t$, writes $a(k)\mathbb{P}(S_k^X = \ell)$ as the integral over $[-\pi a(k), \pi a(k)]$ of

$$\frac{1}{2\pi} e^{-iv(\ell - mk)/a(k)} \left[e^{-imv/a(k)} \phi(v/a(k)) \right]^k.$$

The bracketed kernel converges in $L^1(dv)$ to $\exp\{(-iv)^{3/2}\}$. On the shrinking small-angle region it is bounded by $\exp(-c'|v|^{3/2})$; on the remaining angles the integral is at most $O(a(k)\gamma^k)$ for some $\gamma < 1$. Dominated convergence proves the L^1 assertion. The oscillatory factor has modulus one for every ℓ , so it also proves the uniform bound (107). $\square$

16.2 Central windows and exhaustion of the remaining mass

Let $a_n = a(n/m) = (\kappa n/m)^{2/3}$, interpreting the formula at a real argument, and define

$$x_{n,k} = \frac{m(k - n/m)}{a_n}.$$

For $x_{n,k}$ in any fixed bounded interval, $a(k)/a_n \rightarrow 1$ uniformly and (107) gives

$$\mathbb{P}(S_k^X = n) = a_n^{-1}(g(-x_{n,k}) + o(1)). \quad (109)$$

Also, uniformly on this window,

$$\frac{r_n^k}{e^{-\tau_n}} \rightarrow 1. \quad (110)$$

To check the latter, use $\log r_n = -m\tau_n/n + O(\tau_n^2/n^2)$ and $k = n/m + O(n^{2/3})$. The remaining exponent is $O(\tau_n n^{-1/3} + \tau_n^2/n) = o(1)$.

For fixed $A < B$, a Riemann sum in (109) yields

$$\sum_{A \leq x_{n,k} \leq B} r_n^k \mathbb{P}(S_k^X = n) = \frac{e^{-\tau_n}}{m} \left(\int_A^B g(-x) dx + o(1) \right). \quad (111)$$

The mesh of $x_{n,k}$ is m/a_n , explaining the factor $1/m$.

The contribution in this window having a block greater than $n/2$ is exponentially small by (105). Hence (111) holds with the event $\mathcal{F}_n$ imposed. On the other hand, (92) and (102) imply

$$Z_n(r_n) - B_n(r_n) \sim \frac{e^{-\tau_n}}{m}. \quad (112)$$

This subtraction is legitimate because s is fixed, so both limiting contributions have positive finite relative weights.

Divide (111) by (112). Letting $A \rightarrow -\infty$ and $B \rightarrow \infty$ exhausts mass one because g is a probability density. This proves tightness of the conditional rescaled count as well as convergence on bounded intervals. The limit has density $g(-x)$, which is precisely the law of $-Z$. Together with Section 15, this proves Theorem 12.3.

The exhaustion argument is useful: a local limit theorem alone does not control all possible intermediate ranges. The separately proved giant weight and total coefficient show that no additional range has unaccounted mass at coexistence.

16.3 Largest fluid block and its dependence on the count

Let $\mathcal{P}$ be a Poisson point process on $(0, \infty)$ with intensity

$$\nu(dv) = \frac{v^{-5/2}}{\Gamma(-3/2)} dv.$$

Let V be its largest point and define the compensated sum

$$Z = \lim_{\epsilon \downarrow 0} \left(\sum_{\substack{v \in \mathcal{P} \\ v > \epsilon}} v - \int_{\epsilon}^{\infty} v \nu(dv) \right). \quad (113)$$

The compensated contribution of points below one converges in L^2 because $\int_0^1 v^2 \nu(dv) < \infty$. Above one there are finitely many points almost surely, and their sum has finite mean. The exponential formula gives

$$\mathbb{E} e^{itZ} = \exp \left\{ \int_0^\infty (e^{itv} - 1 - itv) \nu(dv) \right\} = \exp \{ (-it)^{3/2} \}.$$

The last integral follows by two integrations by parts, or the gamma integral with a positive real damping parameter followed by a limit. Thus this is the same stable law as before. Also

$$\mathbb{P}(V \leq y) = \exp \{ -\nu((y, \infty)) \} = \exp \left(-\frac{1}{2\sqrt{\pi} y^{3/2}} \right).$$

We prove the joint limit (75), including its dependence. For $y > 0$, define the defective density g_y by

$$\int_{\mathbb{R}} e^{itx} g_y(x) dx = \Psi_y(t) = \exp \left\{ (-it)^{3/2} - \int_y^\infty e^{itv} \nu(dv) \right\}. \quad (114)$$

The Poisson formula identifies the right side as $\mathbb{E}[e^{itZ} \mathbf{1}_{\{V \leq y\}}]$. It is integrable: its modulus is bounded by $\exp \{ \nu((y, \infty)) - |t|^{3/2}/\sqrt{2} \}$. Consequently g_y is continuous, nonnegative, and has total mass $\mathbb{P}(V \leq y)$.

Truncate the one-block characteristic function at $\lfloor ya_n \rfloor$:

$$\phi_{n,y}(t) = \sum_{j \leq ya_n} q_j e^{itj}.$$

For $k = n/m + O(a_n)$, regular variation and a Riemann sum imply

$$k \sum_{j > ya_n} q_j e^{itj/a_n} \longrightarrow \frac{1}{\Gamma(-3/2)} \int_y^\infty e^{itv} v^{-5/2} dv.$$

Combining this with (108) shows

$$e^{-itm k/a_n} \phi_{n,y}(t/a_n)^k \longrightarrow \Psi_y(t).$$

Convergence is uniform on bounded count windows. The Fourier kernels again converge in L^1 . Near zero, removal of the tail changes the characteristic function by at most C_y/n , and hence

$$|\phi_{n,y}(t/a_n)|^k \leq C'_y \exp(-c'|t|^{3/2})$$

for small fixed $|t|/a_n$. One may obtain this bound by applying $|\phi_{n,y}| \leq |\phi| + C_y/n$ to the exponential bound used in Lemma 16.1. Away from zero, the strict aperiodicity bound still applies for large n .

Fourier inversion therefore gives, uniformly on bounded count windows,

$$a_n \mathbb{P}(S_k^X = n, \max_{i \leq k} X_i \leq ya_n) = g_y(-x_{n,k}) + o(1).$$

After multiplying by r_n^k and taking a Riemann sum, its weight on a fixed interval I of count coordinates is

$$\frac{e^{-\tau_n}}{m} \left(\int_I g_y(-x) dx + o(1) \right).$$

For fixed y this event lies in $\mathcal{F}_n$ for large n , because $a_n = o(n)$. Divide by (112). The already proved conditional tightness of the count permits I to increase to all of $\mathbb{R}$. This yields the stated Fréchet

marginal; bounded rectangles identify the joint law. The count and maximum marginals give joint tightness, completing the proof.

The compensation term in (114) is necessary. Simply discarding the points above y would change the deterministic centering as well as the probability that no such point occurs. The largest fluid block and the stable count fluctuation are thus two observables of one limiting process.

The argument also identifies every large fluid block jointly. For nonnegative continuous f of compact support in $(0, \infty)$, replace the truncated characteristic function by $\sum_j q_j \exp(itj/a_n - f(j/a_n))$. The perturbation is $O(n^{-1})$ and has exactly the same Fourier domination. Its limiting exponent is

$$(-it)^{3/2} - \int_0^\infty (1 - e^{-f(v)}) e^{itv} \nu(dv).$$

The same local inversion, count-window summation, and tightness argument prove joint convergence of the rescaled count and the point measure of block sizes to $(-Z, \mathcal{P})$, with vague convergence on $(0, \infty)$. Here the mark $\exp(-\sum f)$ is at most one, so inserting $\mathcal{F}_n$ changes the central-window expectations by at most the exponentially small giant contamination in (105). In particular, any fixed number of largest fluid blocks converges to the corresponding ranked points. This is a statement about sizes; a functional limit keeping their positions in the ordered sequence is a separate question.

17 Finer laws inside the giant phase

17.1 Stable fluctuations at a Gamma random time

Let N_δ be the count in one independent geometric string. For $t \geq 0$ and $\theta \in \mathbb{R}$, its exact transform is

$$\mathbb{E} \exp \left\{ -t\delta N_\delta + i\theta \frac{T_\delta - mN_\delta}{(\kappa/\delta)^{2/3}} \right\} = \frac{\delta}{1 - re^{-t\delta} \mathbb{E} \exp \{ i\theta(X - m)(\delta/\kappa)^{2/3} \}}. \quad (115)$$

Equation (108) says that the last expectation is $1 + \delta(-i\theta)^{3/2} + o(\delta)$. The denominator divided by δ tends to $1 + t - (-i\theta)^{3/2}$. Squaring for two independent strings gives

$$(1 + t - (-i\theta)^{3/2})^{-2}. \quad (116)$$

This is the joint transform of $(G, G^{2/3}Z)$: conditional on G , the second coordinate has characteristic function $\exp\{G(-i\theta)^{3/2}\}$. Total variation transfer and $\delta_n \rightarrow 0$ permit replacing the residual count $K_n - 1$ by K_n in the first coordinate. This proves (76).

The variables in (72) should therefore not be read as independent. Their leading limits are the same random variable. Equation (76) measures the smaller fluctuation around that common leading value.

17.2 Second-largest block and joint maximum law

Let $F(x) = \mathbb{P}(X \leq x)$ and $B(\delta) = ((2d/3)/\delta)^{2/3}$. By (83),

$$1 - F(B(\delta)x) = \delta x^{-3/2}(1 + o(1)) \quad (x > 0). \quad (117)$$

If V_δ is the largest size in one geometric string, with maximum zero for the empty string, summing the geometric series gives

$$\mathbb{E}[e^{-t\delta N_\delta} \mathbf{1}_{\{V_\delta \leq B(\delta)x\}}] = \frac{\delta}{1 - re^{-t\delta} F(B(\delta)x)} \longrightarrow \frac{1}{1 + t + x^{-3/2}}. \quad (118)$$

The maximum of the two residual strings is the second-largest block of the original object on $\mathcal{C}_n$. Squaring (118) and applying Theorem 12.2 proves (78)–(79).

There is a useful probabilistic reading: conditional on $G = g$, the limiting rescaled second maximum has distribution $\exp(-gx^{-3/2})$. Mixing this Fréchet distribution over $G \sim \text{Gamma}(2, 1)$ produces $(1 + x^{-3/2})^{-2}$. Thus the second maximum and the total residual count are dependent.

17.3 All large residual blocks as a point process

The same computation describes more than the largest residual block. For a nonnegative continuous function f with compact support in $(0, \infty)$, set

$$J_f = \int_0^\infty (1 - e^{-f(x)}) \frac{3}{2} x^{-5/2} dx.$$

Regular variation gives

$$\delta^{-1} \mathbb{E}[1 - e^{-f(X/B(\delta))}] \longrightarrow J_f.$$

Repeating the geometric sum with this factor instead of F yields

$$\mathbb{E} \left[\exp \left\{ -t\delta_n K_n - \sum_{j \geq 2} f(M_{j,n}/B_n) \right\} \mid \mathcal{C}_n \right] \longrightarrow (1 + t + J_f)^{-2}. \quad (119)$$

This specifies the Laplace functionals of the limiting point process. Given G , it is a Poisson point process on $(0, \infty)$ with intensity $G(3/2)x^{-5/2}dx$. This statement concerns points bounded away from zero. It does not assert that uncentered sums of all limiting points are finite; the intensity has a divergent first moment at zero, and centered stable fluctuations require compensation.

17.4 All fixed inverse-logarithmic corrections

Here $(a)_j = a(a+1)\cdots(a+j-1)$ and $(a)_0 = 1$. The expansion of the heavy-tail kernel has a direct interpretation in terms of the gamma residual geometry; the coefficients are not merely formal contour artifacts.

Proposition 17.1 (All fixed orders for the giant weight). *Fix $C_0 > 0$ and an integer $M \geq 0$. Let $r = (1 + m\tau/n)^{-1}$ and $\delta = 1 - r$, with $1 \leq \tau \leq C_0 \log n$. Then, uniformly in this range,*

$$B_n(r) = \frac{rq_n}{\delta^2} \left\{ \sum_{j=0}^M (j+1)(5/2)_j \tau^{-j} + O_{M,C_0}(\tau^{-M-1}) \right\}.$$

In particular the expansion begins with $1 + 5/\tau + 105/(4\tau^2)$, and its prefactor can equivalently be written as $(d/(rm^2))n^{-1/2}\tau^{-2}(1 + O(n^{-1}))$.

Proof. First, the tail estimate gives an elementary first absolute moment bound for centered sums:

$$\mathbb{E}|S_k^X - mk| \leq Ck^{2/3} \quad (k \geq 1).$$

Indeed, take $A = k^{2/3}$ and split each summand into its part at most A and its remaining part. The assumed mass-function bound gives

$$\mathbb{E}[X\mathbf{1}_{X>A}] \leq CA^{-1/2}, \quad \mathbb{E}[X^2\mathbf{1}_{X\leq A}] \leq CA^{1/2}.$$

The centered truncated sum has first absolute moment at most its standard deviation, $\sqrt{CkA^{1/2}}$, and the centered tail sum has first absolute moment at most $2CkA^{-1/2}$. Both are $O(k^{2/3})$.

Under the reference law write $N = N_L + N_R$, so T has law S_N^X . Concavity and independence imply

$$\mathbb{E}_r|T - mN| \leq C\mathbb{E}_r N^{2/3} \leq C\delta^{-2/3}.$$

Let $f(t)$ equal $(1-t)^{-5/2}$ for $0 \leq t \leq 1/2$ and equal $2^{5/2}$ for $t \geq 1/2$. This is bounded and Lipschitz. Its use also controls the discontinuous cutoff in h_n :

$$\mathbb{E}_r|h_n(T) - f(mN/n)| = O\left(n^{-1} + \frac{\delta^{-2/3}}{n} + \mathbb{P}_r(T \geq n/2)\right).$$

Moreover

$$\mathbb{P}_r(T \geq n/2) \leq \mathbb{P}_r(N \geq n/(4m)) + \frac{4\mathbb{E}_r|T - mN|}{n}.$$

For the negative-binomial variable N , the exact tail at integer $L \geq 0$ is

$$\mathbb{P}_r(N \geq L) = r^L(1 + \delta L).$$

Because $n\delta = mr\tau$ and $r \rightarrow 1$ uniformly in the stated window, this proves

$$\mathbb{E}_r|h_n(T) - f(mN/n)| = O\left(n^{-1} + n^{-1/3}\tau^{-2/3} + (1 + \tau)e^{-c\tau}\right)$$

with a fixed $c > 0$.

For every fixed M , the global Taylor-remainder bound

$$\left|f(t) - \sum_{j=0}^M \frac{(5/2)_j}{j!} t^j\right| \leq C_M t^{M+1} \quad (t \geq 0)$$

holds. On $[0, 1/2]$ it is the usual Taylor estimate; on $[1/2, \infty)$ it follows because f is bounded and the displayed polynomial has degree M . The negative-binomial factorial moments are exact:

$$\mathbb{E}_r N^j = (j+1)! \left(\frac{r}{\delta}\right)^j.$$

Converting ordinary moments to factorial moments and using $mr/(n\delta) = 1/\tau$ gives

$$\mathbb{E}_r f(mN/n) = \sum_{j=0}^M (j+1)(5/2)_j \tau^{-j} + O_M(\tau^{-M-1} + n^{-1}).$$

Finally, for $1 \leq \tau \leq C_0 \log n$, each of n^{-1} , $n^{-1/3}\tau^{-2/3}$, and $(1 + \tau)e^{-c\tau}$ is $O_{M,C_0}(\tau^{-M-1})$. Insert the resulting expansion of $\mathbb{E}_r h_n(T)$ into the exact formula $B_n(r) = r q_n \delta^{-2} \mathbb{E}_r h_n(T)$. The equivalent prefactor follows from $r q_n / \delta^2 = (d/(r m^2)) n^{-1/2} \tau^{-2} (1 + O(n^{-1}))$. $\square$

Corollary 17.2 (Explicit residual approximations to every fixed order). *Let R_r denote the product of the two geometric-string reference laws in Theorem 12.2, and write $N = N_L + N_R$. For a fixed integer $M \geq 0$, define*

$$P_M(x) = \sum_{j=0}^M \frac{(5/2)_j}{j!} x^j, \quad \frac{dR_r^{(M)}}{dR_r} = \frac{P_M(mN/n)}{\mathbb{E}_r P_M(mN/n)}.$$

This is a probability measure because the polynomial is positive on $[0, \infty)$ and N has moments of every order. If $\tau \rightarrow \infty$ and $\tau \leq C_0 \log n$, then the actual joint residual-string law $\mathbf{L}_{n,r}$, conditional on $\mathcal{C}_n$, satisfies

$$d_{\text{TV}}(\mathbf{L}_{n,r}, \mathbf{R}_r^{(M)}) = O_{M,C_0}(\tau^{-M-1}).$$

In particular, in the coexistence window this gives an explicit $O((\log n)^{-M-1})$ approximation for every fixed M . Conditionally on their two counts, the block lengths under $\mathbf{R}_r^{(M)}$ still have the independent law q .

Proof. The absolute-value estimates in the preceding proof and its global Taylor bound give

$$\mathbb{E}_r |h_n(T) - P_M(mN/n)| = O_{M,C_0}(\tau^{-M-1}).$$

Indeed the Taylor error has expectation at most $C_M(m/n)^{M+1} \mathbb{E}_r N^{M+1} = O_M(\tau^{-M-1})$. The exact residual density is $h_n(T)/\mathbb{E}_r h_n(T)$, and $\mathbb{E}_r h_n(T) = 1 + O(\tau^{-1})$ is bounded away from zero. For nonnegative integrable functions h, P ,

$$\frac{1}{2} \mathbb{E} \left| \frac{h}{\mathbb{E}h} - \frac{P}{\mathbb{E}P} \right| \leq \frac{\mathbb{E}|h - P|}{\mathbb{E}h}.$$

Applying this inequality proves the assertion. $\square$

18 Centering, finite-size effects, and a reusable principle

18.1 An implicit center that keeps the logarithmic corrections

The explicit b_n is convenient for a theorem, but the $2 \log \log n$ correction can be large at accessible sizes. A useful alternative is the large solution $\hat{b}_n > 2$ of

$$\hat{b}_n^2 e^{-\hat{b}_n} = \frac{d}{m\sqrt{n}}. \quad (120)$$

For sufficiently large n it is unique on $(2, \infty)$ because $x^2 e^{-x}$ is strictly decreasing there. In terms of the lower real Lambert branch,

$$\hat{b}_n = -2W_{-1} \left(-\frac{1}{2} \sqrt{d/m} n^{-1/4} \right). \quad (121)$$

This is the exact balance of the leading fluid and giant terms in (90), not the exact finite- n median of the giant event.

[write] **Added 3 October 2026, batch 85.** The inversion (121) is an instance of `p0:thm:lambert-core` of the repository's transseries volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, and no novelty is claimed for it. Taking logarithms in (120) gives $x - 2 \log x = \Lambda$ with $\Lambda = \frac{1}{2} \log n + \log(m/d)$ (the equation used in the proof of Proposition 18.1); this is the theorem's equation $aX + b \log X = \Lambda$ with $a = 1$, $b = -2$, and its branch rule for $b < 0$ gives two positive solutions once $\Lambda > 2(1 - \log 2)$, the larger one, $X > |b|/a = 2$, on the branch W_{-1} at $z = (a/b)e^{\Lambda/b} = -\frac{1}{2} \sqrt{d/m} n^{-1/4}$. The order-zero inverse (51) of Part I is the same theorem with $a = \log 4$, $b = -5/2$ (the batch-77 note at the end of Section 7).

Proposition 18.1 (Refined center and error scale). *Put $c_0 = \log(m/(4d))$ and $L = \log n$. Then*

$$\hat{b}_n = b_n + \frac{8 \log L + 4c_0}{L} + O\left(\frac{(\log L)^2}{L^2}\right). \quad (122)$$

All the main theorems remain valid with $\tau_n = \hat{b}_n + s$. For this choice, uniformly for bounded s ,

$$\mathbb{P}(\mathcal{C}_n) = \frac{1}{1 + e^{-s}} + O\left(\frac{1}{\log n}\right). \quad (123)$$

With the elementary center $b_n + s$, the same argument gives the weaker bound $O(\log \log n / \log n)$.

Proof. Taking logarithms of (120) gives $x - 2 \log x = L/2 + \log(m/d)$. Substitute the right side of (122); expansion of $\log x$ leaves a residual $O((\log L)^2/L^2)$. The derivative $1 - 2/x$ is bounded away from zero for large x , proving the expansion by the mean value theorem. In particular $\hat{b}_n - b_n \rightarrow 0$, which preserves all the limiting distributions.

For the rate, write $D = dn^{-1/2}/(m^2 \hat{b}_n^2)$. Equation (120) makes $e^{-(\hat{b}_n + s)}/m = De^{-s}$ exactly. Theorem 14.2 gives $B_n = D(1 + O(1/\log n))$, uniformly for bounded s . Proposition 13.2 and (86) give $Z_n = D(1 + e^{-s} + O(1/\log n))$; the relative contour error $O(n^{-1/2} \log^2 n)$ is smaller. Dividing proves (123). The last assertion follows from (122) and smoothness of the logistic function. $\square$

The probability transition occurs over $O(1)$ changes of s , or $O(1/n)$ changes of u around its moving center. The center itself lies below u_c by order $\log n/n$. Confusing this displacement with the narrower transition width obscures the two scales.

18.2 What is specific to tournaments

Theorem 14.2 applies to any positive integer law with a finite mean and a regular power tail. The tournament identities supply the particular exponent $5/2$, the constants, and the uniform coefficient control. The stable exponent $3/2$ comes from the block tail, while Gamma shape two comes from having a string on each side of a unique giant.

For a law $q_j \sim dj^{-1-\alpha}$ with $1 < \alpha < 2$, a two-term renewal estimate of the corresponding form would balance

$$\frac{e^{-\tau}}{m} \quad \text{and} \quad \frac{d}{m^2} n^{1-\alpha} \tau^{-2}.$$

It would have center

$$(\alpha - 1) \log n + 2 \log \log n + \log \frac{m(\alpha - 1)^2}{d}.$$

The giant-deletion theorem would then give the same Gamma law, and the appropriate stable local limit would give count fluctuations $n^{1/\alpha}$ in the linear-block regime. This is a deduction conditional on those two analytic inputs, not a blanket theorem under regular variation alone. It states the reusable architecture without silently assuming a uniform renewal expansion for every law.

19 Computation and reproducibility

The computations accompanying this manuscript have two separate roles: exact checks of the finite combinatorial identities and numerical diagnostics of convergence. The proofs above do not rely on a numerical extrapolation.

19.1 Exact arithmetic checks

The exact recurrences are

$$\begin{aligned} nS_n &= \sum_{j=1}^n N_j S_{n-j}, \\ I_n &= S_n - \sum_{j=1}^{n-1} I_j S_{n-j}, \\ W_0(u) &= 1, \quad W_n(u) = u \sum_{j=1}^n I_j W_{n-j}(u). \end{aligned} \tag{124}$$

For comparison with direct enumeration, the full block-count polynomial is obtained by retaining u symbolically. The giant-event polynomial has the independent marked-block expression

$$\sum_{j > n/2} u I_j [z^{n-j}] (1 - uI(z))^{-2}. \tag{125}$$

The verifier enumerates the integer tuples in (59), cuts each at its equality indices, and compares both polynomials coefficient by coefficient. This checks the count, the block statistic, and the strict inequality defining the giant event.

The completed run examined 125,476 nondecreasing candidate sequences over all lengths through $n = 10$. Every exact comparison passed. At $n = 10$, the block-count polynomial is

$$573u + 433u^2 + 246u^3 + 130u^4 + 55u^5 + 33u^6 + 7u^7 + 8u^8 + u^{10},$$

and the giant-event polynomial is

$$573u + 424u^2 + 225u^3 + 84u^4 + 35u^5.$$

Their values at $u = 1$ are 1486 and 1341. Exact counts through $n = 200$ also supply independent checks of the normalized numerical recurrences.

19.2 Constants and floating diagnostics

If $R_j = [z^j]R(z)$ in (80), then

$$\lambda = \frac{\pi^2}{12} - \log^2 2 + \sum_{j \geq 1} \frac{R_j}{4^j}, \quad \mu = \log 2 + \sum_{j \geq 1} \frac{j R_j}{4^j}. \tag{126}$$

The omitted tails after M are bounded respectively by 2^{-M-1} and $(M+2)2^{-M-1}$. These are analytic truncation bounds; ordinary floating evaluations of the finite sums are still diagnostics unless their rounding is separately enclosed.

Numerically the constants begin

$$\begin{aligned} \lambda &= 0.3302375439859293 \dots, & \mu &= 0.6739526018515949 \dots, \\ u_c &= 3.5555932000209863 \dots, & m &= 1.7223486864283872 \dots \end{aligned}$$

The numerical code works with 4^{-n} -normalized coefficients to avoid exponential overflow. It evaluates $Z_n(r)$ by its positive renewal recurrence and evaluates (101) by a convolution. It compares the

resulting giant probabilities with the limiting logistic curve around $\hat{b}_n$. The default run computes normalized coefficients through $n = 8000$. In the recorded environment the long-double type has 64 significant binary digits. Through $n = 200$, its relative discrepancies from the exact counts are at most $1.1 \cdot 10^{-19}$ for S_n and $1.7 \cdot 10^{-19}$ for I_n . These comparisons are useful numerical checks, not interval error bounds for every subsequent calculation.

The measured center probabilities show why the distinction matters:

n	$\hat{b}_n$	$\mathbb{P}(\mathcal{C}_n)$ at $s = 0$
512	8.1978140113	0.5241164884
2000	9.0845092174	0.5341770504
8000	9.9620885205	0.5498246417

These values do not converge monotonically toward the limiting $1/2$. At $n = 8000$, the giant weight is about 1.73615 times its leading approximation, and the total weight is about 1.57882 times the sum of the two leading approximations. Terms such as $5/\tau$ and $105/(4\tau^2)$ in Proposition 17.1 are still substantial when τ is only about ten. No theorem above claims a small finite- n relative error from the leading terms alone.

For Figure 2, FFT polynomial powers were checked against the positive recurrence: the total-mass relative discrepancy was below $1.4 \cdot 10^{-15}$, and the giant-mass discrepancy below $1.0 \cdot 10^{-17}$. The computed conditional means are approximately 148.45 blocks on $\mathcal{C}_n$ and 468.27 on $\mathcal{F}_n$, whereas the leading scales give 134.45 and 580.60. This quantifies the visible finite-size correction rather than treating the plots as close asymptotic agreement.

The package contains the source, compiled article, verification programs, recorded outputs, and figures. It is independent of a ProveIt checkout. No assertion is made that numerical agreement establishes a total variation estimate or a distributional limit.

[write] **Added 3 October 2026, batch 85.** The preceding paragraph describes the delivered package. In this report its files carry the prefix `02-coex-`: the program `code/verify_and_plot.py` is `code/02-coex-verify_and_plot.py`, the build script is `code/02-coex-build.sh`, the code README is `02-coex-code-README.md` at the report root, the requirements and `provenance.json` are in `data/`, and every `data/` and `figures/` file keeps its name after the prefix. The delivered source, PDF, README and checksum list are not shipped; they survive in the archive of arrival commit `317c1ce2e`. Run as delivered, the program writes into `data/` and `figures/` beside `code/` under the unprefix names, so the report's `README.md` runs it on a scratch copy with an explicit `-output-root`. Such a run at intake (Windows, Python 3.12, the pinned packages) passed every exact assertion with identical exact output; its floating diagnostics agreed with the recorded ones to a relative $7.2 \cdot 10^{-13}$ (coexistence table), $5 \cdot 10^{-15}$ (normalized counts and renewal masses) and an absolute $6.5 \cdot 10^{-14}$ (block-count distribution), but not digit for digit, because on that platform `numpy.longdouble` is ordinary double precision (52 stored mantissa bits against the recorded 63), the platform dependence that the code README discloses. The figure showing the theoretical phase profiles, which the manuscript ships but does not display, is printed as Figure 3.

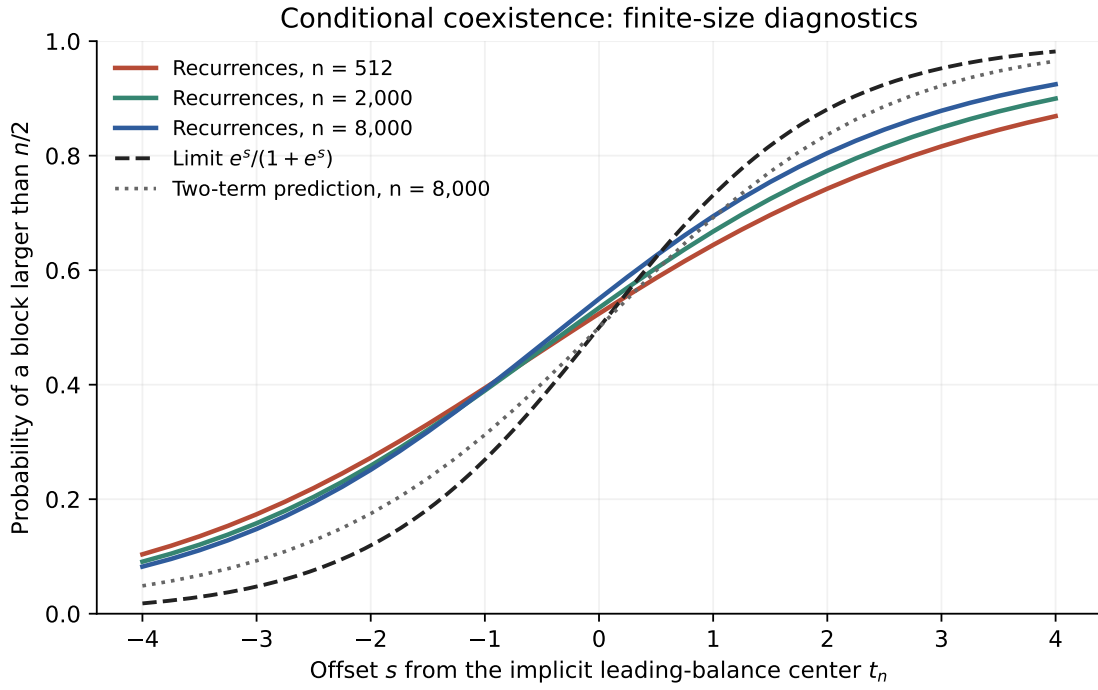


Figure 1: Slow finite-size behavior of the giant probability around the implicit center (120). The limiting logistic curve is shown for comparison. The finite curves are floating-point diagnostics, not interval certificates or evidence of monotone convergence.

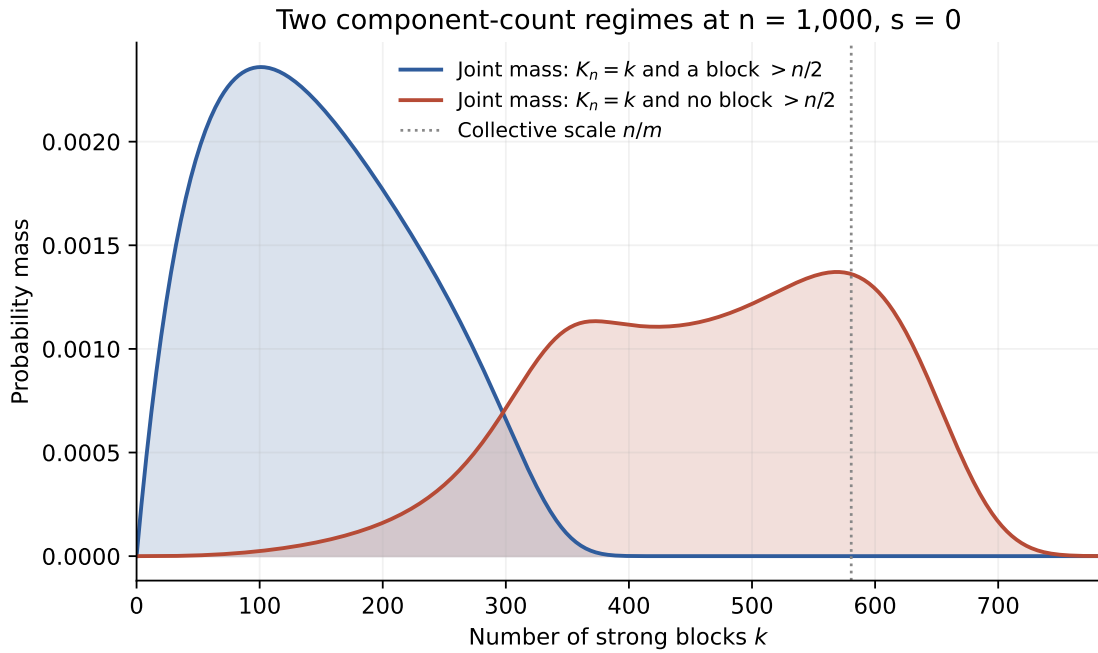


Figure 2: Block-count probabilities at $n = 1000$ and $\tau = \hat{b}_n$, separated by the giant event. The two curves are joint probabilities with their indicated events, so their combined mass is one. This finite distribution shows distinct count ranges and substantial corrections to the limiting scales.

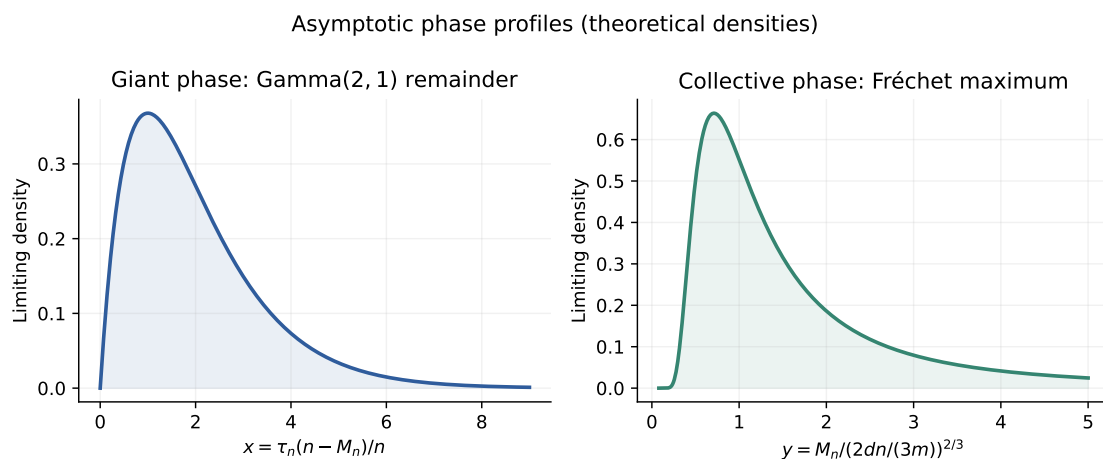


Figure 3: [Added in the write step, batch 85; caption from the delivered code README.] Theoretical limiting densities, not numerical distribution estimates: left, the Gamma(2, 1) density xe^{-x} of the rescaled giant deficit $x = \tau_n(n - M_{1,n})/n$ on $\mathcal{C}_n$ (Theorem 12.2); right, the Fréchet density $\frac{3}{2}y^{-5/2} \exp(-y^{-3/2})$ of the largest block on $\mathcal{F}_n$ in the variable $y = M_{1,n}/(2dn/(3m))^{2/3}$, which is V of (75) after the change of scale from a_n . The curves illustrate the proved limits; they are not an independent numerical test of them.

20 Further research questions

The following questions are not used as assumptions above. They continue the structural description at places where a new argument is actually needed.

- RQ1. A functional limit retaining the order of the blocks.** The joint count and ranked-size limits are now proved. Retain instead positions in the ordered sequence and study the centered cumulative block-size path on the whole unit interval, jointly with the count. Specify the topology, the endpoint conditioning, and the placement of large jumps. This would give a structural bridge to functional Gibbs-partition limit theorems.
- RQ2. A sharp finite-size coexistence center.** Continuity and uniform logistic convergence already imply that, for sufficiently large n , a weight u_n^* near the transition satisfies $\mathbb{P}_{n,u_n^*}(M_{1,n} > n/2) = 1/2$. Prove local uniqueness and derive sharper root corrections beyond (121). The implicit leading balance is not an exact median equation. Monotonicity of this particular event in the block weight should be proved rather than inferred from its limiting profile.
- RQ3. Rare configurations and unbounded residual observables.** Corollary 17.2 gives every fixed inverse-logarithmic order in total variation through an explicit count-polynomial tilt. Determine relative errors on specified rare residual configurations, or develop weighted-norm estimates controlling unbounded observables. These demands are stronger than total variation. For example, the unconditioned reference residual mass has an infinite second moment, whereas the actual residual mass is bounded by $n/2$. The polynomial tilt in the count alone does not remove this distinction.
- RQ4. Quantitative stable approximation.** Obtain explicit bounds for the conditional distribution in (74). The Fourier proof gives convergence but not an optimized rate. The exact next Puiseux coefficients may produce local corrections, but control outside fixed central windows must be retained.
- RQ5. Uniform growing shifts and the two pure regimes.** Extend the mixture theorem to $s = s_n \rightarrow \pm\infty$, identifying the largest ranges for which the phase-conditioned laws remain uniform. The compact- s subtraction in (112) is not uniform when one of the contributions becomes negligible. A proof must replace it with a relative estimate for the smaller event.
- RQ6. The critical and supercritical sides.** At $u = u_c$ the renewal mass is of order one, while for fixed $u > u_c$ a pole occurs inside the original convergence radius. Give a joint structural theorem connecting the logarithmic subcritical window, the critical law, and the moving supercritical pole. The present article does not analyze a negative τ .
- RQ7. General weights and slowly varying tails.** Prove the required local two-term coefficient estimate for broader classes $q_j = j^{-1-\alpha}L(j)$, and determine how L changes the center. Compare the exact hypotheses and conclusions with the Gibbs-partition and variable-box-number literature before asserting a new universality theorem.
- RQ8. Certified computation and exact sampling.** Use interval arithmetic for the constants and normalized recurrence to enclose finite-size phase probabilities. The product-string approximation also suggests a sampler on $\mathcal{C}_n$ with an explicit error budget; implement a rejection correction using $h_n(T)$ and prove its complexity.
- RQ9. Proof-assistant formalization.** Formalize the equality-index decomposition, the marked giant identity, and the total variation lemma first. They use finite combinatorics, geometric

series, and elementary expectation bounds. Then formalize the stable local limit and the common-contour estimate as separate analytic dependencies. Merely checking the integer table would not formalize any of the limit theorems.

[write] **Added 5 October 2026, batch 98.** RQ6 is answered in part by Part III (unrefereed, AI-assisted, not formalized). The critical law is its endpoint $t = 0$: the count limit X_0 of Theorem 27.2 at $\sigma = 0$ is the law of $-Z$ in (74), and the maximum limit of Theorem 28.2 at $\sigma = 0$ is the law of V in (75), with the same scale (Part III's b_n is a_n here), now without conditioning and at weights $u \geq u_c$. The moving supercritical pole is treated for $0 \leq t \leq t_0$ (Theorem 24.1), with tempered-stable, Gaussian and Gumbel regimes (Theorems 27.2–29.1). The joint structural theorem that RQ6 asks for, connecting the logarithmic subcritical window, the critical law and the supercritical pole in one statement, is not proved: it is Part III's RQ4, still open, and the sentence “The present article does not analyze a negative τ ” remains true of Part II. RQ4 is not answered either: Part III's $O(n^{-1/3})$ characteristic-function rate (184) concerns the unconditioned count at $u \geq u_c$, not the conditional law on $\mathcal{F}_n$ in (74).

Part III

Beyond criticality: pole–branch expansions and tempered-stable crossovers

Summary. A tournament score sequence has a unique ordered decomposition into strong score sequences. Weighting each strong block by u produces a critical weight u_c . The existing ProveIt report (Parts I and II of this report) analyzes the subcritical side and explicitly leaves the supercritical moving-pole problem open. We develop that side from the established exact generating function.

The principal analytic result is an exact pole contribution plus a branch-cut expansion to every fixed inverse-power order, uniformly from criticality through a fixed small supercritical interval. A computable kernel resolves the coalescence at $nt = O(1)$, where t is the excess exponential growth rate. A different scale, $tn^{2/3} = O(1)$, controls the block statistics: the centered block count has an explicit tempered 3/2-stable limit, with an $O(n^{-1/3})$ characteristic-function error on compact parameter sets, and becomes Gaussian when $nt^{3/2} \rightarrow \infty$. Large blocks converge to a tempered Poisson point process; their maximum changes from the critical power-law law to a Gumbel law. We also obtain convergent Puiseux inversion of the free energy, a quadratic inverse density response, and Lambert- W and incomplete-gamma centerings for the maximum.

The leading unweighted enumeration, exact generating function, and general stable/renewal mechanisms are prior work, not new claims. The contribution is the supercritical specialization, uniform correction generator, and the detailed crossover proofs. The manuscript is unrefereed and not formally verified; numerical checks are reproducible but are not interval certificates.

Deliverable and scope, as delivered. **Deliverable.** Self-contained proofs after one explicitly credited exact-enumeration input; LaTeX source, compiled PDF, exact finite checks, numerical diagnostics, and replay scripts. **Scope.** Marginal count and point-process theorems are proved. Joint count–extreme independence, growing truncation orders, and effective finite-size probability bounds are not claimed.

[write] **Added 5 October 2026, batch 98: provenance of Part III.** Part III is a single manuscript, printed in full: manuscript 03 of batch 98 of ProveIt’s incoming-reports intake, *Beyond Criticality: Pole–Branch Expansions and Tempered-Stable Crossovers for Tournament Score Sequences* (“OEIS research / A000571 & A351822”, dated 4 October 2026; a 1,078-line source and a 27-page PDF; its title page and PDF metadata name the author as “Research manuscript prepared with AI assistance”). It arrived as the archive `tournament_supercritical.zip` (wrapper directory `tournament_supercritical/`, main file `article.tex`) in commit 2172df76a and was placed in commit 9353a7171 (batch 98A) as an addition to this report. It pins the repository at commit 106eb191a2a47ab6281e2c26c7f4fe3d28872b83 (a merge commit of 4 October 2026, an ancestor of the placement) and continues this report; neither `article.tex` nor `README.md` of the report changed between the pin and the placement, so the manuscript read exactly the text printed as Parts I–II. Its code, data, build scripts and source notes are shipped with the prefix `03-supercrit-`; its own L^AT_EX source (printed here), delivery README, PDF and checksum list are not shipped and survive in the archive of the arrival commit. The manuscript is unrefereed and was prepared with AI assistance; its proofs are ordinary mathematical proofs, nothing in it is formalized in Lean or Rocq, and its place in this report confers no formal status.

What it answers. Part III answers Part I’s second further question (“Supercritical crossover”, Section 9) on the fixed small supercritical interval $0 \leq t \leq t_0$, that is for weights $u = r(t)/p$ between u_c and $r(t_0)/p$, with $t_0 > 0$ not made effective: the moving pole is separated from the branch point (Lemma 24.2) and its residue plus a branch integral expanded to every fixed order give Theorem 24.1. It answers Part II’s RQ6 (“The

critical and supercritical sides”, Section 20) in part: the critical law and the supercritical side are treated, but the joint theorem that RQ6 asks for, connecting them with the subcritical logarithmic window in one family, is not proved and is restated as Part III’s RQ4. Dated notes at those two questions say so.

Dictionary. Part III uses Part II’s normalization: $r = up$, $Z_n(r) = 4^{-n}W_n(r/p)$, and m, κ, d, q_j, Q as in (63)–(65) (m is Part I’s a_c). Under Part II’s parametrization $r = (1 + m\tau/n)^{-1}$, Part III’s argument $a_n(t) = nA_r/m$ of (152) equals $-\tau$ exactly, so the supercritical side $a \geq 0$ is $\tau \leq 0$, which Parts I and II exclude. Part III’s $b_n = (\kappa n/m)^{2/3}$ is Part II’s stable scale a_n , not the coexistence centre b_n of Parts I and II; the notation note below lists every such clash.

Overlap with Parts I–II, printed and credited. The strong-block decomposition of Section 22 is CDFS’s, proved by the same argument as Section 11 and Section 1. In Lemma 22.1 the constants m, κ, d and the tail (141) re-derive Lemma 13.1 and (82); its a_2 is Part I’s e_2 of (17) (because $\rho^2/2 = 1/32$), and its β equals $-m\ell$ with Part I’s $\ell = d_4$ of (40); κ_1 is new. The first two terms of the critical expansion (164) are (29) of Proposition 3.2 ($1/m = 1/a_c$ and $\kappa/m = k = 2/(3\mu)$), as the manuscript itself says; its $n^{-3/2}$ coefficient $\kappa\mathcal{K}_1(0)/(\pi m^2)$ is new (Part I prints only $O(n^{-3/2})$; a dated note after Proposition 3.2 says so). The large- a series of Proposition 25.1 is the series of (39) and of Proposition 17.1 with $\tau = -a$ (because $\mathcal{K}_0(a)/\pi$ and Part I’s $H(\tau)$ have the leading factor $\Gamma(5/2)/\pi = 3/(4\sqrt{\pi})$), and the branch amplitude $rd/(r-1)^2$ of (167) is Part I’s subcritical amplitude (28) continued to $r > 1$ (with $u = r/p$ and $d = e^{-\lambda}/(2\sqrt{\pi}p)$). At $\sigma = 0$ the limits of Theorems 27.2 and 28.2 are the laws of $-Z$ and V in Theorem 12.3, with the same scale (Part III’s b_n is Part II’s a_n); Part II proves them in a different regime, conditionally on $\mathcal{F}_n$ at the subcritical coexistence weight, while Part III’s are unconditional at weights $u \geq u_c$. Its Lambert- W centre and integer-threshold separation are instances of theorems of the repository’s transseries volume (note in Section 26). Everything else is new to the repository: Theorems 24.1, 26.1, 27.2 (for $\sigma > 0$), 27.3, 28.2 (for $\sigma > 0$) and 29.1, Proposition 23.1, Lemmas 23.2, 24.2, 27.1 and 28.1, Corollary 24.3, the kernels $\mathcal{K}_0, \mathcal{K}_1$ with the closed form (162), and the inverse formulas of Section 26. Priority beyond the repository is discussed in the note at the end of Section 21.

Editorial changes. Section k of the manuscript is Section $k + 20$ here, and its Appendices A, B and C are Appendices C, D and E, after Part II’s two appendices, with “(Part III)” added to their titles; the Part’s title is shortened to “tempered crossovers” in the table of contents only. Every label carries the prefix `tss:sc:`. The title page and table of contents are dropped; the abstract and the title page’s “Deliverable” and “Scope” lines are printed as the two paragraphs above. The manuscript’s references to “the existing ProveIt report” and its citation `Repo` (whose pin and the README lines 120–170 it checked are recorded in this note) now point to Parts I and II, in the summary and in Section 21; its three OEIS entries are the shared [6] entry, and Grabchak’s paper [11] is added to the bibliography. The manuscript’s macro `\Pp` is printed with Part II’s `\Prob` (the same $\mathbb{P}$), and its five table fragments are read from the shipped `data/03-supercrit-` files. The [write] notes and Remark 29.2 are new. No statement, proof or number of the manuscript was altered.

[write] **Added 5 October 2026, batch 98: notation of Part III.** Part III keeps the manuscript’s symbols. These mean the same as in Part II: $\lambda, \mu, \rho, p, u_c, A, F, R, N_n$ (A145855), $S_n, I_n, I(z), W_n(u), Q, q_j, m, \kappa, d, r = up, Z_n(r), K_n$ (the number of blocks) and $\Gamma(-3/2)$. Every other symbol below has the meaning in the middle column inside Part III; the right column gives the tempting false reading from Parts I–II. The manuscript’s own list is Appendix C.

symbol	meaning in Part III	not: meaning in Parts I–II
$t, r(t), h$	the tilt or excess growth rate, $t \geq 0$; $r(t) = 1/Q(e^{-t})$; $h(t) = \log r(t)$, and in Theorem 26.1 also its value $h = \log r$	$t = \sqrt{1-4z}$; Part I’s free weight parameter of $\mathcal{M}_{n,M}(t)$, Part II’s Fourier variable; $h = \log 4$ (Part I), $h_n(T)$ and a Chernoff h (Part II)
τ	only in Lemma 27.1: the local solution of $h(\tau) = \log r$, close to $t \geq 0$	the subcritical parameter of Parts I–II, with $\tau = -a_n(t) \approx -nt$: opposite sign and scale
$A_r, a = a_n(t)$	$A_r = 1 - r^{-1}$ and $a_n(t) = nA_r/m = -\tau$	$A(z)$; $a_n(\tau) = u_n(\tau)c$ of Part I (Part II’s rm); Part II’s stable scale a_n
b_n	the stable block scale $(\kappa n/m)^{2/3}$, Part II’s a_n	the coexistence centre b_n ; Part I’s Puiseux coefficients b_j

symbol	meaning in Part III	not: meaning in Parts I–II
$H(v), B(v)$	the real analytic factors of the boundary values of Q on the cut (142)	$H(\tau)$, the entire kernel of Parts I–II; Part I's odd series $B(t)$
$\mathcal{K}_j(a)$	the cut-integral kernels (153), (154), (159); $\mathcal{K}_0(a)/\pi$ continues Part I's H to $\tau = -a$ up to an elementary term (note in Section 31)	Part I's Hankel kernels $\mathcal{K}_{\alpha,b}(\tau)$ of (41)
$E(\epsilon, x), \epsilon, \varepsilon$	the coefficient-kernel series (158) in the formal variable ϵ ; ε a sector margin in Lemma 27.1	Part I's even part $E(v)$, resolvent error $E_n(w)$ and $\varepsilon = n^{-1/2}$; Part II's exponential variables E
J, C_j, c_J, C, R_0	a fixed truncation order; generic constants; the outer radius of Lemma 24.2	Part I's compact interval J , uniform terms $C_m(\tau)$, divisor remainder $R(z)$; Part II's J, C, R_j
v, δ	the cut variable $z = 1 + v$; also $v = h/m$ in Theorem 26.1; δ the cut length in the proof of Lemma 22.1	Part I's $v = t^2$; Part II's $\delta_n = 1 - r_n$
q_t, m_t, v_t, h_2	the tilted block law, its mean and variance, the t^2 coefficient of h	—
β, κ_1, a_2	local coefficients of Q at 1; $a_2 = e_2$ and $\beta = -m\ell$ in Part I's notation	Part I's $\ell = d_4$ and e_2 are the same numbers under other names
$U_t(n), V_{k,t}(N)$	the renewal visit probability (147) and its convolution powers (191)	Part II's uniform variable U and largest Poisson point V
N	an integer argument of $V_{k,t}$	N_n (A145855); Part II's string counts N_δ, N
X_n, X_σ, Y_n	centred and scaled block counts in Theorem 27.2	Part II's block lengths X, X_i
σ, σ_n	the tempering parameter, $\sigma_n = t_n b_n \rightarrow \sigma$	Part I's sign $\sigma = \pm$ (all or strong)
s	$\sqrt{1-z}$ in the proof of Lemma 24.2; the variable of the generating function (185); $t = s/n$ in Sections 25 and 29	the coexistence shift s and Part I's Hankel variable $s = nw$
S, T	bounds for σ_n and $ \xi $ in Theorem 27.2	$S(z), S_n$; Part II's lengths $T, T_{L,n}$
$L_{i,n}, M_n$	the block lengths in their order, and their maximum (Part II's $M_{1,n}$)	Part I's truncation order M
L, ℓ	$L = \log B$ in (177), an order in Proposition 25.1, a cutoff in (197); $\ell = \log 4 + t$ in Section 26 and the number of marked blocks in Lemma 28.1	Part I's $L = \log Y$ and $\ell = d_4 \simeq 1.2257$; Part II's $L = \log n$
$w, \widehat{w}, w_n, B, B_n$	$w = 1 - z$ in Lemma 22.1; the Lambert core and incomplete-gamma centres (176), (178); B and B_n the large-tail intensity parameters (175), (198)	Part I's $w = 1 - 4z$; Part II's second-maximum scale B_n and giant weight $B_n(r)$; Part I's $B(t)$
θ, Δ	the block density $1/m_t$ and $m\theta - 1$	—
g, y, e, c	the local inverse of h (proof of Theorem 27.3); a target in (174) and an integration variable in (178); a coefficient in the proof of Theorem 26.1; generic constants	Part I's $g_\ell(\alpha)$, $y = 4z$, $e, c = e^{-\lambda}\mu$; Part II's density g
ν_σ	the tempered Lévy intensity of (193)	Part II's ν and string law ν_δ

New macros: \Kc ($\mathcal{K}$), and the operators Li and PPP; the manuscript's unused \supp is not defined here.

21 The problem, the source gap, and the contributions

Let S_n be the number of nondecreasing score sequences of tournaments on n vertices. This is OEIS A000571 [6]. A score sequence records the sorted vertex scores; it is not a labeled tournament. Different tournaments with the same sorted scores contribute only once. Let I_n count strong score sequences of positive length. These are A351822 with its empty object omitted, so that $I_0 = 0$ in our generating functions [6].

The exact enumeration was clarified by Claesson, Dukes, Franklín, and Stefánsson (CDFS) [1]. Kolesnik [2] established the leading constants for all and strong score sequences, as well as the limiting unweighted number of irreducible subscores. Bassan, Donderwinkel, and Kolesnik [3] give a further treatment through the Lévy–Khintchine method. In particular, the leading asymptotic of A000571 is not an open problem being solved here.

The directly continued repository item is *Tournament Score Sequences and Weighted Component Asymptotics*, Parts I–II, now Parts I and II of this report. Its README explicitly excludes the supercritical moving pole, and identifies the supercritical crossover as Part I’s second further question and Part II’s RQ6. We checked this statement at commit

106eb191a2a47ab6281e2c26c7f4fe3d28872b83.

The snapshot was retrieved on 4 October 2026 in Pacific time (5 October UTC). The report’s subcritical coexistence results are not imported as assumptions below. Instead, we return to CDFS’s exact formula and derive the analytic input needed on the other side.

21.1 What is added

The new development relative to that inspected item has four parts. First, Theorem 24.1 gives a uniform pole–branch expansion with an explicit error at every fixed order. It covers the critical endpoint and fixed sufficiently small supercritical weights in one formula. Second, Theorems 27.2 and 27.3 give the tempered-stable and Gaussian block-count regimes, and identify their transition scale. Third, Theorems 28.2 and 29.1 give the large-block and maximum laws. Finally, Section 26 derives inverses adapted to this side of the transition.

These statements answer the specified repository question in a neighborhood of the critical weight. They do not assert that every formulation of a supercritical model has been exhausted. Exponential tilting, renewal conditioning, singularity analysis, and tempered stable distributions are established methods; for the last topic see, for example, Grabchak [11]. Our literature check is bounded, not a proof of worldwide publication priority.

[write] **Added 5 October 2026, batch 98: priority at the critical endpoint.** The manuscript’s bounded search (Appendix D) cites neither Banderier–Flajolet–Schaeffer–Soria [12] nor Flajolet–Sedgewick [13, Section IX.11, “Non-Gaussian continuous limits”], the sources named at placement for limit laws of *critical composition schemes* and the map–Airy law. The write checked them, as far as they were accessible, before keeping any $\sigma = 0$ statement as new. At $u = u_c$ the weighted construction $1/(1 - u_c I(z)) = 1/(1 - Q(z/\rho))$ is such a scheme: the outer function $1/(1 - y)$ is singular at $y = 1 = Q(1)$, exactly where the inner function has its singularity, of exponent $3/2$ by (138). For inner exponents between 1 and 2 the number of components in a critical composition has a stable limit law of that index, for exponent $3/2$ the *map–Airy* law, which BFSS analysed (their Theorem 12, “partially ... without a precise statement for the right renormalizations”, in the account of Banderier–Kuba–Wallner [14, Section 1], whose companion paper on this range is announced there as in preparation). Outcome: the stable $3/2$ limit of the centred block count at the critical weight itself ($t_n = 0$, the case $\sigma = 0$ of Theorem 27.2) is an instance of that known mechanism and is *not* claimed as new here; within the repository it is also Part II’s law of $-Z$ (Theorem 12.3). Whether the $\sigma = 0$ maximum law of Theorem 28.2 (a Fréchet-type law) appears in the critical-composition literature was not determined, so it too is claimed only relative to the repository. Neither of the recent phase-transition papers of Banderier, Kuba, Wallner and Wagner [14, 15] perturbs the weight near criticality: both work at fixed weights at, below or above the critical value. The uniform pole–branch expansion, the window $t_n b_n \rightarrow \sigma$ with $t_n > 0$ (including the intermediate range $n^{-1} \ll t_n \ll n^{-2/3}$ at $\sigma = 0$), the tempered laws for $\sigma > 0$, the Gaussian side with its rate and the Gumbel regime were not found in the sources checked; they remain new relative to the repository, under the manuscript’s own bounded-search caveat above. Limits of this check: the text of [13]

was not consulted at the write (only its section title), BFSS was read through the account in [14], and the announced companion of [14] could not be checked.

21.2 Two scales, not one

The natural parameter $t \geq 0$ is defined in Section 23. Its meaning is especially simple: $\log 4 + t$ is the exponential growth rate of the weighted count. Put

$$b_n = \left(\frac{\kappa n}{m} \right)^{2/3}, \quad (127)$$

where $m, \kappa > 0$ are explicit constants. The two transitions are

observable	transition scale	consequence
coefficient correction	$nt = O(1)$	the pole and the nearby branch endpoint require a uniform kernel
block count and large blocks	$tb_n = O(1)$	a tempered-stable law and an exponentially tempered Poisson intensity
Gaussian regime	$tb_n \rightarrow \infty, t \rightarrow 0$	a central limit theorem and a continuously scaled Gumbel maximum

In particular, $n^{-1} \ll t \ll n^{-2/3}$ is a genuine intermediate region: the pole already dominates the coefficients, but the scaled block fluctuations still have the critical stable limit.

22 Exact enumeration and the analytic block law

22.1 The ordered strong-block decomposition

A nondecreasing sequence $s_1, \dots, s_n$ is a score sequence precisely when

$$\sum_{j=1}^k s_j \geq \binom{k}{2} \quad (1 \leq k < n), \quad \sum_{j=1}^n s_j = \binom{n}{2}. \quad (128)$$

The usual bounds $0 \leq s_j \leq n-1$ may be included. A nonempty sequence is strong when all proper inequalities are strict. These characterizations and their role in the decomposition are reviewed by CDFS [1].

Equality positions in (128), including 0 and n , determine the blocks. If consecutive equality positions are $a < b$, subtract a from $s_{a+1}, \dots, s_b$. The resulting sequence has total $\binom{b-a}{2}$ and satisfies the prefix inequalities, strictly at proper prefixes because no equality position was skipped. Its first term is nonnegative by the original inequality at $a+1$; its last term is at most $b-a-1$ by the inequality at $b-1$. Thus it is a strong score sequence. Conversely, concatenate strong sequences while adding the total length of preceding blocks to every entry of the next block. This produces a score sequence whose equality positions are exactly the block boundaries. The decomposition is unique.

Write

$$S(z) = \sum_{n \geq 0} S_n z^n, \quad I(z) = \sum_{n \geq 1} I_n z^n. \quad (129)$$

The decomposition gives

$$S(z) = \frac{1}{1 - I(z)}, \quad \sum_{n \geq 0} W_n(u) z^n = \frac{1}{1 - uI(z)}, \quad (130)$$

where $W_n(u)$ is the sum of u^K over score sequences of length n , and K is the number of strong blocks. For $u > 0$ we use the probability measure assigning a score sequence weight $u^K/W_n(u)$.

22.2 The credited enumeration input

Let N_n denote A145855. The exact identity used here is

$$A(z) = \sum_{n \geq 1} \frac{N_n}{n} z^n, \quad S(z) = \exp A(z), \quad I(z) = 1 - \exp(-A(z)), \quad (131)$$

$$N_n = \frac{1}{2n} \sum_{d|n} (-1)^{n+d} \varphi(n/d) \binom{2d}{d}. \quad (132)$$

The identity $S = \exp A$ in (131), with both S and A ordinary generating functions, is the CDFS enumeration theorem; the divisor evaluation is attributed there and in A145855 to Alekseyev [1, 6]. We do not reprove the CDFS bijection yielding (131). All subsequent asymptotic and probability arguments are supplied below.

For exact computations, (131) immediately gives

$$nS_n = \sum_{j=1}^n N_j S_{n-j}, \quad nI_n = N_n - \sum_{j=1}^{n-1} N_j I_{n-j}. \quad (133)$$

22.3 Separating the singular part

Put $\rho = 1/4$. Separate the $d = n$ summand in (132):

$$A(z) = F(z) + R(z), \quad F(z) = \frac{1}{2} \sum_{n \geq 1} \binom{2n}{n} \frac{z^n}{n^2}. \quad (134)$$

Every proper divisor d of n satisfies $d \leq n/2$. Since $\varphi(n/d) \leq n$ and $\binom{2d}{d} \leq 4^d$, the sum of the absolute values of the proper-divisor contributions to $[z^n]R$ is at most 2^{n-1} . Consequently R is analytic for $|z| < 1/2$. In the scaled variable z/ρ , its radius is at least 2.

Set $y = (1 - \sqrt{1 - 4z})/2$, taking the square root positive on $0 < z < \rho$. Termwise differentiation, followed by matching the value at $z = 0$, proves

$$F(z) = \text{Li}_2(y) - \frac{1}{2} \log^2(1 - y), \quad zF'(z) = -\log \frac{1 + \sqrt{1 - 4z}}{2}. \quad (135)$$

One can check the first identity directly by differentiating and using $z = y(1 - y)$. In particular,

$$\lambda = A(\rho) = \frac{\pi^2}{12} - \log^2 2 + R(\rho), \quad \mu = \rho A'(\rho) = \log 2 + \rho R'(\rho). \quad (136)$$

Both constants are positive. Define

$$\begin{aligned} p &= 1 - e^{-\lambda}, & u_c &= p^{-1}, & Q(z) &= \frac{I(\rho z)}{p} = \sum_{j \geq 1} q_j z^j, & q_j &= \frac{I_j 4^{-j}}{p}, \\ m &= \frac{e^{-\lambda} \mu}{p}, & \kappa &= \frac{2e^{-\lambda}}{3p}, & d &= \frac{\kappa}{\Gamma(-3/2)} = \frac{3\kappa}{4\sqrt{\pi}}. \end{aligned} \quad (137)$$

Thus $q_j \geq 0$, $Q(1) = 1$, and $Q'(1) = m$. Moreover $q_1, q_3 > 0$, so the support has span one.

Lemma 22.1 (Local expansion and tail). *There is a slit neighborhood of 1 in which Q has a convergent expansion in $\sqrt{1-z}$. With $w = 1 - z$,*

$$Q(1 - w) = 1 - mw + \kappa w^{3/2} + \beta w^2 + \kappa_1 w^{5/2} + O(w^3), \quad (138)$$

where

$$a_2 = -\frac{1}{2} \log 2 - \frac{1}{4} + \frac{1}{2} \rho^2 R''(\rho), \quad (139)$$

$$\beta = \frac{e^{-\lambda}}{p} \left(a_2 - \frac{\mu^2}{2} \right), \quad \kappa_1 = \frac{e^{-\lambda}}{p} \left(\frac{8}{15} + \frac{2\mu}{3} \right). \quad (140)$$

Furthermore,

$$q_j = dj^{-5/2}(1 + O(j^{-1})). \quad (141)$$

The tail has a full expansion in integral inverse powers of j to every fixed order.

Proof. Equation (135) gives

$$\frac{d}{dw} F(\rho(1 - w)) = \frac{\log((1 + \sqrt{w})/2)}{1 - w}.$$

The right side has a convergent power series in $\sqrt{w}$ near zero. Integration and addition of the analytic function R give

$$A(\rho(1 - w)) = \lambda - \mu w + \frac{2}{3} w^{3/2} + a_2 w^2 + \frac{8}{15} w^{5/2} + O(w^3).$$

Expanding $1 - \exp(-A)$ proves the asserted constants.

For completeness the tail can be obtained without an unquoted transfer theorem. On the upper side of the cut $z = 1 + v$, $v > 0$, the principal value has $w = -v - i0$. Thus there are real analytic H, B such that

$$Q(1 + v + i0) = 1 + vH(v) + iv^{3/2}B(v), \quad H(v) = m + \beta v + O(v^2), \quad B(v) = \kappa - \kappa_1 v + O(v^2). \quad (142)$$

The lower value is its conjugate. A Cauchy contour extended to a fixed slit disk gives

$$q_j = \frac{1}{\pi} \int_0^\delta (1 + v)^{-j-1} v^{3/2} B(v) dv + O(R_0^{-j})$$

for some $R_0 > 1$. Substitution $v = x/j$ and a fixed-order Taylor expansion under an exponentially dominated integral yield $\kappa \Gamma(5/2) j^{-5/2} / \pi = dj^{-5/2}$ and every subsequent order. The same domination argument is detailed for the more demanding resolvent in Section 24. $\square$

Numerically, with the qualifications in Section 30,

$$\begin{aligned} u_c &= 3.5555932000209863 \dots, & m &= 1.7223486864283872 \dots, \\ \kappa &= 1.7037288000139909 \dots, & d &= 0.7209195316186591 \dots, \\ \beta &= -2.1110749090341058 \dots, & \kappa_1 &= 2.5112154976301175 \dots \end{aligned} \quad (143)$$

23 Exponential tilting and the free-energy parameter

Use the normalized component weight $r = up$ and normalized count

$$Z_n(r) = 4^{-n} W_n(r/p) = [z^n] \frac{1}{1 - rQ(z)}. \quad (144)$$

The critical value is $r = 1$. For $t \geq 0$ define

$$r(t) = \frac{1}{Q(e^{-t})}, \quad h(t) = \log r(t), \quad q_t(j) = r(t)e^{-tj}q_j. \quad (145)$$

Then q_t is a probability mass function on the positive integers. Let

$$m_t = h'(t), \quad v_t = -h''(t) \quad (t > 0), \quad m_0 = m. \quad (146)$$

Differentiating a normalized Laplace transform shows that m_t and v_t are the mean and variance of q_t , respectively. In particular $m_t > 0$, and $v_t > 0$ because the law is not concentrated at one point.

Let $U_t(n)$ be the probability that a renewal process with independent increments of law q_t ever visits n , including $U_t(0) = 1$. Positivity of the increments makes the events of visiting n at different renewal numbers disjoint. Therefore

$$U_t(n) = \sum_{k \geq 0} q_t^{*k}(n), \quad 0 \leq U_t(n) \leq 1. \quad (147)$$

Proposition 23.1 (Exact change of measure). *For every $n \geq 0$ and $t \geq 0$,*

$$Z_n(r(t)) = e^{nt} U_t(n). \quad (148)$$

Moreover, under the score-sequence measure with weight $u = r(t)/p$, the ordered list of block lengths has exactly the law of the q_t -renewal increments conditioned to visit n .

Proof. For a composition $j_1 + \cdots + j_k = n$,

$$\prod_{\ell=1}^k q_t(j_\ell) = e^{-nt} 4^{-n} (r(t)/p)^k \prod_{\ell=1}^k I_{j_\ell}.$$

Summation proves the coefficient identity; division by its total proves the conditional law. The conditioning permits every possible number of increments. It is not conditioning a fixed number of independent blocks. $\square$

Lemma 23.2 (Small- t free energy). *As $t \downarrow 0$,*

$$h(t) = mt - \kappa t^{3/2} + h_2 t^2 + O(t^{5/2}), \quad (149)$$

$$h_2 = \frac{m^2 - m}{2} - \beta,$$

$$m_t = m - \frac{3}{2}\kappa\sqrt{t} + 2h_2 t + O(t^{3/2}), \quad (150)$$

$$v_t = \frac{3}{4}\kappa t^{-1/2} - 2h_2 + O(\sqrt{t}). \quad (151)$$

All these expansions arise from convergent power series in $\sqrt{t}$ on a slit neighborhood; differentiated remainders there have the corresponding orders.

Proof. Substitute $w = 1 - e^{-t} = t - t^2/2 + O(t^3)$ into (138), and expand $-\log Q(e^{-t})$. The coefficient of t^2 is $(m^2 - m)/2 - \beta$. Analyticity in $\sqrt{t}$ justifies termwise differentiation. $\square$

24 A uniform pole–branch expansion to every fixed order

For $r = r(t)$ set

$$A_r = 1 - r^{-1} = 1 - Q(e^{-t}), \quad a = a_n(t) = \frac{nA_r}{m}. \quad (152)$$

For a sufficiently small fixed $t_0 > 0$, $a \geq 0$ and $a \asymp nt$ uniformly on $0 \leq t \leq t_0$.

Define the leading kernel

$$\mathcal{K}_0(a) = \int_0^\infty \frac{e^{-x} x^{3/2}}{(a+x)^2} dx, \quad a \geq 0, \quad (153)$$

and the first correction

$$\begin{aligned} \mathcal{K}_1(a) = \int_0^\infty \frac{e^{-x} x^{3/2}}{(a+x)^2} & \left(\frac{x^2}{2} - x - \frac{\kappa_1}{\kappa} x - \frac{2\beta}{m} \frac{x^2}{a+x} \right. \\ & \left. - \frac{\kappa^2}{m^2} \frac{x^3}{(a+x)^2} \right) dx. \end{aligned} \quad (154)$$

They are finite at $a = 0$; in particular $\mathcal{K}_0(0) = \sqrt{\pi}$.

Theorem 24.1 (Uniform supercritical expansion). *There are $t_0 > 0$ and $R_0 > 1$ such that, for every fixed integer $J \geq 1$, uniformly for $0 \leq t \leq t_0$,*

$$\begin{aligned} Z_n(r(t)) = \frac{e^{nt}}{m_t} + \frac{\kappa}{\pi r(t) m^2} n^{-1/2} & \left\{ \sum_{j=0}^{J-1} \frac{\mathcal{K}_j(a_n(t))}{n^j} + O_J \left(\frac{n^{-J}}{(1+a_n(t))^2} \right) \right\} \\ & + O_J(R_0^{-n}). \end{aligned} \quad (155)$$

Every kernel is constructively specified by (159) below, and

$$|\mathcal{K}_j(a)| \leq C_j (1+a)^{-2} \quad (a \geq 0). \quad (156)$$

The constants t_0, R_0 can be chosen independently of J ; remainder constants may depend on J . No assertion is made for J growing with n .

24.1 The exact branch integral

We first justify a contour representation uniformly in the moving pole.

Lemma 24.2 (Local pole and contour separation). *For all real $r \geq 1$ sufficiently close to 1, the only pole of $(1 - rQ(z))^{-1}$ in a suitable fixed slit disk $|z| < R_0$, cut along $[1, R_0)$, is the simple pole at $z = e^{-t}$, with $r = r(t)$; at $r = 1$ it merges with the endpoint. For $r > 1$,*

$$Z_n(r) = \frac{e^{nt}}{m_t} + \frac{1}{\pi r} \int_0^{R_0-1} \frac{(1+v)^{-n-1} v^{3/2} B(v)}{(A_r + vH(v))^2 + v^3 B(v)^2} dv + O(R_0^{-n}). \quad (157)$$

After shrinking $R_0 - 1$ if necessary, $H(v) \geq m/2$ and $B(v) \geq \kappa/2$ on the integration interval. Formula (157) remains valid at $r = 1$ by a limit.

Proof. For $|z| \leq 1$, positivity gives $|Q(z)| \leq Q(|z|) \leq 1$. Equality $Q(z) = 1$ forces $|z| = 1$ and $z^j = 1$ at every supported j ; since $q_1 > 0$, this forces $z = 1$. Compactness away from 1, followed by

continuity into a slightly larger slit disk, separates all other points from zeros of $1 - rQ$ when r is close to 1.

Near 1, put $s = \sqrt{1 - z}$. The map

$$T(s) = s \sqrt{\frac{1 - Q(1 - s^2)}{s^2}} = \sqrt{m} s (1 + O(s))$$

is analytic with nonzero derivative at zero. The equation $1 - Q(z) = A_r$ is $T(s)^2 = A_r$. The inverse-function theorem gives two local solutions in the s -plane. For real $A_r > 0$, exactly the one near $+\sqrt{A_r/m}$ is in the principal-square-root domain. It is the real solution $z = e^{-t}$. By taking the neighborhood smaller, there are no other local solutions. Together with the preceding compactness argument, this proves the separation assertion and the uniform outer-contour bound.

The residue contribution to coefficient extraction at $z_0 = e^{-t}$ is

$$\frac{z_0^{-n}}{r z_0 Q'(z_0)} = \frac{e^{nt}}{m_t}.$$

On the upper side of the cut, (142) gives

$$\Im \frac{1}{1 - rQ(1 + v + i0)} = \frac{v^{3/2} B(v)}{r \{(A_r + vH(v))^2 + v^3 B(v)^2\}}.$$

A Cauchy contour pushed to the two sides of the slit and the outer circle yields (157). The outer-circle contribution is $O(R_0^{-n})$ uniformly, because its denominator is bounded away from zero.

At $A_r = 0$, the branch-integral integrand is bounded near zero by a constant times $v^{-1/2}$. The same bound holds for all $A_r \geq 0$ sufficiently small. Dominated convergence, continuity of each fixed coefficient in r , and $m_t \rightarrow m$ extend the identity to the endpoint. $\square$

24.2 The correction generator

Introduce a formal variable ϵ and the power series

$$E(\epsilon, x) = \exp \left\{ x - (\epsilon^{-1} + 1) \log(1 + \epsilon x) \right\}. \quad (158)$$

The apparent singularity at $\epsilon = 0$ is removable, and

$$E(\epsilon, x) = 1 + \epsilon(x^2/2 - x) + \epsilon^2(x^2 - 5x^3/6 + x^4/8) + \dots$$

For H, B in (142), define

$$\mathcal{K}_j(a) = \int_0^\infty e^{-x} x^{3/2} [\epsilon^j] \frac{\{B(\epsilon x)/\kappa\} E(\epsilon, x)}{[a + xH(\epsilon x)/m]^2 + \epsilon x^3 B(\epsilon x)^2/m^2} dx. \quad (159)$$

Here coefficient extraction is performed before integration. For every fixed j it uses only finitely many coefficients of H and B . It produces a finite rational combination of integrals of the form

$$\int_0^\infty \frac{e^{-x} x^{b+1/2}}{(a+x)^c} dx, \quad (160)$$

with integrable behavior at zero for the combinations actually produced. They can also be evaluated through incomplete gamma functions, but the positive-axis integrals are more convenient for uniform estimates. Direct coefficient extraction gives (153) and (154).

24.3 Proof of the uniform remainder

Proof of Theorem 24.1. In (157), substitute $v = x/n$ and $A_r = ma/n$. The branch contribution becomes

$$\frac{\kappa}{\pi r m^2} n^{-1/2} \int_0^{n(R_0-1)} e^{-x} x^{3/2} \frac{\{B(x/n)/\kappa\} E(1/n, x)}{[a + xH(x/n)/m]^2 + n^{-1}x^3 B(x/n)^2/m^2} dx.$$

We give details of why expansion under this integral is uniform even when $a = 0$ or a grows proportionally to n .

First, for real $a, x \geq 0$ and $0 \leq x/n \leq R_0 - 1$, the exact denominator is comparable to $(a + x)^2$. Indeed H stays between two positive constants, and the extra term is nonnegative. Relative to $(a + x)^2$, every perturbation produced by $H(x/n) - m$ is $O(x/n)$, and

$$\frac{x^3/n}{(a + x)^2} \leq \frac{x}{n}.$$

Choose the fixed slit length small enough that these relative perturbations have modulus less than a fixed number below one. Taylor expansion of the rational denominator to order $J - 1$ then has a remainder bounded by

$$C_J n^{-J} (a + x)^{-2} (1 + x)^{c_J}$$

for some finite integer c_J , with the harmless polynomial allowing all numerator derivatives.

For E , its logarithm is an analytic series in ϵx with leading term $\epsilon(x^2/2 - x)$. Taylor's formula, or Cauchy estimates on a smaller ϵx -disk, bounds its order- J remainder by $C_J n^{-J} (1 + x)^{c'_J} e^{c_J x}$, with $c < 1$ after reducing the slit length. The same bound applies to the finite product with the analytic numerator and denominator expansions. Multiplication by e^{-x} leaves exponential domination.

Finally,

$$\frac{1}{(a + x)^2} \leq \frac{C}{(1 + a)^2} \max(1, x^{-2}) \quad (a, x > 0).$$

Each expansion term is bounded near $x = 0$ by a polynomial times $x^{-1/2}$: a relative denominator correction contributes at least one additional power of x , not a negative one. Thus the integrated remainder is at most $C_J n^{-J} (1 + a)^{-2}$. The same estimate proves (156). Extending each of the finitely many coefficient integrals from $n(R_0 - 1)$ to infinity changes it by an exponentially small quantity, which is also dominated by the stated algebraic remainder, uniformly for $a \leq Cn$. Combining these bounds with the exact outer-contour error proves (155). $\square$

Corollary 24.3 (A uniform renewal estimate). *Uniformly for $0 \leq t \leq t_0$,*

$$U_t(n) = \frac{1}{m_t} + O\left(e^{-nt} n^{-1/2} (1 + nt)^{-2}\right) + O\left(e^{-nt} R_0^{-n}\right). \quad (161)$$

In particular $U_t(n) = m_t^{-1} + O(n^{-1/2})$ uniformly, and $U_t(n)$ has a positive lower bound for all sufficiently large n , uniformly in t .

Proof. Divide Theorem 24.1 with $J = 1$ by e^{nt} , use (156), and use $a \asymp nt$ and the compact positive range of m_t . $\square$

25 Matching the regimes and interpreting the transseries

25.1 A closed form for the coalescence kernel

For $a > 0$, the leading kernel also has the closed form

$$\mathcal{K}_0(a) = \sqrt{\pi}(1+a) - \pi\sqrt{a} \left(a + \frac{3}{2} \right) e^a \operatorname{erfc}(\sqrt{a}). \quad (162)$$

To prove it, write $I(a) = \int_0^\infty e^{-x} x^{-1/2} (x+a)^{-1} dx$. A Laplace representation of $(x+a)^{-1}$ gives $I(a) = \pi a^{-1/2} e^a \operatorname{erfc}(\sqrt{a})$. The identity $x^2/(x+a)^2 = 1 - 2a/(x+a) + a^2/(x+a)^2$ yields $\mathcal{K}_0(a) = \sqrt{\pi} - 2aI(a) - a^2I'(a)$, hence (162). For large a , the integral form is numerically safer because the two terms in (162) nearly cancel.

A useful endpoint check is

$$\mathcal{K}_0(a) = \sqrt{\pi} - \frac{3\pi}{2}\sqrt{a} + 4\sqrt{\pi}a + O(a^{3/2}). \quad (163)$$

The $\sqrt{a}$ term in the branch correction cancels the leading $\sqrt{t}$ term of $1/m_t$ as $a \sim nt \downarrow 0$. This is required by the analyticity of each fixed coefficient in the weight: a moving-pole approximation by itself would produce a spurious square-root singularity.

25.2 The critical endpoint

At $t = 0$, $r = 1$ and $a = 0$. The first terms of (155) are

$$Z_n(1) = \frac{1}{m} + \frac{\kappa}{m^2\sqrt{\pi}} n^{-1/2} + \frac{\kappa\mathcal{K}_1(0)}{\pi m^2} n^{-3/2} + O(n^{-5/2}). \quad (164)$$

In particular there is no n^{-1} term. More generally, apart from the constant pole term, only the powers $n^{-j-1/2}$ occur. This critical behavior is a consistency check against the preceding repository report, not a newly discovered critical law.

The first endpoint correction can be read explicitly from (154):

$$\mathcal{K}_1(0) = \frac{1}{2}\Gamma(5/2) - \left(1 + \frac{\kappa_1}{\kappa} + \frac{2\beta}{m} + \frac{\kappa^2}{m^2} \right) \Gamma(3/2). \quad (165)$$

For a moving $t = s/n$ with s in a compact subset of $[0, \infty)$, it is safest to retain the exact arguments $m_t, r(t), a_n(t)$ in (155); replacing them by limiting values introduces additional half-power terms. In particular $a_n(s/n) = s + O(n^{-1/2})$, but that replacement alone is not a valid all-orders simplification.

25.3 The large-argument kernel and fixed supercritical weight

Proposition 25.1 (Large- a expansion). *For every fixed integer $L \geq 1$,*

$$\mathcal{K}_0(a) = \frac{\Gamma(5/2)}{a^2} \left\{ \sum_{\ell=0}^{L-1} \frac{(-1)^\ell (\ell+1)(5/2)_\ell}{a^\ell} + O_L(a^{-L}) \right\} \quad (a \rightarrow \infty). \quad (166)$$

The opening coefficients in braces are 1, -5 , $105/4$, $-315/2$.

Proof. Expand $(1 + x/a)^{-2}$ to order $L - 1$. Taylor's integral remainder on $x/a \geq 0$ is bounded in absolute value by $(L + 1)(x/a)^L$, because the absolute value of its L th derivative is at most $(L + 1)!$. Integration against $e^{-x}x^{3/2}$ gives the result. $\square$

For each fixed $t \in (0, t_0]$, Theorem 24.1 and Proposition 25.1 give

$$Z_n(r) = \frac{e^{nt}}{m_t} + \frac{rd}{(r-1)^2} n^{-5/2} (1 + O_t(n^{-1})). \quad (167)$$

The factor $r/(r-1)^2$ follows from $a = n(r-1)/(rm)$ and $d = \kappa\Gamma(5/2)/\pi$. It is also what direct expansion of $(1 - rQ)^{-1}$ about the branch point gives; the sign is positive. At any fixed supercritical weight the algebraic branch contribution is exponentially small *relative* to the pole, although not small on the absolute scale of Z_n .

Equation (155) is a two-scale asymptotic expansion, not just a leading equivalent. For fixed $t > 0$, its transmonomials include e^{nt} and $n^{-5/2-j}$; relative to the leading term they include $e^{-nt}n^{-5/2-j}$. Near $t = 0$ the kernels keep these contributions meaningful through coalescence. No convergence of the inverse- n expansion is asserted. The convergent local expansion in $\sqrt{t}$, discussed next, is a different statement.

26 Inverse growth, inverse density, and extreme-value centering

26.1 Convergent Puiseux inversion of the free energy

Theorem 26.1 (Weight-to-growth inversion). *For small $h = \log r > 0$, put $v = h/m$. The solution of $h = h(t)$ satisfies*

$$t = v + \frac{\kappa}{m}v^{3/2} + \left(\frac{3\kappa^2}{2m^2} - \frac{h_2}{m} \right) v^2 + O(v^{5/2}). \quad (168)$$

It has a convergent Puiseux expansion in $\sqrt{v}$ near zero, with every coefficient determined recursively by substitution into $h(t)$.

Proof. Write $t = s^2$. By Lemma 23.2,

$$\sqrt{h(s^2)/m} = s - \frac{\kappa}{2m}s^2 + O(s^3)$$

is an analytic function of s with derivative one at zero. Its local analytic inverse gives s as a convergent power series in $\sqrt{v}$; squaring gives the convergent Puiseux expansion for t . To calculate the displayed terms, substitute $t = v + cv^{3/2} + ev^2 + O(v^{5/2})$ into (149). The coefficients of $v^{3/2}$ and v^2 give $c = \kappa/m$ and $e = 3\kappa^2/(2m^2) - h_2/m$. $\square$

Thus a small supercritical weight increment can be translated into its actual growth rate without repeatedly solving a nearly singular coefficient problem. The branch $t > 0$ is specified; neither a negative real branch nor an arbitrary branch of a complex inverse is implicit in the statement.

26.2 A quadratic inverse density law

For fixed $t \in (0, t_0]$, differentiation of the simple-pole asymptotic in a complex neighborhood of the weight gives the thermodynamic block density

$$\theta(t) = \lim_{n \rightarrow \infty} \frac{\mathbb{E}K_n}{n} = \frac{1}{m_t}. \quad (169)$$

The differentiation is justified by the isolated simple pole and exponentially smaller remainder at fixed t ; it does not assume differentiation of an uncontrolled real asymptotic. Lemma 23.2 gives

$$\theta(t) = \frac{1}{m} + \frac{3\kappa}{2m^2}\sqrt{t} + O(t). \quad (170)$$

Let $\Delta = m\theta - 1$ measure the excess density in dimensionless units. Since $\Delta = (3\kappa/(2m))\sqrt{t} + O(t)$ and the leading coefficient is positive, local inversion yields

$$h = \log(up) = \frac{4m^3}{9\kappa^2}\Delta^2 + O(\Delta^3). \quad (171)$$

This is an inverse response at the critical boundary, not a finite- n confidence interval for estimating u . All higher coefficients again follow from an analytic inverse in $\sqrt{t}$. The corresponding susceptibility diverges:

$$\frac{d\theta}{dh} \sim \frac{3\kappa}{4m^3} t^{-1/2}. \quad (172)$$

26.3 A fixed-weight size inverse

At fixed $t \in (0, t_0]$, let $\ell = \log 4 + t$. Equation (167) gives

$$\log W_n(r/p) = \ell n - \log m_t + O_t(e^{-nt}n^{-5/2}). \quad (173)$$

For a large target y , the leading continuous model inverse is

$$x_0(y) = \frac{\log y + \log m_t}{\log 4 + t}. \quad (174)$$

Here $W_n(r/p)$ is a weighted count and need not be an integer. It is nevertheless strictly increasing in n : adjoining a final singleton strong block gives an injection whose weight is multiplied by $u = r/p > 1$.

A rigorous integer-threshold interpretation must retain a separation condition. If the distance of $x_0(y)$ from the nearest integer exceeds a sufficiently large constant times $e^{-tx_0(y)}x_0(y)^{-5/2}$, then the first integer n with $W_n(r/p) \geq y$ is $\lceil x_0(y) \rceil$. This follows by applying (173) at the two neighboring integers; monotonicity handles all other integers. Without separation from an integer, the leading model does not certify rounding. This is the same model-versus-staircase distinction emphasized in the repository's transseries-inversion program.

[write] **Added 5 October 2026, batch 98.** The “transseries-inversion program” is the repository’s transseries volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; the manuscript names no label, and no novelty is claimed for these general steps. The separation condition just stated is part (2) of `p0:thm:staircase` there, applied to the model inverse $x_0(y)$ with the error η a constant times $e^{-tx_0(y)}x_0(y)^{-5/2}$; its exact-rounding part (1)

is not used. The Lambert core (176) below is `p0:thm:lambert-core` for $w + \frac{5}{2} \log w = L$ with $L = \log B$, that is $a = 1$, $b = 5/2 > 0$, whose branch rule gives the unique positive solution on the branch W_0 at $z = (a/b)e^{L/b} = \frac{2}{5}B^{2/5}$ (Part I's inverse (51) and Part II's centre (121) are the same theorem with $b = -5/2$ and $b = -2$ on the branch W_{-1}). The convergent inversion of Theorem 26.1 is the analytic inverse function theorem in $\sqrt{t}$, as its proof says. The Lean lemmas named in the batch-77 note of Section 7 concern an arbitrary monotone function and give Part III no formal status.

26.4 The centering inverse needed for maxima

For a large positive B consider

$$Be^{-w}w^{-5/2} = 1. \quad (175)$$

Its unique positive solution is

$$w = \frac{5}{2}W_0\left(\frac{2}{5}B^{2/5}\right), \quad (176)$$

where W_0 is the positive real branch of Lambert's function. This is an exact inverse of the specified model, not a new inversion method. With $L = \log B$,

$$w = L - \frac{5}{2} \log L + \frac{25}{4} \frac{\log L}{L} + O\left(\frac{(\log L)^2}{L^2}\right). \quad (177)$$

Substituting into $w + (5/2) \log w = L$ proves the displayed expansion and supplies the recursion for further logarithmic terms.

A more accurate continuous tail model replaces (175) by

$$B \int_{\hat{w}}^{\infty} e^{-y} y^{-5/2} dy = 1. \quad (178)$$

Its unique positive solution can be computed from an upper incomplete gamma function. Repeated integration by parts gives, for every fixed order,

$$\int_w^{\infty} e^{-y} y^{-5/2} dy \sim e^{-w} w^{-5/2} \left(1 - \frac{5}{2w} + \frac{35}{4w^2} - \frac{315}{8w^3} + \cdots\right). \quad (179)$$

The remainder after any fixed number of integrations is bounded by the next alternating-sign integral term. Inverting this expansion about the exact Lambert core gives

$$\hat{w} = w - \frac{5}{2w} + \frac{95}{8w^2} + O(w^{-3}). \quad (180)$$

The incomplete-gamma center and the core center differ by $o(1)$ and hence give the same limiting Gumbel distribution. Their finite-size accuracy can be quite different. Neither continuous center removes the lattice effect at a fixed positive t .

27 The block-count crossover

We next prove the probability limits, rather than inferring them from the coefficient scale. Throughout, K_n is sampled with weight $u_n = r(t_n)/p$. Exact centering below uses the tilted mean m_{t_n} , not the unknown finite- n mean of K_n .

27.1 A complex continuation lemma

Characteristic functions require a small complex perturbation of the weight. The needed extension is local and can be proved by the same contour.

Lemma 27.1 (Complex weights near the critical ray). *Fix $\varepsilon > 0$. Let r be complex and close to 1, and suppose $A_r = 1 - r^{-1}$ lies in $|\arg A_r| \leq \pi - \varepsilon$. Let τ be the local solution of $h(\tau) = \log r$ obtained by the principal continuation from positive τ . In a sufficiently small neighborhood, uniformly in this sector,*

$$[z^n] \frac{1}{1 - rQ(z)} = \frac{e^{n\tau}}{h'(\tau)} + O_\varepsilon \left(n^{-1/2} (1 + |nA_r|)^{-2} \right) + O_\varepsilon(R_0^{-n}). \quad (181)$$

The endpoint $r = 1$ is included by a limit. The pole need not lie strictly inside the unit circle after a complex perturbation.

Proof. The inverse map T in Lemma 24.2 gives the unique pole on the principal sheet also for A_r in the indicated sector: the root corresponding to $+\sqrt{A_r}$ stays in the principal-square-root domain, while the other local root does not. The outer-contour separation persists by compactness.

The upper and lower boundary values of Q on the cut still have the form $1 + vH(v) \pm iv^{3/2}B(v)$. Subtracting the two resolvents, rather than taking the imaginary part of a complex- r resolvent, gives exactly the same rational jump as in (157). For $a = nA_r/m$ in a sector away from the negative real axis,

$$|a + x| \geq c_\varepsilon(|a| + x) \quad (x \geq 0).$$

After making the slit smaller, its analytic perturbations preserve this lower bound for the product of the two denominator factors. The leading-kernel estimate in the proof of Theorem 24.1 therefore applies with $|a|$ in place of a . The residue is $e^{n\tau}/h'(\tau)$, which proves (181). The extension does not require $\Re\tau > 0$: small characteristic-function perturbations of the critical endpoint may have $\Re\tau < 0$ of order $1/n$, and the fixed outer contour still surrounds their poles. $\square$

27.2 The tempered-stable window

Theorem 27.2 (Tempered 3/2-stable block count). *Let b_n be (127), and suppose $t_n \geq 0$ with $t_n b_n \rightarrow \sigma \in [0, \infty)$. Then*

$$X_n = \frac{m_{t_n}}{b_n} \left(K_n - \frac{n}{m_{t_n}} \right) \xrightarrow{d} X_\sigma, \quad (182)$$

where

$$\mathbb{E} e^{i\xi X_\sigma} = \exp \left\{ (\sigma + i\xi)^{3/2} - \sigma^{3/2} - \frac{3}{2} i\xi \sqrt{\sigma} \right\}. \quad (183)$$

Powers use the principal branch. More precisely, for every fixed $S, T < \infty$, uniformly for $0 \leq t_n b_n \leq S$ and $|\xi| \leq T$,

$$\left| \mathbb{E} e^{i\xi X_n} - \exp \left\{ (\sigma_n + i\xi)^{3/2} - \sigma_n^{3/2} - \frac{3}{2} i\xi \sqrt{\sigma_n} \right\} \right| = O_{S,T}(n^{-1/3}), \quad \sigma_n = t_n b_n. \quad (184)$$

This is a characteristic-function estimate on compact frequency sets, not a Kolmogorov-distance or total-variation bound.

Proof. First center and scale using m , defining $Y_n = (mK_n - n)/b_n$. The exact probability generating function is

$$\mathbb{E}e^{sK_n} = \frac{Z_n(r(t_n)e^s)}{Z_n(r(t_n))}. \quad (185)$$

For bounded real ξ , put $s = i\xi m/b_n$, and let t' solve

$$h(t') = h(t_n) + i\xi m/b_n.$$

The local inverse and (149), uniformly for $\sigma_n \in [0, S]$, give

$$t' = t_n + \frac{i\xi}{b_n} + \frac{\kappa}{m} \left\{ \left(t_n + \frac{i\xi}{b_n} \right)^{3/2} - t_n^{3/2} \right\} + O_{S,T}(b_n^{-2}). \quad (186)$$

One way to verify the uniform remainder is to use the convergent inverse from Theorem 26.1: the omitted analytic-in-square-root terms are $O(|h|^2)$ in a fixed closed sector. Substitution of $h(t_n)$ then gives (186); the $3/2$ -power has bounded derivative on the resulting compact scaled sector. This remains valid as both σ_n and ξ approach zero.

The perturbed A_r lies in a sector bounded away from the negative real axis. Lemma 27.1 applies, and

$$\frac{Z_n(r(t_n)e^{i\xi m/b_n})}{Z_n(r(t_n))} = \frac{m_{t_n}}{h'(t')} \exp\{n(t' - t_n)\} (1 + O_{S,T}(n^{-1/2})).$$

To justify the relative error, note that $\Re(n(t' - t_n))$ stays bounded on these compact scaled parameter sets after the purely imaginary linear term is removed. Thus the numerator pole cannot be exponentially smaller than its real-weight normalization. The residue ratio is $1 + O_{S,T}(b_n^{-1/2})$.

Since

$$\frac{n\kappa}{mb_n^{3/2}} = 1, \quad nb_n^{-2} = O(n^{-1/3}), \quad b_n^{-1/2} = O(n^{-1/3}),$$

we obtain

$$\mathbb{E}e^{i\xi Y_n} = \exp\{(\sigma_n + i\xi)^{3/2} - \sigma_n^{3/2}\} + O_{S,T}(n^{-1/3}). \quad (187)$$

Finally,

$$X_n = \frac{m_{t_n}}{m} Y_n + \left(\frac{m_{t_n}}{m} - 1 \right) \frac{n}{b_n}, \quad \left(\frac{m_{t_n}}{m} - 1 \right) \frac{n}{b_n} = -\frac{3}{2} \sqrt{\sigma_n} + O_S(n^{-1/3}).$$

The scale factor differs from 1 by $O_S(n^{-1/3})$. Using (187) at the correspondingly perturbed frequency proves (184). The limiting characteristic function is continuous at zero and is a probability characteristic function by the representation below, so convergence in distribution follows. $\square$

The exponent in (183) has the Lévy representation

$$\log \mathbb{E}e^{i\xi X_\sigma} = \frac{1}{\Gamma(-3/2)} \int_0^\infty (e^{-i\xi x} - 1 + i\xi x) e^{-\sigma x} x^{-5/2} dx. \quad (188)$$

The integral is absolutely convergent at both endpoints, also when $\sigma = 0$. For $\sigma > 0$, differentiate twice in ξ , evaluate the ordinary gamma integral, and integrate back using the zero value and zero first derivative at $\xi = 0$; dominated convergence then gives the $\sigma = 0$ identity. This proves that X_σ is centered and spectrally negative, with an exponentially tempered $3/2$ -stable Lévy measure. At $\sigma = 0$ it is the stable law; for $\sigma > 0$ its variance is $3/(4\sqrt{\sigma})$. These properties describe the limiting law and do not assert convergence of all finite- n moments.

27.3 The Gaussian side of the second transition

Theorem 27.3 (Gaussian block count). *Suppose $t_n > 0$, $t_n \rightarrow 0$, and $nt_n^{3/2} \rightarrow \infty$, equivalently $t_n b_n \rightarrow \infty$. Then*

$$\frac{K_n - n/m_{t_n}}{\sqrt{nv_{t_n}/m_{t_n}^3}} \xrightarrow{d} N(0, 1). \quad (189)$$

On every compact set of real frequencies, the difference of characteristic functions from $e^{-\xi^2/2}$ is $O((nt_n^{3/2})^{-1/2})$.

Proof. Let g be the local inverse of h about a positive $t = t_n$. Its derivatives satisfy

$$g'(h(t)) = \frac{1}{m_t}, \quad g''(h(t)) = \frac{v_t}{m_t^3}, \quad g'''(h(t)) = O(t^{-3/2}).$$

The last estimate follows by differentiating the convergent local expansion in Lemma 23.2; it is uniform on a complex disk of radius a fixed multiple of t about t .

Put $s_n = \sqrt{nv_{t_n}/m_{t_n}^3}$ and perturb the weight by $\exp(i\xi/s_n)$. The induced displacement in t divided by t is $O((nt^{3/2})^{-1/2})$, so it stays in that analytic disk. Taylor's formula gives

$$n\{g(h(t) + i\xi/s_n) - t\} = \frac{i\xi n}{m_t s_n} - \frac{\xi^2}{2} + O_T((nt^{3/2})^{-1/2}).$$

The logarithm of the residue-amplitude ratio is $O_T(n^{-1/2}t^{-1/4})$, smaller than the displayed error when t is small. The complex pole estimate and (161) show that branch errors are exponentially suppressed, since $nt \rightarrow \infty$. Substitution into (185), followed by the stated centering, proves the characteristic-function estimate and the limit. $\square$

The finite- σ distributions in (183) are non-Gaussian. Thus replacing $t_n b_n = O(1)$ by a Gaussian approximation would miss an entire family of limiting laws. Conversely, the divergence of v_t as $t \downarrow 0$ explains why the ordinary fixed-weight central limit scale is not uniform at criticality.

28 Large blocks: a tempered Poisson point process

Let $L_{1,n}, \dots, L_{K_n,n}$ denote the block lengths in their original order, and let $M_n = \max_i L_{i,n}$. The order is not retained in the point processes below. We begin with an exact identity that makes the renewal conditioning explicit.

Lemma 28.1 (Ordered marked blocks). *For a fixed positive integer ℓ and bounded functions $f_1, \dots, f_\ell$ on the positive integers,*

$$\begin{aligned} \mathbb{E} \sum_{i_1, \dots, i_\ell \text{ distinct}} \prod_{s=1}^{\ell} f_s(L_{i_s, n}) \\ = \frac{\ell!}{U_t(n)} \sum_{j_1 + \dots + j_\ell \leq n} \left\{ \prod_{s=1}^{\ell} q_t(j_s) f_s(j_s) \right\} V_{\ell+1, t}(n - j_1 - \dots - j_\ell), \end{aligned} \quad (190)$$

where

$$V_{k, t}(N) = [z^N](1 - Q_t(z))^{-k}, \quad Q_t(z) = \sum_{j \geq 1} q_t(j) z^j. \quad (191)$$

For each fixed $\ell \geq 0$, uniformly on $0 \leq t \leq t_0$,

$$V_{\ell+1,t}(N) = \frac{N^\ell}{\ell! m_t^{\ell+1}} + O_\ell((N+1)^{\ell-1/2}) \quad (N \rightarrow \infty). \quad (192)$$

Also $V_{\ell+1,t}(N) \leq \binom{N+\ell}{\ell}$.

Proof. Distinguishing ℓ labeled blocks separates an ordered renewal string into $\ell + 1$ unmarked strings. There are $\ell!$ possible orders of the labels, giving the numerator in (190); division by the visit probability $U_t(n)$ imposes conditioning. Summing over labeled lengths makes all label orders identical after relabeling, which explains the factor even when the f_s are different.

For the asymptotic, write $U_t(j) = m_t^{-1} + E_t(j)$ with $|E_t(j)| \leq C(j+1)^{-1/2}$, by Corollary 24.3, enlarging C to cover $j = 0$. Since $V_{\ell+1,t}$ is the convolution of $\ell + 1$ copies of U_t , the all-constant product contributes $m_t^{-\ell-1} \binom{N+\ell}{\ell}$. For $\ell \geq 1$, a telescoping product estimate bounds the difference by a constant times

$$\sum_{j=0}^N (j+1)^{-1/2} (N-j+1)^{\ell-1} = O_\ell((N+1)^{\ell-1/2}).$$

All other factors are uniformly bounded. The difference between the binomial coefficient and $N^\ell/\ell!$ is $O_\ell((N+1)^{\ell-1})$. For $\ell = 0$ the result is the renewal estimate itself. Finally $U_t(j) \leq 1$ in (147) bounds its convolution by the number of weak compositions, proving the last assertion. $\square$

Theorem 28.2 (Tempered large-block process). *Suppose $t_n b_n \rightarrow \sigma \in [0, \infty)$. In the vague topology on point measures on $(0, \infty)$,*

$$\sum_{i=1}^{K_n} \delta_{L_{i,n}/b_n} \xrightarrow{d} \text{PPP}(\nu_\sigma), \quad \nu_\sigma(dx) = \frac{1}{\Gamma(-3/2)} e^{-\sigma x} x^{-5/2} dx. \quad (193)$$

Consequently, for every $x > 0$,

$$\mathbb{P}(M_n/b_n \leq x) \longrightarrow \exp \left\{ -\frac{1}{\Gamma(-3/2)} \int_x^\infty e^{-\sigma y} y^{-5/2} dy \right\}. \quad (194)$$

At $\sigma = 0$ this becomes $\exp\{-1/(2\sqrt{\pi}x^{3/2})\}$.

There is also a finite-order refinement. Fix ℓ , a compact interval $[a, b] \subset (0, \infty)$, and Lipschitz functions f_s supported there. Uniformly for $0 \leq \sigma_n = t_n b_n \leq S$, the mixed factorial moment of the scaled process equals

$$\prod_{s=1}^{\ell} \int f_s d\nu_{\sigma_n} + O(n^{-1/3}). \quad (195)$$

The implied constant may depend on the functions, their support, S , and ℓ .

Proof. Uniformly for j/b_n in a compact subset of $(0, \infty)$, the tail (141) and $q_t(j) = r(t)e^{-tj}q_j$ imply the vague convergence of finite measures

$$\frac{n}{m_{t_n}} \sum_{j \geq 1} q_{t_n}(j) \delta_{j/b_n} \longrightarrow \nu_\sigma. \quad (196)$$

Indeed the Riemann-sum prefactor is

$$\frac{nr(t_n)d}{m_{t_n}b_n^{3/2}} = \frac{mr(t_n)d}{m_{t_n}\kappa} \longrightarrow \frac{1}{\Gamma(-3/2)}.$$

If $t_nb_n = \sigma_n$ stays bounded, this prefactor differs from its limit by $O(n^{-1/3})$, since $m_t - m = O(\sqrt{t})$ and $r(t) - 1 = O(t)$. The $O(j^{-1})$ tail correction and the Riemann-sum error for Lipschitz test functions are $O(b_n^{-1})$. This proves the corresponding measure approximation with error $O(n^{-1/3})$ on a fixed compact support.

For $j_s/b_n \in [a, b]$, the sum $J = j_1 + \dots + j_\ell$ is $O(b_n) = o(n)$. Lemma 28.1 and the uniform renewal estimate give

$$\frac{\ell!V_{\ell+1,t_n}(n-J)}{U_{t_n}(n)} = \left(\frac{n}{m_{t_n}}\right)^\ell (1 + O_\ell(b_n/n + n^{-1/2})).$$

Inserting this into (190) and applying the preceding Riemann-sum estimate proves (195). The limit is the product of the intensity integrals, precisely the mixed factorial moments of a Poisson point process.

Here is a tightness and uniqueness justification for that moment argument. On any compact set away from zero, first moments are bounded. For each fixed order the bound $V_{\ell+1,t}(N) \leq \binom{N+\ell}{\ell}$, the positive lower bound on $U_t(n)$, and boundedness of $n \sum q_t(j)$ over the compact range bound the corresponding factorial moments. The limiting Poisson moments determine the finite collection of counts on disjoint continuity intervals (their exponential generating function is finite near zero). Thus the mixed moment convergence gives those finite-dimensional count laws, and the first-moment bound gives local tightness. These characterize the vague point-process limit.

To pass from compact sets to the largest point, the $\ell = 1$ identity and $U_t \leq 1$ give

$$\mathbb{E}\#\{i : L_{i,n} > Lb_n\} \leq Cn \sum_{j>Lb_n} q_t(j) \leq C'L^{-3/2} \quad (197)$$

uniformly for bounded t_nb_n . This tends to zero as $L \rightarrow \infty$. The limiting intensity has no atoms, so the void probability above x is the exponential in (194). Finally, evaluating the integral at $\sigma = 0$ gives the stated critical formula. $\square$

Remark 28.3 (What the point process does not say). The intensity has infinite mass at zero. Accordingly the convergence is local away from zero, and the total number of blocks is not a continuous functional of this point process. Theorems 27.2 and 28.2 are separate marginal results. No joint convergence or independence between X_n and the extreme blocks is inferred from them. Their natural coupling is a further question in Section 31.

29 From tempered extremes to a Gumbel maximum

Theorem 29.1 (The supercritical Gumbel regime). *Suppose $t_n > 0$, $t_n \rightarrow 0$, and $nt_n^{3/2} \rightarrow \infty$. Define*

$$B_n = \frac{nr(t_n)d t_n^{3/2}}{m_{t_n}}, \quad w_n = \frac{5}{2}W_0 \left(\frac{2}{5}B_n^{2/5} \right). \quad (198)$$

Then $B_n \rightarrow \infty$, $w_n \rightarrow \infty$, and, vaguely on $\mathbb{R}$,

$$\sum_{i=1}^{K_n} \delta_{t_n L_{i,n} - w_n} \xrightarrow{d} \text{PPP}(e^{-x} dx). \quad (199)$$

In particular,

$$\boxed{\mathbb{P}(t_n M_n - w_n \leq x) \longrightarrow \exp(-e^{-x}) \quad (x \in \mathbb{R}).} \quad (200)$$

Replacing w_n by the incomplete-gamma center $\hat{w}_n$ of (178), with $B = B_n$, leaves the same limits.

Proof. Since $r(t_n) \rightarrow 1$, $m_{t_n} \rightarrow m$, and $d > 0$, the condition implies $B_n \rightarrow \infty$. By construction, $B_n e^{-w_n} w_n^{-5/2} = 1$. For $x = t_n j - w_n$ in a compact interval, $j = (w_n + x)/t_n \rightarrow \infty$, and the intensity per unit x is asymptotically

$$\begin{aligned} \frac{n}{m_{t_n}} \frac{q_{t_n}(j)}{t_n} &= B_n e^{-w_n - x} (w_n + x)^{-5/2} (1 + O(j^{-1})) \\ &= e^{-x} \left(1 + \frac{x}{w_n}\right)^{-5/2} (1 + o(1)) \longrightarrow e^{-x}. \end{aligned}$$

The lattice spacing in x is $t_n \rightarrow 0$, so Riemann sums converge to this continuous intensity.

The relevant lengths are $o(n)$: $w_n = O(\log B_n)$ and $w_n/(nt_n) \rightarrow 0$, because $nt_n^{3/2} \rightarrow \infty$ and $t_n \rightarrow 0$. The marked-block proof of Theorem 28.2 therefore applies with these shifted, shrinking-lattice coordinates. Its convolution factor is again $(n/m_{t_n})^\ell (1 + o(1))$ on every fixed compact interval of x . This proves the Poisson limit.

For passage to the maximum, use (190) with one marked block and the tail estimate

$$n \sum_{j > (w_n + L)/t_n} q_{t_n}(j) \leq C e^{-L}$$

for fixed large L and all sufficiently large n . It follows either by comparing the monotone exponential-power tail with its integral, or from (179); the relative lattice error tends to zero because $t_n \rightarrow 0$. The expected number of points above L is thus uniformly small as $L \rightarrow \infty$. The limiting void probability above x is $\exp(-e^{-x})$. Finally (180) gives $\hat{w}_n - w_n = o(1)$, so replacing the center has no effect on the limit. $\square$

The condition $t_n \rightarrow 0$ is essential to this continuous formulation. If $t > 0$ is held fixed, $tM_n - w_n$ remains on a lattice of fixed span; a non-oscillatory continuous Gumbel limit cannot be asserted in this form. Likewise, the Gaussian count theorem and the Gumbel theorem do not by themselves imply their asymptotic independence.

Remark 29.2 ([write], added 5 October 2026, batch 98: no continuous limit on a lattice of fixed span). The manuscript states without proof, here and in RQ6 of Section 31, that at a fixed $t > 0$ a continuous Gumbel claim “would be false”. It is true, in the following stronger form, which uses only that M_n is integer-valued. Let $t > 0$, let M_n be integer-valued random variables, let (c_n) be any real sequence, and let G be a distribution function that is continuous and strictly increasing on $\mathbb{R}$, such as $G(x) = \exp(-e^{-x})$. Then $F_n(x) = \mathbb{P}(tM_n - c_n \leq x)$ does not converge to $G(x)$ at every $x \in \mathbb{R}$, not even along a subsequence.

Proof. The variable $tM_n - c_n$ takes values in the lattice $t\mathbb{Z} - c_n$, so F_n is constant on each interval $[tk - c_n, t(k+1) - c_n)$, $k \in \mathbb{Z}$. Let $\alpha_n \in (-t, 0]$ be the left end of the one containing 0. Given a subsequence, pass to a further subsequence along which $\alpha_n \rightarrow \alpha \in [-t, 0]$, and put $x_1 = \alpha + t/4$, $x_2 = \alpha + 3t/4$. For all large n in it, $\alpha_n < x_1 < x_2 < \alpha_n + t$, hence $F_n(x_1) = F_n(x_2)$. Convergence at x_1 and x_2 would give $G(x_1) = G(x_2)$, contradicting strict monotonicity. $\square$

The same proof applies along any subsequence on which the spans t_n stay at least $t_* > 0$: the constancy interval containing 0 then contains an interval of length $t_*/2$ whose left end lies in

$[-t_*/2, 0]$, and the argument runs with $t_*/2$ in place of t . Hence a limit G of $t_n M_n - c_n$ that is continuous and strictly increasing forces $t_n \rightarrow 0$, the hypothesis of Theorem 29.1. The remark settles only the negative statement; the positive formulation, a lattice or oscillatory extreme-value law at fixed $t > 0$, remains RQ6.

29.1 A diagnostic of the distinction between the two transitions

In the pole-coalescence window $t = s/n$, $tb_n = O(n^{-1/3}) \rightarrow 0$. Thus the coefficient correction changes with s , but the scaled count and maximum have their *untempered* critical laws. In the tempered-statistics window $t = \sigma/b_n$ with fixed $\sigma > 0$, we instead have $nt \asymp n^{1/3} \rightarrow \infty$. The coefficient is already overwhelmingly represented by its simple pole, while the block statistics still change nontrivially with σ . This establishes mathematically why the two scales must not be conflated.

30 Exact checks, numerical diagnostics, and reproducibility

The programs accompanying this article distinguish exact integer assertions from floating-point diagnostics. They make no network requests. All decimal tables below are generated from the shipped CSV files by `code/make_tables.py`; they are not hand-entered.

30.1 Finite exact checks

The script `code/reproduce.py` computes (132) and (133) with Python integers through $n = 180$. Its initial values match the three OEIS entries in the stated conventions. It independently enumerates all nondecreasing tuples satisfying (128) for $1 \leq n \leq 10$. For each tuple it counts equality positions and compares the resulting complete component-count polynomial with (130). Every comparison passes. Thus the checks test not only the total count but also the parameter to which the probability theorems apply.

The coefficient recurrence after scaling by 4^{-n} is evaluated separately in double precision, and its values through $n = 30$ agree with the exact integer values to the programmed tolerance. A SymPy audit verifies exactly the coefficients in (149), (168), (154), and (171). None of these finite checks substitutes for the contour or probability proofs.

30.2 Constants and stable numerical evaluation

To compute λ, μ, β and the other constants, the implementation sums the rapidly convergent proper-divisor part R , not the slowly convergent full boundary series for A . The coefficient bound in Section 22 gives

$$|[z^n]R(\rho z)| \leq 2^{-n-1}.$$

Consequently all fixed derivative tails have explicit bounds obtained by summing a polynomial in n times 2^{-n-1} . The recorded run uses 180 terms and 60 decimal digits. Its decimal arithmetic is not outward-rounded; the data file records numerical values, not certified enclosing intervals.

For $t > 0$, put $s = \sqrt{1 - e^{-t}}$, $z = \rho e^{-t}$, and $M = zA'(z)$. The implementation uses

$$m_t = \frac{M}{e^{A(z)} - 1}, \quad v_t = \frac{(z\partial_z)^2 A(z)}{e^{A(z)} - 1} - \frac{M^2 e^{A(z)}}{(e^{A(z)} - 1)^2}, \quad (201)$$

with

$$M = -\log((1+s)/2) + zR'(z), \quad (z\partial_z)^2 A(z) = \frac{1-s}{2s} + (z\partial_z)^2 R(z).$$

This avoids numerical differentiation. The substitution $x = y^2$ removes the integrable endpoint singularity from $\mathcal{K}_0(0)$ and is also used for $\mathcal{K}_1$.

30.3 Accuracy of the pole–branch formula

Table 3 compares the normalized renewal value $U_t(n)$ from its finite convolution recurrence with the pole alone, with the pole plus $\mathcal{K}_0$, and with the additional $\mathcal{K}_1/n$ term. Errors are approximation minus recurrence value. Using U_t instead of Z_n avoids multiplication by a potentially large e^{nt} .

n	nt	pole error	with $\mathcal{K}_0$	with $\mathcal{K}_1$
256	0	-2.02×10^{-2}	9.94×10^{-6}	9.50×10^{-9}
256	1	-8.25×10^{-4}	9.89×10^{-7}	2.76×10^{-9}
256	4	-8.78×10^{-6}	2.03×10^{-8}	8.90×10^{-12}
1024	0	-1.01×10^{-2}	1.24×10^{-6}	2.96×10^{-10}
1024	1	-4.03×10^{-4}	1.22×10^{-7}	8.47×10^{-11}
1024	4	-4.12×10^{-6}	2.45×10^{-9}	1.64×10^{-13}
4096	0	-5.06×10^{-3}	1.55×10^{-7}	9.38×10^{-12}
4096	1	-1.99×10^{-4}	1.51×10^{-8}	2.74×10^{-12}
4096	4	-1.99×10^{-6}	3.00×10^{-10}	5.76×10^{-14}

Table 3: Absolute errors for $U_t(n)$ at the coefficient-coalescence scale. The last column at $nt = 4$, $n = 4096$ is close enough to floating-point accumulation error that it should not be read as an independently certified remainder.

At criticality, the $\mathcal{K}_0$ error falls as $n^{-3/2}$ and the $\mathcal{K}_1$ error as $n^{-5/2}$ over the displayed sizes, in agreement with the analytic expansion. For $nt > 0$, the exact values $a_n(t), r(t), m_t$ are retained; replacing them by their limits would invalidate this comparison at the indicated precision.

30.4 Tempered count and maximum diagnostics

Table 4 evaluates the centered characteristic function in Theorem 27.2 at frequency 1. The error decreases much more slowly than the coefficient error, consistent with the distinct $n^{-1/3}$ distributional correction scale. The table is evidence of consistency, not the proof of that rate.

n	σ	$\Re\phi_n(1)$	$\Im\phi_n(1)$	$ \phi_n(1) - \phi_\sigma(1) $
256	0	0.381898	0.433788	0.113691
256	0.5	0.729158	0.136592	0.119906
256	2	0.870128	0.050907	0.106057
1024	0	0.380056	0.392753	0.072624
1024	0.5	0.693124	0.115900	0.078387
1024	2	0.839215	0.042280	0.074047
4096	0	0.378532	0.366390	0.046221
4096	0.5	0.669442	0.101192	0.050514
4096	2	0.816392	0.034474	0.049941

Table 4: Tempered-window characteristic functions at frequency 1, with $t = \sigma/b_n$.

For maxima, the finite- n CDF is computed exactly at the generating-function level by setting $q_t(j) = 0$ for $j > \lfloor xb_n \rfloor$ in the numerator renewal recurrence. This operation counts the event that every block respects the cutoff, not an independent-block approximation. The subsequent recurrence arithmetic is floating point.

n	σ	x	finite- n CDF	limiting CDF
256	0	1	0.810173	0.754202
256	0.5	1	0.905816	0.893878
256	2	1	0.985485	0.985938
1024	0	1	0.787737	0.754202
1024	0.5	1	0.897967	0.893878
1024	2	1	0.984807	0.985938
4096	0	1	0.777633	0.754202
4096	0.5	1	0.896940	0.893878
4096	2	1	0.985382	0.985938

Table 5: Maximum-block CDFs at $x = 1$ in the tempered window. The lattice cutoff and conditioning both matter at these finite sizes. Additional $x = 2$ rows are in `data/maximum_cdf.csv`.

30.5 Gaussian and Gumbel diagnostics

Table 6 uses $t = n^{-1/4}$, which is asymptotically in the Gaussian/Gumbel regime. The variance ratio is the recurrence variance of K_n divided by nv_t/m_t^3 . The characteristic-function error is for the centered, variance-model-scaled count at frequency 1. The final two columns give the finite- n probability that tM_n is at most the Lambert core w_n or the incomplete-gamma center $\hat{w}_n$, respectively; the limiting value is $e^{-1} = 0.367879\dots$

n	variance ratio	CF error	core CDF	refined CDF
256	0.983698	0.060855	0.705897	0.360445
1024	0.994893	0.033471	0.666909	0.442421
4096	0.998366	0.018749	0.623533	0.387653

Table 6: Gaussian-count and maximum diagnostics. The crude Gumbel center has a large finite-size error; the refined center helps but still shows lattice oscillations.

In particular, the core maximum CDF is still about 0.624 at $n = 4096$. This is not close to its limiting 0.368. The analytic reason is that w_n is only logarithmically large, and replacing the incomplete-gamma tail by its leading term can be inaccurate. The refined column is not monotone, and no finite- n confidence bound is claimed.

30.6 Inverse checks

Table 7 compares the three displayed truncations of (168) with a high-precision root of the exact analytic model $h(t) = h$. The “exact-model” column is a numerical root, not a rational or interval-exact quantity.

h	exact-model t	order v error	through $v^{3/2}$	through v^2
1.00×10^{-2}	6.24×10^{-3}	-4.32×10^{-4}	5.36×10^{-6}	1.34×10^{-6}
1.00×10^{-3}	5.94×10^{-4}	-1.38×10^{-5}	4.45×10^{-8}	4.31×10^{-9}
1.00×10^{-4}	5.85×10^{-5}	-4.37×10^{-7}	4.15×10^{-10}	1.37×10^{-11}
1.00×10^{-5}	5.82×10^{-6}	-1.38×10^{-8}	4.06×10^{-12}	4.33×10^{-14}

Table 7: Numerical errors in the free-energy inverse. The last column has the predicted $O(h^{5/2})$ scale.

30.7 Replay commands and computational scope

From the package root, the recorded checks can be reproduced with

```
python -m pip install -r requirements.txt
python code/reproduce.py --max-n 4096 --dps 60
python code/symbolic_audit.py
python code/make_tables.py
./build.sh
```

The numerical and symbolic scripts accept `-output` to redirect their data; by default they overwrite the corresponding files in the package’s `data/` directory. Run on a copy to preserve the recorded outputs. The table formatter reads and writes the package’s own data directory. The numerical recurrence uses $O(N^2)$ arithmetic operations and $O(N)$ working storage, apart from the exact diagnostic arrays and small output tables. The all-orders proof does not depend on an unlimited symbolic implementation: the supplied numerical code evaluates $\mathcal{K}_0, \mathcal{K}_1$, while (159) specifies every fixed further order.

[write] **Added 5 October 2026, batch 98.** This subsection and Section 30 describe the delivered package. In this report its files carry the prefix `03-supercrit-`: `code/reproduce.py`, `code/symbolic_audit.py` and `code/make_tables.py` are `code/03-supercrit-reproduce.py` and so on, the two build scripts are in `code/`, `requirements.txt` and every `data/` file are in `data/` with the prefix, and the source notes are `03-supercrit-SOURCE_NOTES.md` at the report root. The delivered `article.tex`, `article.pdf`, `README` and checksum list are not shipped (they survive in the archive of arrival commit 2172df76a), so the build scripts, which compile an `article.tex` beside themselves, do not build this report, and `make_tables.py` reads and writes unprefixed names in the `data/` directory beside its own. The report’s `README.md` therefore reruns the programs on a scratch copy with the delivered layout. Such a rerun at the write (5 October 2026; Windows, Python 3.13.5 with NumPy 2.3.5, mpmath 1.3.0, SymPy 1.14.0; 21 s) exited 0: the exact CSV, the symbolic audit and the inverse errors were identical to the recorded files up to line endings, `verification.json` differed only in its interpreter, platform and timing fields, and the floating CSVs agreed to at most $9 \cdot 10^{-13}$ absolute ($5.6 \cdot 10^{-10}$ for the Gaussian/Gumbel diagnostics). Four last digits of Table 3 change on rerun, among them the entry near 10^{-13} at $n = 4096$, $nt = 4$ (5.76 becomes $6.56 \cdot 10^{-14}$), which the caption already marks as floating-point noise.

31 Further research questions

The following questions are not resolved by the preceding theorems. They are proposed in order of proximity to the proofs, rather than as claims that no related result exists in the broader literature.

RQ1. Joint count–extreme coupling. Prove joint convergence of X_n and the large-block point measure throughout the tempered window. The shared Lévy density suggests a

compensated-jump coupling, not independence. One must control the contribution of blocks whose scaled sizes tend to zero, which the vague point-process theorem discards.

- RQ2. Gaussian count versus Gumbel maximum.** Determine whether the Gaussian count fluctuation and the centered maximum become independent when $nt^{3/2} \rightarrow \infty$, and whether the answer is uniform as that quantity grows slowly. A joint complex-marked resolvent, not two marginal theorems, is needed.
- RQ3. Effective probability bounds.** Convert the compact-frequency error in (184) and the factorial-moment error in (195) into explicit Kolmogorov or suitable point-process distances. This requires control of growing frequencies or a coupling, not a direct relabeling of the present errors.
- RQ4. A full two-sided theory.** Combine the positive- t pole-branch formula with the repository's subcritical coexistence window in one analytic family. The continuation across the branch cut must account for the appearance or disappearance of a pole contribution. Uniformity across that change should be proved, not inferred by substituting negative t into a real probability law.
- RQ5. Growing order and exponentially smaller sectors.** Estimate the growth of $\mathcal{K}_j(a)$ as $j \rightarrow \infty$, locate an optimal truncation scale, and determine whether a Borel or hyperasymptotic treatment is useful. The fixed- J theorem alone neither proves a Gevrey bound nor identifies all exponentially small singularity contributions.
- RQ6. Fixed-supercritical lattice extremes.** For each fixed $t > 0$, formulate the correct subsequential or oscillatory extreme-value law. The lattice spacing no longer vanishes, so a continuous Gumbel claim with no phase correction would be false in the present normalization.
- RQ7. Explicit constants and certified calibration.** Make the zero-free slit size, remainder constants, and decimal values effective using interval arithmetic. Use these to certify an inverse weight from a prescribed finite-size count or expected block density, and to certify integer-threshold rounding away from or near the exceptional small neighborhoods in Section 26.
- RQ8. Other OEIS block classes and other tail exponents.** For exact sequence constructions with $q_j \sim dj^{-1-\alpha}$, $1 < \alpha < 2$, test and prove analogues of the two scales nt and $tn^{1/\alpha}$. The proof would need an appropriate complex continuation and a uniform cut estimate for each class; a formal replacement of $3/2$ by α is not yet a theorem.
- RQ9. Formalization.** Begin with the equality-index decomposition, the exact component polynomial, and the marked-block identity. Next formalize the local algebraic coefficient generator and the Puiseux inverse recursion. Contour deformation, uniform asymptotic remainders, and triangular probability limits require substantially more analytic infrastructure. No Lean or Rocq formalization is claimed for this manuscript.

31.1 Further questions and research recorded at the write

[write] **Added 5 October 2026, batch 98.** Under the repository's standing rule (4 October 2026) every claim of the manuscript that it does not prove in full is recorded here, with its source, its argument sketch and what is missing; none was found to be false. The write's own additions are marked as such.

- F1. Full tail expansion (Lemma 22.1).** The lemma asserts that q_j has an expansion in integral inverse powers of j “to every fixed order”. *Sketch (manuscript):* the cut integral for q_j with $v = x/j$ and a fixed-order Taylor expansion under an exponentially dominated integral, “the same domination argument” as in the proof of Theorem 24.1. *Missing:* the written domination at every order. Only the leading term $dj^{-5/2}(1+O(j^{-1}))$ is used later, and it is also Lemma 13.1 of Part II.
- F2. The sector bound in Lemma 27.1.** For complex r the proof asserts that, “after making the slit smaller”, analytic perturbations preserve the lower bound $|a+x| \geq c_\varepsilon(|a|+x)$ for the product of the two denominator factors $A_r + vH(v) \pm iv^{3/2}B(v)$. *Missing:* a written bound, uniform in the sector, for the relative perturbations $v(H(v) - m)/(A_r + mv)$ and $v^{3/2}B(v)/(A_r + mv)$ of the complex factors (the real-weight proof of Theorem 24.1 bounds the real sum of squares instead). Theorems 27.2 and 27.3 and item F3 depend on this lemma.
- F3. The block density (169).** The manuscript justifies $\theta(t) = \lim \mathbb{E}K_n/n = 1/m_t$ in one sentence (“differentiation of the simple-pole asymptotic in a complex neighborhood of the weight”). *[write] Proof, conditional on Lemma 27.1.* Fix $t \in (0, t_0]$ and write $\mathbb{E}K_n = r \partial_r \log Z_n(r)$ at $r = r(t)$. On a small closed disk D about $r(t)$, A_r stays in the sector of Lemma 27.1, the pole parameter $\tau(r)$, defined by $h(\tau(r)) = \log r$, is analytic with $\Re \tau \geq t/2$, and $h'(\tau(r)) \neq 0$. The lemma gives $Z_n(r) = e^{n\tau(r)} h'(\tau(r))^{-1} (1 + \epsilon_n(r))$ with ϵ_n analytic on D and $\sup_D |\epsilon_n| = O(e^{-nt/2})$, because $|e^{n\tau}| \geq e^{nt/2}$ while the remainders are $O(n^{-1/2}) + O(R_0^{-n})$. Cauchy’s estimate on the half-radius disk gives $\partial_r \log(1 + \epsilon_n) = O(e^{-nt/2})$ at $r(t)$. Since $\tau'(r) = 1/(rh'(\tau(r)))$,
- $$\mathbb{E}K_n = \frac{n}{h'(t)} - \frac{h''(t)}{h'(t)^2} + O(e^{-nt/2}) = \frac{n}{m_t} + \frac{v_t}{m_t^2} + O(e^{-nt/2}),$$
- which proves (169) with a bounded correction. *Missing:* only item F2.
- F4. Effectivity and the generator (Theorem 24.1).** t_0 and R_0 are not effective (the manuscript’s RQ7), and the generator (159) is said to produce integrals with “integrable behavior at zero for the combinations actually produced”, checked explicitly for $\mathcal{K}_0$ and $\mathcal{K}_1$. *[write] assessment:* the proof of Theorem 24.1 supplies the reason, since every relative correction it expands is $O(\epsilon x)$ or $O(\epsilon x^2/(a+x)) \leq O(\epsilon x)$, so each coefficient of ϵ^j is bounded near $x = 0$ by a polynomial times $x^{3/2}/(a+x)^2 \leq x^{-1/2}$. *Missing:* explicit constants (RQ7) and a term-by-term listing of the elementary integrals (160) beyond $j = 1$.
- F5. Why the $\sqrt{a}$ term must cancel (Section 25).** The manuscript says the cancellation of the $\sqrt{t}$ term of $1/m_t$ by the $\sqrt{a}$ term of $\mathcal{K}_0$ is “required by the analyticity of each fixed coefficient in the weight”. The cancellation itself holds: by (150) the pole term contributes $+\frac{3\kappa}{2m^2}\sqrt{t}$, and by (163) the branch term contributes $\frac{\kappa}{\pi m^2}n^{-1/2}(-\frac{3\pi}{2}\sqrt{a}) = -\frac{3\kappa}{2m^2}\sqrt{a/n}$, with $a/n = t + O(t^{3/2})$. *Missing:* the “required by analyticity” reasoning is a heuristic, since $Z_n(r)$ is a polynomial in r but the asymptotic terms are not shown to be individually analytic in r .
- F6. Continuation of Part I’s expansion to $\tau < 0$ ([write], bearing on RQ4 and on Part II’s RQ6).** *An exact identity.* For $a > 0$, with Part I’s entire kernel H of (32),

$$H(-a) = \frac{\mathcal{K}_0(a)}{\pi} + e^a \left(a^{3/2} + \frac{3}{2}a^{1/2} \right). \quad (202)$$

Proof. By (162) the right side equals $\pi^{-1/2}(1 + a + \sqrt{\pi a}(a + \frac{3}{2})e^a \operatorname{erf} \sqrt{a})$, and $H(-a) = {}_1F_1(2; \frac{1}{2}; a)/\sqrt{\pi}$. The classical series $\operatorname{erf} x = \frac{2}{\sqrt{\pi}}e^{-x^2} \sum_{j \geq 0} 2^j x^{2j+1}/(2j+1)!!$ and $(\frac{1}{2})_{j+1} =$

$(2j+1)!!/2^{j+1}$ give $M_1(a) := {}_1F_1(1; \frac{1}{2}; a) = 1 + \sqrt{\pi a} e^a \operatorname{erf} \sqrt{a}$. Kummer's relation $zM'(\alpha, b, z) = \alpha\{M(\alpha+1, b, z) - M(\alpha, b, z)\}$ at $\alpha = 1$ gives ${}_1F_1(2; \frac{1}{2}; a) = M_1(a) + aM_1'(a)$, and $aM_1'(a) = a + \sqrt{\pi} e^a \operatorname{erf} \sqrt{a} (\frac{1}{2}\sqrt{a} + a^{3/2})$. Adding, ${}_1F_1(2; \frac{1}{2}; a) = 1 + a + \sqrt{\pi a} (a + \frac{3}{2})e^a \operatorname{erf} \sqrt{a}$. $\square$ (At $a = 0.01, 0.3, 1, 3, 8$ both sides agree to the 40-digit working precision of a check at the write.)

A formal consistency. Put $\tau = -a$ in Part I's two-term formula (31), which is then evaluated outside its window; with Part II's dictionary its weight is $r = (1 - ma/n)^{-1}$, so $A_r = ma/n$ and $a_n(t) = a$ in Part III. Re-expand (155) with $J = 1$ at bounded a , using (i) $nt = a + \frac{\kappa}{m}a^{3/2}n^{-1/2} + O(n^{-1})$, from inverting $a = n(1 - Q(e^{-t}))/m$ with (138); (ii) $1/m_t = 1/m + \frac{3\kappa}{2m^2}\sqrt{t} + O(t)$ from (150); (iii) $r = 1 + O(n^{-1})$. The order- $n^{-1/2}$ terms are $\frac{\kappa}{m^2}\{e^a(a^{3/2} + \frac{3}{2}a^{1/2}) + \mathcal{K}_0(a)/\pi\}$ from Part III and $\frac{\kappa}{m^2}H(-a)$ from Part I, equal by (202). So Part III's first two terms reproduce Part I's at $\tau = -a$, with a discrepancy that the bookkeeping makes $O_C(n^{-1})$ for $0 \leq a \leq C$. Numerical evidence: at $a = 1$, with exact integer coefficients and 30-digit arithmetic (an intake computation, not shipped), $n\{Z_n(r) - \text{Part I's two-term model at } \tau = -1\}$ is 7.31, 7.09, 6.95, 6.84 for $n = 250, 500, 1000, 2000$. *Open question.* Prove that Part I's every-order expansion, Theorem 5.1 with its entire coefficients $C_m(\tau)$, holds uniformly on $-C \leq \tau \leq C \log n$, and identify how its Hankel contour, which excludes the pole $s = -\tau$ only for $\tau \geq 0$, must be modified when that pole crosses to $s = a > 0$ and contributes the residue e^a . *Missing:* the uniform bookkeeping of (i)–(iii) written as a proof (it rests on the unrefereed Theorem 24.1), and any statement beyond two terms. This is a special case of RQ4.

F7. RQ1 and Part II. The joint count–extreme coupling asked for in RQ1 is proved in Part II in another regime: the joint limit $(-Z, V)$ of (75), and the joint limit of the count and the point measure of large blocks (Section 16.3), conditionally on $\mathcal{F}_n$ at the subcritical coexistence weight. At $\sigma = 0$ and $t_n = 0$ the same coupling may transfer; RQ1 stays open for $\sigma > 0$ and for the unconditioned law.

F8. Priority of the critical-composition statements. Whether BFSS's Theorem 12, the announced companion of [14], or Flajolet–Sedgewick's Section IX.11 already contain the $\sigma = 0$ maximum law, the tempered window or the intermediate range $n^{-1} \ll t \ll n^{-2/3}$ is an open bibliographic question (note at the end of Section 21).

32 Conclusion

The supercritical moving-weight question has a clean resolution once one distinguishes a coefficient problem from a fluctuation problem. Exact exponential tilting isolates the moving pole, and a positive branch integral supplies a uniform, every-fixed-order correction expansion. The count statistics do not change at the pole-coalescence scale: their transition occurs later, at $tn^{2/3}$, where the $3/2$ -stable law is tempered. Beyond that scale the block count is Gaussian and the maximum has a shrinking-lattice Gumbel law. Convergent Puiseux inversion and tail-adapted centering make these formulas usable in the inverse direction.

The substantive new claim is this specified supercritical development relative to the inspected repository report. Established enumeration theorems are credited; computations are finite checks; the distinction between proved results, model inverses, and remaining joint or effective questions is maintained throughout.

[write] **Added 5 October 2026, batch 98.** Appendices A and B belong to Part II; Appendices C–E, after them, belong to Part III, whose sections precede these appendices.

A Notation and normalization checks

Symbol	Meaning
S_n, I_n	All score sequences and nonempty irreducible score sequences.
$S_{n,k}$	Number with exactly k irreducible blocks.
A, N_j	Logarithmic generating function and its divisor coefficients.
p	$I(1/4) = 1 - e^{-\lambda}$; not a phase probability.
$u, r = up$	Weight per irreducible block and its critical normalization.
q_j, Q	Probability mass function of a critical block length and its pgf.
m	Mean of q ; $m = \mu e^{-\lambda}/p$.
d	Tail amplitude: $q_j \sim dj^{-5/2}$.
κ	Puiseux and stable coefficient: $\kappa = d\Gamma(-3/2)$.
$Z_n(r)$	Normalized total weight $4^{-n}W_n(r/p)$.
$B_n(r)$	Unnormalized weight of the event $M_{1,n} > n/2$.
B_n	Second-maximum scale in (77), always without a parameter argument.
δ_n	$1 - r_n$, asymptotic to $m\tau_n/n$.
$b_n, \hat{b}_n$	Elementary and implicit leading-balance centers.
s, π_s	Bounded displacement from the center and limiting giant probability.
$K_n, M_{j,n}$	Number of blocks and ranked sizes.
a_n	$(\kappa n/m)^{2/3}$, used to normalize the fluid count.
G, Z	Gamma(2, 1) and the standard stable law (73).

Some checks prevent tempting normalization errors. First,

$$\Gamma(-3/2) = \frac{4\sqrt{\pi}}{3} > 0, \quad \frac{d}{m} = \frac{1}{2\mu\sqrt{\pi}}, \quad \frac{\kappa}{m} = \frac{2}{3\mu}.$$

Second, the block-count scale in the fluid theorem is a_n/m , not a_n ; inversion of a mean- m sum supplies the extra factor. Third, on $\mathcal{C}_n$,

$$\delta_n K_n \approx \frac{m\tau_n}{n} K_n, \quad \frac{\delta_n}{m}(n - M_{1,n}) \approx \frac{\tau_n}{n}(n - M_{1,n}).$$

Both converge to the same Gamma variable. Finally, the weights π_s and $1 - \pi_s$ are probabilities only after the exact large-block event has been identified.

B Dependency and validation record

The theorem dependencies are short enough to state without a large diagram. The exact identity (62) and the decomposition (60) yield the renewal model. The analytic split proves the local expansion and tail. The local expansion yields both the uniform two-term coefficient estimate and the Fourier local limit theorem. Independently, the tail and the finite mean yield the giant-deletion total

variation theorem. The total weight and the giant weight identify the mixture. Central local-limit windows exhaust its other component. The residual transforms then give all finer giant-phase laws. The exact enumerative theorem is the only substantial outside combinatorial input whose full proof is not reproduced. Standard complex integration, Fourier inversion, weak-convergence criteria, and the strong law of large numbers are used in their usual forms. No unproved probabilistic large-deviation estimate is inserted. The full source and the output records make the arithmetic and typesetting reproducible; analytic correctness rests on the proofs.

C Notation and normalization dictionary (Part III)

symbol	meaning
S_n, I_n	all and strong score-sequence counts; $S_0 = 1, I_0 = 0$
N_n	A145855, the exact divisor sum in (132)
A, F, R	logarithm of S , its central-binomial singular part, and its analytic proper-divisor remainder
$W_n(u)$	weighted score count with weight u per strong block; not Lambert's function
ρ	the unweighted coefficient radius, $1/4$
λ, μ	$A(\rho)$ and $\rho A'(\rho)$
p, u_c	$I(\rho) = 1 - e^{-\lambda}$ and the critical block weight p^{-1}
Q, q_j	critical probability generating function $I(\rho z)/p$ and its mass function
m, κ, d	critical block mean; coefficient of $(1 - z)^{3/2}$ in Q ; mass-tail constant
β, κ_1	coefficients of $(1 - z)^2$ and $(1 - z)^{5/2}$ in Q
r	normalized weight up , critical at 1
$t, h(t)$	supercritical free-energy increment and $\log r(t)$
q_t, m_t, v_t	exponentially tilted mass, its mean, and its variance
$Z_n(r)$	normalized weighted count $4^{-n}W_n(r/p)$
$U_t(n)$	renewal visit probability, $e^{-nt}Z_n(r(t))$
$A_r, a_n(t)$	$1 - r^{-1}$ and nA_r/m ; neither is the function $A(z)$
H, B	analytic real and imaginary factors of the boundary values of Q , (142)
$\mathcal{K}_j$	uniform cut-integral correction kernels
b_n	stable block scale $(\kappa n/m)^{2/3}$
$K_n, L_{i,n}, M_n$	block number, ordered block lengths, and maximum length
σ_n	$t_n b_n$, the tempering parameter
B_n, w_n	large-tail intensity parameter and Lambert-core maximum center; B_n is not $B(v)$
W_0	principal real Lambert- W function on positive arguments

D Source audit and limitations (Part III)

The source search covered the three OEIS entries, the CDFS exact-enumeration paper, Kolesnik's asymptotic paper, the Bassan–Donderwinkel–Kolesnik treatment, the cited tempered-stable reference, and the current repository report's scope and proofs as available during this session. It was not an exhaustive MathSciNet, zbMATH, or journal-archive search. The assertion that the supercritical question is unanswered in the continued repository item is supported by its pinned README; that is narrower than a claim that no general theorem in probability theory could imply a related result.

The principal exact input is (131)–(132). Its attribution is independent of the unrefereed repository manuscript. We have rederived the slit expansion, all analytic estimates used here, the renewal transformation, the inverse formulas, and the probability limits. The critical endpoint and leading heavy-tail constant overlap established or preceding results and are used as checks, not as priority claims.

All probability statements concern the measure on *score sequences* with weight u^K , not a uniformly random labeled tournament. The maximum theorem at a positive fixed weight would require a

lattice qualification and is deliberately not stated. The two point-process theorems are local in their indicated state spaces. The block-count and extreme-value limits are not claimed jointly. The O constants are not numerically certified. The coefficient expansion is for each fixed truncation order. The Puiseux inverse is genuinely convergent locally, but that fact should not be transferred to the inverse- n expansion.

E Package contents (Part III)

The package includes `article.tex`, `article.pdf`, a build script and dependency list, a source-provenance note, and three Python scripts. The `data/` directory contains exact OEIS terms through 80, the full finite-check log through 180 with independent component-polynomial enumeration through 10, symbolic-audit output, five numerical CSV tables, and their LaTeX display fragments. The recorded numerical run reaches $n = 4096$. The build does not need network access, an external bibliography database, or external figure files. Font files are not distributed.

[write] **Added 5 October 2026, batch 98.** In this report the package is shipped as described in the note at the end of Section 30: everything except the delivered `article.tex` (printed as Part III), `article.pdf`, `README` and `SHA256SUMS.txt` (a checksum list, verified at placement and retired), with the prefix `03-supercrit-`. The “full finite-check log” is `data/03-supercrit-run.log`, a transcript of the program’s standard output, which the program prints but does not write; likewise `data/03-supercrit-symbolic_run.log`. `data/03-supercrit-build_quality.json` records the delivered PDF’s build (27 pages, three pdfLaTeX passes) and the SHA-256 of the delivered source and PDF; its “no LaTeX warnings” omits the pdfTeX warning about a duplicate destination `page.1` that the title page causes in a rebuild of the delivered source.

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